Optimal Multistage Selection

Evidence from Graduate Admissions

Jeremy Rosen

Georgetown University

October 17, 2026

Motivation

- **Economic Problem:** Selecting best options under costly evaluation
  - Determining each option's true quality costs decision-maker time and effort
  - Low-cost screening of all options $\Rightarrow$ high-cost evaluation of surviving options
  - **Examples:** Admissions and hiring, clinical trials, matching platforms
- **Why Graduate Admissions:** Benefits graduate programs and future applicants
  - Graduate programs invest time and money into students' development
  - Shrinking PhD cohorts (Angrist et al. 2026) $\Rightarrow$ each admission matters more
  - Programs may be undervaluing applicants with certain characteristics
  - Future such applicants may receive currently-withheld opportunities

Research Questions

1. Are admissions committees optimizing, or are they making mistakes?
   - Goal is to accept applicants with highest success probabilities who will matriculate
   - Per literature, measure success with program completion and job placement ranking
   - Predictors of acceptance only (success only) may be overweighted (underweighted)
2. If they are making mistakes, how can they make better admissions decisions?
   - In turn, by changing decision rules, how much better of a cohort can they admit?
   - Look for applicants who were rejected, would have matriculated, and succeeded

★ Methods apply to undergraduate/professional admissions, hiring, and other settings

Methods and Results

- Build structural models of admissions committee's decision problem that support:
  - Statistical tests of whether committee's decision rule aligns with desired objective
  - Counterfactual simulations quantifying gain from deviating to model's decision rule
- Estimate model parameters with admissions data from a major research university
  - On most dimensions, estimated success probabilities align with committee's decisions
  - Reweighting some applicant characteristics could have increased admitted cohort's average success probability by 4 percentage points without reading more files

Contribution to Literature

- **Graduate Admissions:** Regress acceptance and success on applicant characteristics
  - Krueger & Wu (2000); Athey et al. (2007); Stock et al. (2006–2015); Bai et al. (2022)
- **Optimal Multistage Selection:**
  - **Information Acquisition:** Weitzman (1979); Sims (2003); Matějka & McKay (2015)
  - **Optimality:** Cho & Rust (2010); Hoffman et al. (2018); Kleinberg et al. (2018); Rambachan (2024); Anderson, Rosen, Rust & Wong (2025)
  - **Multistage Systems:** Baron et al. (2026); Lessem & Miller (2026)

★ **This Paper:** Builds and estimates multistage selection models of admissions, which relate acceptance effects to success effects through the committee's problem

Synthetic Data Protocol (IRB-Approved)

1. **Administrative office receives data from university departments**
   - Algorithmically extracts variables from textual files, such as recommendation letters
2. **Synthetic data (Patki et al. 2016) used for model development**
   - Estimate joint distribution of variables, and simulate synthetic dataset from it
   - Rows do not correspond to real subjects, preventing identity disclosure
   - Cells with < 5 comparable observations binned to prevent attribute disclosure
3. **Administrative office executes final estimation code on actual data**
   - Only summary statistics and parameter estimates are returned to researcher

★ **Contribution:** Privacy-protecting framework for researchers studying sensitive data

Synthetic Data Example

Outline

1. Regression Analysis
2. Structural Models
3. Structural Estimation
4. Optimality and Deviation Gains

Data Description

- **Sample:** Applicants to research university's economics PhD program from 2015–2019
- **Outcomes:** Measures of admission and success
  - **Admission:** Read File $\Rightarrow$ Accept $\Rightarrow$ Matriculate $\Rightarrow$ Success
  - **Success:** Completion and placement category/quality for **all** applicants, with academia-equivalent rankings for non-academic placements (Krueger & Wu 2000)
- **Covariates:** Objective vs. subjective variables
  - **Objective:** Demographics and performance measures, such as GRE scores
  - **Subjective:** Textual processing of recommendation letters, statements of purpose, CVs, and supplemental forms using dictionaries to construct variables

Variables | Inference vs. Prediction | Sample Selection

Regression Results

- **Method:** Logistic regressions for acceptance, matriculation, and success
  - **Success:** *Top 100 Placement* $= 1$ for 20% of applicants vs. 34% of matriculants
- **Results:** Different independent variables best predict each dependent variable
  - **Accept:** *Committee Score* is best predictor. Other predictors: demographics, *GRE Quantitative Percentile*, *Work Experience*, *CV Coding* (negative)
  - **Matriculate:** Predictors of acceptance often negatively predict matriculation. Best predictor of matriculation is *Open House* attendance
  - **Success:** *PhD Program Rank* is best predictor. Rank held equal, other predictors: *Age*, *TOEFL/IELTS*, *Work Experience*, *Essay Positive Terms (%)*

Accept/Matriculate/Success AMEs | Summary Statistics | Logistic Regression Results | Lasso Logistic Regression Results

Acceptance vs. Success AMEs

Dependent Variable	Accept	T100 Placement
Age	-0.0001	0.0053*
Female	0.0163**	-0.0032
International Asian	-0.0214***	-0.0127
GRE Quantitative Percentile	0.0011**	0.0006
TOEFL/IELTS	-0.0022	0.0409*
Work Experience	0.0140**	0.0358**
Committee Score	0.0788***	0.0007
CV Coding	-0.0044*	0.0125
Essay Economics Terms (%)	0.0001	-0.0104
Essay Positive Terms (%)	-0.0013	0.0123*

» Diversity Objective » Insufficiency of Comparing AMEs
» Regressions vs. Model

- Admissions literature regresses acceptance and success on variables in applications, and qualitatively compares the effects
- While such comparisons are of interest, they are insufficient to assess optimality
- Need model of admissions to relate acceptance effects to success effects

Outline

1 Regression Analysis

2 Structural Models

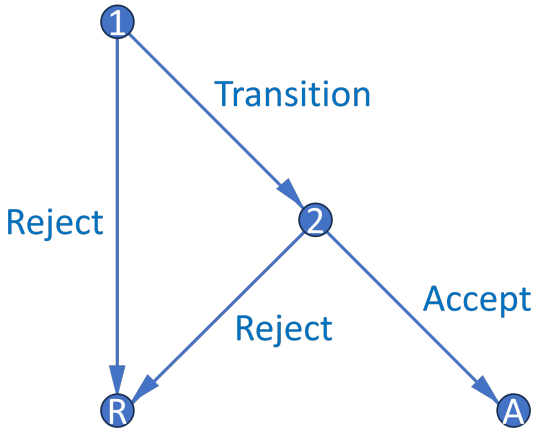
3 Structural Estimation

4 Optimality and Deviation Gains

Why Structural Models?

- **Reason 1:** Test whether committee's decision rule aligns with desired objective
  - Maximize admitted cohort's quality, considering matriculation probabilities
- **Reason 2:** Find counterfactual decision rule in which committee optimizes
  - Quantify gains from deviating from actual to counterfactual decision rule

Two-Round Selection



- **Round 1:** See only objective variables. Transition to Round 2 or reject
- **Round 2:** See survivors' objective and subjective variables. Accept or reject
- Assume for now number of applicants transitioned and accepted is fixed
- Matriculation model will account for success and matriculation probabilities

Portfolio Choice Problem

For each applicant n, choose $d_{1,n} \in \{\text{Transition}, \text{Reject}\}$ and $d_{2,n} \in \{\text{Accept}, \text{Reject}\}$:

Round 2:

$$\max_{(d_{2,1},\ldots,d_{2,N_T}) \in \{A,R\}^{N_T}} \sum_{n=1}^{N_T} \underbrace{P(y_n = S | x_n, z_n)}_{\text{Success probability}} \underbrace{\mathbb{1}(d_{2,n} = A)}_{\text{Acceptance}}, \text{ where: } \sum_{n=1}^{N_T} \underbrace{\mathbb{1}(d_{2,n} = A)}_{\text{Acceptance}} \leq N_A,$$

Round 1:

$$\max_{(d_{1,1},\ldots,d_{1,N}) \in \{T,R\}^{N}} \sum_{n=1}^{N} \underbrace{EV_{2,n}(x_n)}_{\mathbb{E}[\text{Round 2 Value}]} \underbrace{\mathbb{1}(d_{1,n} = T)}_{\text{Transition}}, \text{ where: } \sum_{n=1}^{N} \underbrace{\mathbb{1}(d_{1,n} = T)}_{\text{Transition}} \leq N_T,$$

where x = objective variables, z = subjective variables, $y \in \{\text{Success}, \text{Failure}\}$, N = number of applicants, (N_T, N_A) = max number of files read/acceptances

Portfolio Choice to Cutoff Rule

- How to estimate model, given so many possible combinations of applicants?
- Assuming no complementarities, portfolio choice problem has cutoff rule solution
- **Round 2:** Accept transitioned applicant if success probability exceeds Round 2 cutoff

$$d^*_{2,n} = \begin{cases} A & \text{if } P(y_n = S|x_n, z_n) > \text{Cutoff}_2 \\ \{A, R\} & \text{if } P(y_n = S|x_n, z_n) = \text{Cutoff}_2 \\ R & \text{if } P(y_n = S|x_n, z_n) < \text{Cutoff}_2 \end{cases}$$

- **Round 1:** Transition applicant if Round 2 expected value exceeds Round 1 cutoff

$$d^*_{1,n} = \begin{cases} T & \text{if } EV_{2,n}(x_n) > \text{Cutoff}_1 \\ \{T, R\} & \text{if } EV_{2,n}(x_n) = \text{Cutoff}_1 \\ R & \text{if } EV_{2,n}(x_n) < \text{Cutoff}_1 \end{cases}$$

Lagrangians

Cutoff Rule Model

- **Goal:** Use choices and outcomes data to estimate success probabilities, while respecting constraint on number of files committee can read

- **Legend:** Estimated model objects vs. Choices and outcomes data

Cutoff Rule Model, Round 2

$$v_2(x, z, t, \overline{k}, \epsilon_2) = \max_{d_2 \in \{A,R\}} u_2(x, z, t, \overline{k}, d_2) + \epsilon_2(d_2), \text{ where:}$$

$$u_2(x, z, t, \overline{k}, d_2) = \underbrace{P(y = S|x, z, t, \overline{k})\mathbb{1}(d_2 = A)}_{\text{If accept, get } P(\text{Success})} + \underbrace{P(y_2^* = S|t)\mathbb{1}(d_2 = R)}_{\text{If reject, get Round 2 Cutoff}}$$

- **States:** (x, z) = objective and subjective variables, t = year, $\overline{k}$ = fixed program rank, (ϵ_1, ϵ_2) = Type 1 extreme value preference shocks with scale parameter σ
- **Choices:** $d_1 \in \{\text{Transition}, \text{Reject}\}$, $d_2 \in \{\text{Accept}, \text{Reject}\}$
- **Outcomes:** $y \in \{\text{Success}, \text{Failure}\}$, (y_1^*, y_2^*) = marginal outcomes

Cutoff Rule Model, Round 1

$$v_1(x, t, \overline{k}, \epsilon_1) = \max_{d_1 \in \{T, R\}} u_1(x, t, \overline{k}, d_1) + \epsilon_1(d_1), \text{ where:}$$

$$u_1(x, t, \overline{k}, d_1) = \underbrace{\mathbb{E}[v_2(x, z, t, \overline{k}, \epsilon_2)|x, t, \overline{k}]\mathbb{1}(d_1 = T)}_{\text{If transition, get } \mathbb{E}\text{[Round 2 Value]}} + \underbrace{L(y_1^* = S|t)\mathbb{1}(d_1 = R)}_{\text{If reject, get Round 1 Cutoff}}$$

- **States:** (x, z) = objective and subjective variables, t = year, $\overline{k}$ = fixed program rank, (ϵ_1, ϵ_2) = Type 1 extreme value preference shocks with scale parameter σ
- **Choices:** $d_1 \in \{\text{Transition}, \text{Reject}\}$, $d_2 \in \{\text{Accept}, \text{Reject}\}$
- **Outcomes:** $y \in \{\text{Success}, \text{Failure}\}$, (y_1^*, y_2^*) = marginal outcomes

Conditional Choice Probabilities

- **Round 2:** Higher payoff from accepting $\Rightarrow$ higher probability of accepting

$$P(d_2 = A|x, z, t, \overline{k}) = \frac{\exp(u_2(\cdot, A)/\sigma)}{\exp(u_2(\cdot, A)/\sigma) + \exp(u_2(\cdot, R)/\sigma)}$$

- **Round 1:** Higher payoff from reading file $\Rightarrow$ higher probability of reading file

$$P(d_1 = T|x, t, \overline{k}) = \frac{\exp(u_1(\cdot, T)/\sigma)}{\exp(u_1(\cdot, T)/\sigma) + \exp(u_1(\cdot, R)/\sigma)}$$

Dynamic Parameterization | Static Model | Maximum Likelihood Estimation

Portfolio Choice with Matriculation

Round 2:

$$\max_{(d_{2,1},\ldots,d_{2,N_T})\in\{A,R\}^{N_T}} \sum_{n=1}^{N_T} \underbrace{P(y_n = S|x_n, z_n)}_{\text{Success probability}} \underbrace{P(m_n = M|x_n, z_n)}_{\text{Matriculation probability}} \underbrace{\mathbb{1}(d_{2,n} = A)}_{\text{Acceptance}},$$

$$\text{where: } \sum_{n=1}^{N_T} \underbrace{P(m_n = M|x_n, z_n)}_{\text{Matriculation probability}} \underbrace{\mathbb{1}(d_{2,n} = A)}_{\text{Acceptance}} \leq N_M,$$

Round 1:

$$\max_{(d_{1,1},\ldots,d_{1,N})\in\{T,R\}^{N}} \sum_{n=1}^{N} \underbrace{EV_{2,n}(x_n)}_{\mathbb{E}[\text{Round 2 Value}]} \underbrace{\mathbb{1}(d_{1,n} = T)}_{\text{Transition}}, \text{ where: } \sum_{n=1}^{N} \underbrace{\mathbb{1}(d_{1,n} = T)}_{\text{Transition}} \leq N_T,$$

where (x, z) = variables, $y \in \{\text{Success}, \text{Failure}\}$, $m \in \{\text{Matriculate}, \text{Decline}\}$, N = number of applicants, (N_T, N_M) = max number of files read/matriculants

Cutoff Rule with Matriculation

$$\max_{(d_{2,1},\ldots,d_{2,N_T})\in\{A,R\}^{N_T}} \sum_{n=1}^{N_T} \underbrace{P(y_n = S|x_n, z_n)}_{\text{Success probability}} \underbrace{P(m_n = M|x_n, z_n)}_{\text{Matriculation probability}} \underbrace{\mathbb{1}(d_{2,n} = A)}_{\text{Acceptance}},$$

$$\text{where: } \sum_{n=1}^{N_T} \underbrace{P(m_n = M|x_n, z_n)}_{\text{Matriculation probability}} \underbrace{\mathbb{1}(d_{2,n} = A)}_{\text{Acceptance}} \leq N_M,$$

Theorem 1

For large N_T, Round 2 problem has a cutoff rule solution on **success** *probabilities.*

- Likely matriculants more valuable, but also more expensive ⇒ effects cancel
- Matriculation probabilities matter in Round 1, especially for low-ranked programs

Lagrangians with Matriculation

Cutoff Rule with Matriculation, Round 2

$$v_2(x, z, t, \overline{k}, \epsilon_2) = \max_{d_2 \in \{A,R\}} u_2(x, z, t, \overline{k}, d_2) + \epsilon_2(d_2), \text{ where:}$$

$$u_2(x, z, t, \overline{k}, d_2) = \underbrace{P(y = S|x, z, t, \overline{k})\mathbb{1}(d_2 = A)}_{\text{If accept, get } P(\text{Success})} + \underbrace{P(y_2^* = S|t)\mathbb{1}(d_2 = R)}_{\text{If reject, get Round 2 Cutoff}}$$

- **States:** (x, z) = objective and subjective variables, t = year, $\overline{k}$ = fixed program rank, (ϵ_1, ϵ_2) = Type 1 extreme value preference shocks with scale parameter σ
- **Choices:** $d_1 \in \{\text{Transition}, \text{Reject}\}$, $d_2 \in \{\text{Accept}, \text{Reject}\}$
- **Outcomes:** $y \in \{\text{Success}, \text{Failure}\}$, $m \in \{\text{Matriculate}, \text{Decline}\}$, (y_1^*, y_2^*, m_1^*) = marginal outcomes

Cutoff Rule with Matriculation, Round 1

$$v_1(x, t, \overline{k}, \epsilon_1) = \max_{d_1 \in \{T,R\}} u_1(x, t, \overline{k}, d_1) + \epsilon_1(d_1), \text{ where:}$$

$$u_1(x, t, \overline{k}, d_1) = \underbrace{\mathbb{E}[v_2(x, z, t, \overline{k}, \epsilon_2)|x, t, \overline{k}]\mathbb{1}(d_1 = T)}_{\text{If transition, get } \mathbb{E}[\text{Round 2 Value}]} + \underbrace{L(y_1^* = S, m_1^* = M|t)\mathbb{1}(d_1 = R)}_{\text{If reject, get Round 1 Cutoff}}$$

- **States:** (x, z) = objective and subjective variables, t = year, $\overline{k}$ = fixed program rank, (ϵ_1, ϵ_2) = Type 1 extreme value preference shocks with scale parameter σ
- **Choices:** $d_1 \in \{\text{Transition}, \text{Reject}\}$, $d_2 \in \{\text{Accept}, \text{Reject}\}$
- **Outcomes:** $y \in \{\text{Success}, \text{Failure}\}$, $m \in \{\text{Matriculate}, \text{Decline}\}$, (y_1^*, y_2^*, m_1^*) = marginal outcomes

Importance of Matriculation

$$\lim_{\sigma \to 0} EV_2 = \mathbb{E}_{\widetilde{z}}[\max\{P(y = S|x, \widetilde{z}, t, \overline{k}) - P(y_2^* = S|t), 0\} P(m = M|x, \widetilde{z}, t)|x, t, \overline{k}]$$

- Matriculation matters when expected success probability $\gg$ Round 2 cutoff
- **Upshot:** Low-ranked programs must consider matriculation in Round 1

Outline

1 Regression Analysis

2 Structural Models

3 Structural Estimation

4 Optimality and Deviation Gains

Maximum Likelihood Estimation

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N} \underbrace{f\left(z_n | x_n, t_n; \theta_F\right)}_{\text{Density of } z|x,t} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)}(1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}}$$

$$\prod_{n=1}^{N} \underbrace{P_{d_{1,n}}(\theta_T)^{\mathbb{1}(d_{1,n}=T)}\left(1 - P_{d_{1,n}}(\theta_T)\right)^{\mathbb{1}(d_{1,n}=R)}}_{\text{Round 1 conditional choice probabilities}} \prod_{n=1}^{N_T} \underbrace{P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)}(1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Round 2 conditional choice probabilities}}$$

- In unrestricted likelihood $\mathcal{L}^U$, parameterize CCPs with logistic specifications
- In restricted likelihood $\mathcal{L}^R$, use model CCPs, which depend on structural parameters
- $\mathcal{L}^R$ aligns success probabilities implied by outcomes with those implied by CCPs

Conditional Choice Probabilities | Matriculation-Dynamic Likelihood | Empirical Considerations

Cutoff Probability Identification

- Identify each year's cutoffs by how selective committee is that year
  - Fewer acceptances and stronger applicants increase cutoff
  - More acceptances and weaker applicants decrease cutoffs
- While selectivity identifies cutoffs, sorting improves restricted likelihood

Theorem 2

Cutoff probability estimate is such that number of acceptances N_A equals expected N_A.

For any cutoff and N_A, likelihood is highest when committee accepts best N_A applicants.

Restricted Estimates

	Round 1 Cutoff	Round 2 Cutoff	Sigma
2015	0.3731	36.58%	
	(0.3067, 0.4539)	(29.81%, 43.92%)	
⋮	⋮	⋮	
2019	0.3887	38.14%	
	(0.3268, 0.4624)	(31.89%, 44.81%)	
Sigma			0.0020
			(0.0013)

- Round 1 cutoffs relative to EV_2. Round 2 cutoffs relative to success probabilities
- Cutoffs stable across years. $\hat{\sigma}$ small $\Rightarrow$ distorted restricted success probabilities

Dynamic Estimates | Static Estimates | Myopic Estimates

Restricted Estimates with Matriculation

	Round 1 Cutoff	Round 2 Cutoff	Sigma
2015	0.0071 (0.0026, 0.0194)	36.35% (29.47%, 43.84%)	
⋮	⋮	⋮	
2019	0.0095 (0.0035, 0.0255)	37.41% (31.04%, 44.26%)	
Sigma			0.0020** (0.0010)

- With matriculation, EV_2's functional form yields lower Round 1 cutoffs

⏩ Matriculation-Dynamic Estimates

Outline

1. Regression Analysis
2. Structural Models
3. Structural Estimation
4. Optimality and Deviation Gains

Deviation Gains

- Suppose committee adopted model's decision rule with $\sigma = 0$ and N_T unchanged
- Would average success probability of acceptance/matriculation set have increased?
  1. Select applicants with model, and determine its acceptance/matriculation set
  2. Compute average success probability of its acceptance/matriculation set
  3. Compare average success probability to that of set chosen by committee's rule
- Plug $\hat{\theta}_S^U$ into model because $\hat{\theta}_S^R$ may be distorted to rationalize suboptimal behavior

» Deviation Gains Procedure

Deviation Gains Test

- Is deviation gain significant? Do Z-test of $H_0 \colon g(\theta_S^0) = 0$ vs. $H_1 \colon g(\theta_S^0) > 0$
  1. Gain is approximately normal, so test statistic is standardized gain. **Or:**
  2. Compute average success probabilities across B perturbations $\tilde{\theta}_{S,b}^U$ of $\hat{\theta}_S^U$

Theorem 3

Fraction of perturbations with model's $\overline{P(Success)} \leq$ *committee's converges to p-value.*

- Model's set must be exogenously chosen, e.g. train and test on different data
  - If not, gain will be biased upward, and test will be biased toward rejection
  - In bigger dataset, such as hiring at large company, holdout set is feasible

Deviation Gains Correction

- Observed deviation gain is combination of true gain and "winner's curse"

$$g(\hat{\theta}_S^U) = \underbrace{\overline{P}_{\mathcal{A}^R(\hat{\theta}_S^U)}(\hat{\theta}_S^U)}_{\text{Model}} - \underbrace{\overline{P}_{\mathcal{A}^U}(\hat{\theta}_S^U)}_{\text{Committee}}$$

$$= \underbrace{\overline{P}_{\mathcal{A}^R(\hat{\theta}_S^U)}(\hat{\theta}_S^U) - \overline{P}_{\mathcal{A}^R(\theta_S^0)}(\hat{\theta}_S^U)}_{\text{Winner's curse}} + \underbrace{\overline{P}_{\mathcal{A}^R(\theta_S^0)}(\hat{\theta}_S^U) - \overline{P}_{\mathcal{A}^U}(\hat{\theta}_S^U)}_{\text{True gain at } \hat{\theta}_S^U}$$

- Subtract estimate of winner's curse from observed deviation gain

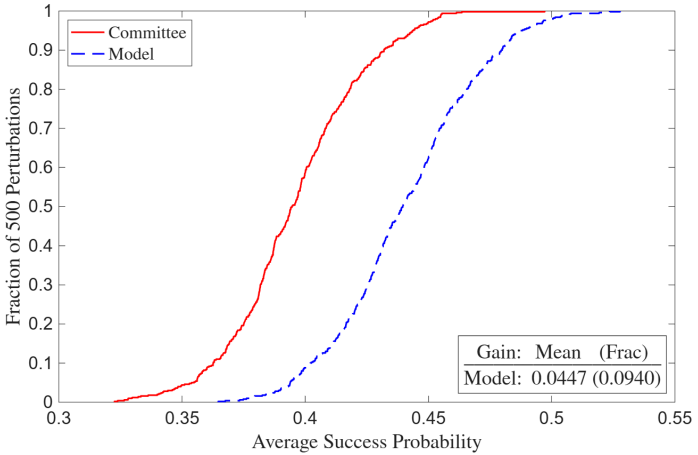
$$\widehat{\text{Winner's curse}} = \frac{1}{B}\sum_{b=1}^{B}\left[\overline{P}_{\mathcal{A}^R(\tilde{\theta}_{S,b}^U)}(\tilde{\theta}_{S,b}^U) - \overline{P}_{\mathcal{A}^R(\hat{\theta}_S^U)}(\tilde{\theta}_{S,b}^U)\right]$$

Theorem 4

Correction is exact in some cases and removes at least $\sqrt{2}-1$ of curse in expectation.

Deviation Gains Interpretation

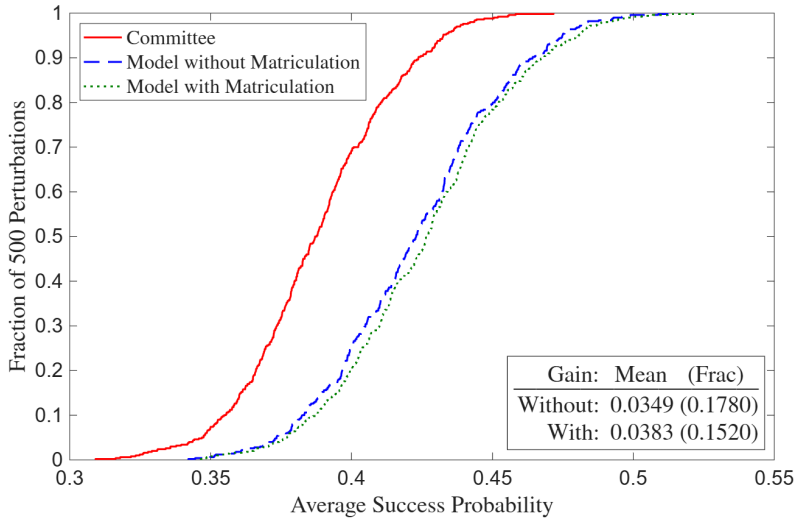
Deviation Gains (Acceptance Set)



- Random selection from applicant pool yields $\overline{P(\text{Success})} = 35\%$

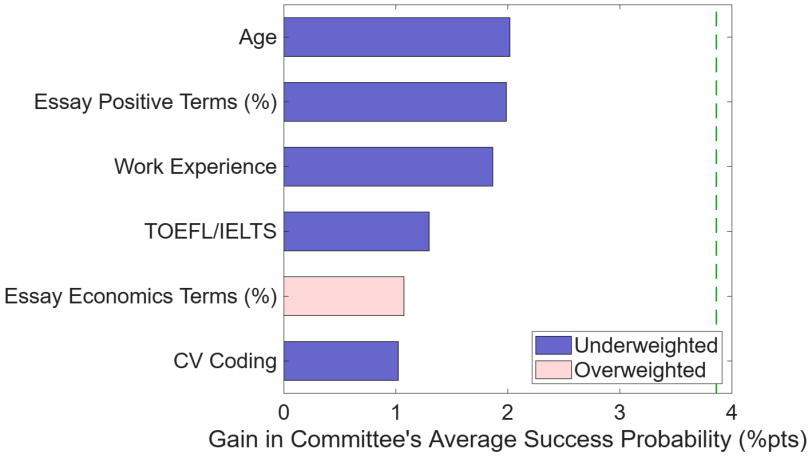
Full Deviation Gains (Acceptance Set) | Uncorrected Deviation Gains (Acceptance Set) | Deviation Gains Robustness

Deviation Gains (Matriculation Set)



Full Deviation Gains (Matriculation Set) | Uncorrected Deviation Gains (Matriculation Set) | Deviation Gains Robustness

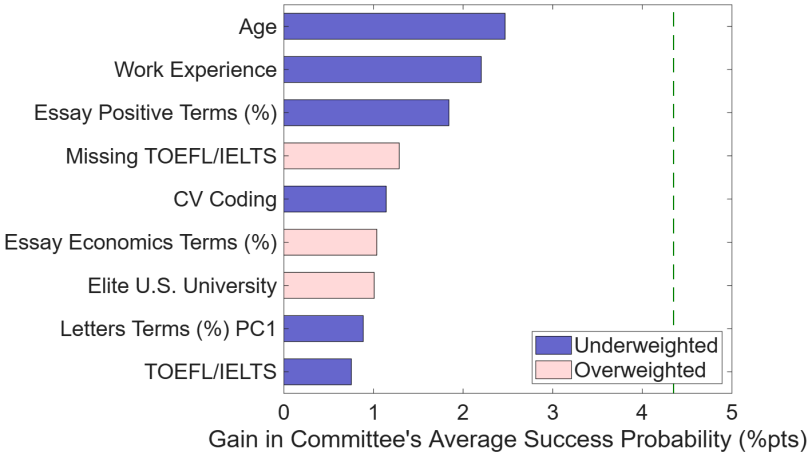
Deviation Gains Attribution (Matriculation Set)



1. Experience
2. English Proficiency
3. Essay Subtext
4. Industry Background

Full Deviation Gains Attribution (Matriculation Set)
Deviation Gains Attribution Table (Matriculation Set)

Deviation Gains Attribution (Acceptance Set)



1. Experience
2. English Proficiency
3. Essay/Letters Subtext
4. Industry Background
5. Undergraduate

» Full Deviation Gains Attribution (Acceptance Set)
» Deviation Gains Attribution Table (Acceptance Set)

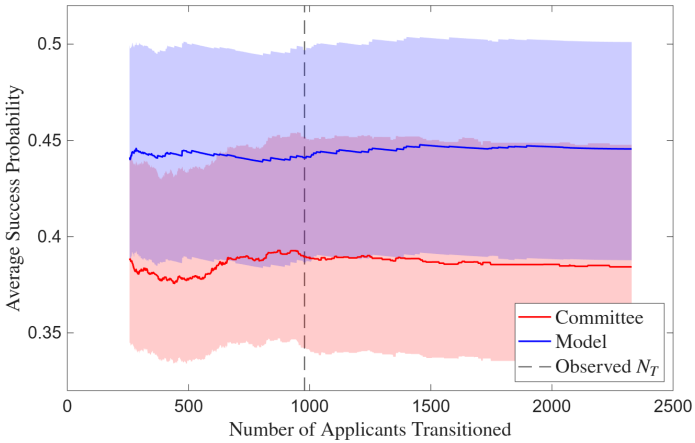
Regressions vs. Model

Dependent Variable	Accept	T100 Placement
Age	-0.0001	0.0053*
Female	0.0163**	-0.0032
International Asian	-0.0214***	-0.0127
GRE Quantitative Percentile	0.0011**	0.0006
TOEFL/IELTS	-0.0022	0.0409*
Work Experience	0.0140**	0.0358**
Committee Score	0.0788***	0.0007
CV Coding	-0.0044*	0.0125
Essay Economics Terms (%)	0.0001	-0.0104
Essay Positive Terms (%)	-0.0013	0.0123*

» Acceptance vs. Success AMEs

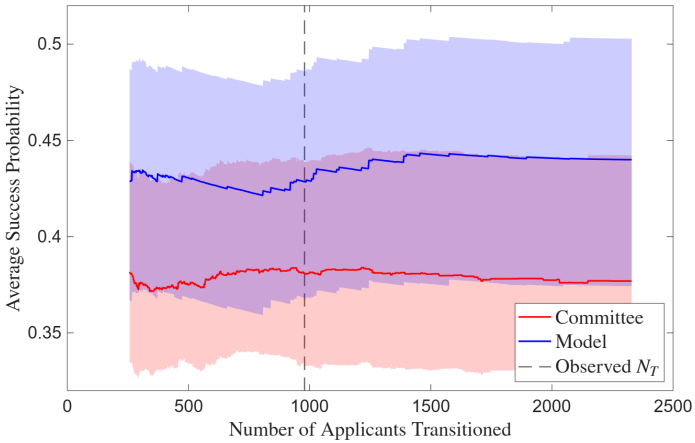
- Blue = underweighted, red = overweighted, white = neither
- AMEs don't reveal if committee deviates from optimal ranking near the cutoff

Benefit of Reading More Files (Acceptance Set)



- Suppose committee's curve is increasing and concave
- $\max_{n \in [N_A, N]} V N_A \overline{P}(n) - cn$
- **FOC:** $V = c/(N_A \overline{P}'(N_T))$
- V is dollar value of an additional expected success

Benefit of Reading More Files (Matriculation Set)



- Suppose committee's curve is increasing and concave
- $\max_{n \in [N_A, N]} VN_M\overline{P}(n) - cn$
- **FOC:** $V = c/(N_M\overline{P}'(N_T))$
- V is dollar value of an additional expected success

Conclusion

- **Economic Problem:** Selecting best options under costly evaluation
- **Application:** Graduate admissions using synthetic data protocol to protect privacy
- **Contribution:** Build models to assess optimality and quantify deviation gains
  1. On most dimensions, estimated success probabilities align with committee's decisions
  2. Reweighting some applicant characteristics could have increased admitted cohort's average success probability by 4 percentage points without reading more files
  3. Success is hard to predict, but even modest gains in selection are worthwhile

★ Methods apply to admissions, hiring, and other settings, especially those with more data, more predictable outcomes, or less effective decision-makers

⏩ Extensions

Thank You!

A1: Synthetic Rows $\neq$ Actual Applicants

Transition	Waitlist	Accept	Matriculate	Success	GRE Quantitative Percentile	Gender	Demographic 1	Demographic 2	Advanced Math	Committee Score	Program Rank
0	0	0	0	0	70	1	0	1	0	·	27
0	0	0	0	0	92	1	1	0	1	·	150
0	0	0	0	0	85	1	1	0	0	·	80
0	0	0	0	0	72	1	0	0	1	·	51
0	0	0	0	0	74	0	1	0	0	·	150
0	0	0	0	0	79	0	0	0	1	·	26
1	1	0	0	1	97	0	0	1	1	8	43
0	0	0	0	0	76	1	0	0	0	·	17
1	1	0	0	0	75	0	0	0	0	3	70
0	0	0	0	0	75	0	0	1	0	·	150



Transition	Waitlist	Accept	Matriculate	Success	GRE Quantitative Percentile	Gender	Demographic 1	Demographic 2	Advanced Math	Committee Score	Program Rank
1	1	0	0	0	92	0	0	1	1	8	31
0	0	0	0	0	83	0	0	0	1	·	60
0	0	0	0	1	94	0	1	0	1	·	19
1	1	1	1	0	90	0	0	1	1	5	37
0	0	0	0	1	86	0	0	1	0	·	52
1	0	1	0	1	91	1	1	0	0	2	29
1	1	0	0	0	86	0	0	0	1	5	71
1	1	0	0	0	98	0	1	0	0	5	21
1	1	0	0	1	88	1	1	0	1	4	37
0	0	0	0	0	92	1	1	0	0	·	150

⏩ Synthetic Data Protocol

A1: Dependent Variables

Admission	Success
$\mathbb{1}$**(Transition)**	$\mathbb{1}$**(Complete Program)**
$\mathbb{1}$(Accept)	$\mathbb{1}$**(Academic Placement)**
$\mathbb{1}$**(Waitlist)**	$\mathbb{1}$**(Government Placement)**
$\mathbb{1}$**(Open House)**	$\mathbb{1}$**(Consulting Placement)**
$\mathbb{1}$(Matriculate)	$\mathbb{1}$**(Tech Placement)**
$\mathbb{1}$**(Attend Program)**	$\mathbb{1}$**(Other Placement)**
Program Rank	$\mathbb{1}$**(No Placement)**
	Placement Rank
	$\mathbb{1}$**(Top 100 Placement)**

- Economics-only variables bold, non-economics-only variables italic
- U.S. News PhD program rankings, IDEAS/RePEc placement rankings
  - For non-tenure track, use weighted average of university's and maximum rankings
  - For non-academia, use average of ChatGPT, Claude, and Gemini's rankings

Data Description

A1: Objective Variables

Demographic	Performance
Year	GRE Verbal Percentile
𝟙*(Covid Year)*	GRE Quantitative Percentile
Age	GRE Analytic Percentile
𝟙(Female)	Undergraduate GPA
𝟙**(U.S.)**	TOEFL/IELTS
𝟙*(U.S. African/Hispanic/Native)*	𝟙(Elite U.S. University)
𝟙*(U.S. Asian)*	𝟙(Elite U.S. Liberal Arts College)
𝟙*(U.S. Other)*	𝟙(Other U.S. University/College)
𝟙(International Asian)	𝟙(Elite Foreign University)
𝟙(International Other)	𝟙(Other Foreign University)
	𝟙(Graduate Degree)
	𝟙(Work Experience)
	#Professor Recommenders
	#Prolific Recommenders
	Concentration Dummy Variables

⏩ Data Description

A1: Subjective Variables

Score/Recommendation Letters	Supplement/CV/Essay
Committee Score	**#Math Courses**
Letters Length	**$\mathbb{1}$(Advanced Math)**
Positive Terms (%) (YS 2012)	**CV Length**
Negative Terms (%) (YS 2012)	**CV Coding**
Standout Terms (%) (BEMS 2022)	**CV $\mathbb{1}$(Publication/Working Paper)**
Ability Terms (%) (BEMS 2022)	**Essay Length**
Research Terms (%) (BEMS 2022)	**Essay $\mathbb{1}$(Specific Professor)**
Grindstone Terms (%) (BEMS 2022)	**Essay Economics Terms (%)**
Teaching Terms (%) (BEMS 2022)	**Essay Research Terms (%)**
Communal Terms (%) (BEMS 2022)	**Essay Positive Terms (%)**
Agentic Terms (%) (MHM 2009)	**Essay Negative Terms (%)**
	Field Dummy Variables

- Dictionaries from Madera et al. (2009); Young & Soroka (2012); Bai et al. (2022)
- *#Math Courses* and *Advanced Math* objective after 2020

Data Description

A1: Academia-Equivalent Rankings (Government)

- **European Institutions:** European Central Bank (40), HM Treasury (80)
- **Foreign Central Banks:** Banco Central do Brasil (93), Banco de México (93), Bank of Canada (80), Bank of England (67), Bank of Israel (93), Bank of Italy (80), Bank of Japan (80), Bank of Spain (80), Banque de France (80), Central Bank of Chile (93), Central Bank of Colombia (93), Central Bank of Korea (93), Central Bank of Turkey (93), Norges Bank (80), Reserve Bank of Australia (80), Reserve Bank of India (93), Reserve Bank of New Zealand (93), Sveriges Riksbank (80), Swiss National Bank (80)
- **Multilaterals/International:** African Development Bank (93), Asian Development Bank (93), Bank for International Settlements (53), European Bank for Reconstruction and Development (93), European Investment Bank (93), Inter-American Development Bank (93), International Monetary Fund (40), Statistics Canada (107), World Bank (40)
- **Policy Research Institutes:** American Enterprise Institute (93), American Institutes for Research (107), Becker Friedman Institute (80), Brookings Institution (80), Center for Global Development (93), Center for Migration Studies in New York (112), Center for Strategic and International Studies (107), Cowles Foundation (80), Harris School Policy Labs (93), Hoover Institution (80), IDinsight (112), Innovations for Poverty Action (93), J-PAL (80), MDRC (107), National Bureau of Economic Research (67), Peterson Institute for International Economics (80), Pew Research Center (107), RAND Corporation (80), Resources for the Future (93), Urban Institute (107)
- **U.S. Executive/Cabinet:** Department of Commerce (107), Department of Justice (80), Department of Labor (107), Department of the Treasury (80)
- **U.S. Executive Policy Offices:** Council of Economic Advisers (80), Office of Management and Budget (93)
- **U.S. Federal Reserve System:** Board of Governors (40), Atlanta (80), Boston (80), Chicago (80), Cleveland (80), Dallas (80), Kansas City (80), Minneapolis (80), New York (67), Philadelphia (80), Richmond (80), San Francisco (80), St. Louis (80)
- **U.S. GSE/Housing Finance:** Fannie Mae (120), Freddie Mac (120)
- **U.S. Independent Agencies:** Commodity Futures Trading Commission (93), Federal Deposit Insurance Corporation (93), Federal Energy Regulatory Commission (93), Federal Housing Finance Agency (93), Federal Trade Commission (80), Office of Financial Research (93), Securities and Exchange Commission (93)
- **U.S. Legislative Committees:** Congressional Budget Office (80), Congressional Research Service (107), Joint Committee on Taxation (93), Joint Economic Committee (107)
- **U.S. Science and Research:** National Science Foundation (107)
- **U.S. Statistical Agencies:** Bureau of Economic Analysis (93), Bureau of Labor Statistics (93), Census Bureau (93)

» Data Description

A1: Academia-Equivalent Rankings (Consulting/Tech/Other)

- **Consulting:** Aecon Consulting (130), AlixPartners (125), Analysis Group (120), Bain & Company (120), Bates White (120), Berkeley Research Group (120), Boston Consulting Group (120), Brattle Group (120), Charles River Associates (120), Coleman Research (130), Compass Lexecon (120), Cornerstone Research (120), Deloitte Economic and Financial Advisory (125), Econic Partners (130), Economists Incorporated (120), Edgeworth Economics (120), Ernst & Young Economic Advisory (125), Frontier Economics (120), FTI Consulting (120), Guidehouse (130), KPMG Economics & Valuation (125), Matrix Economics (130), McKinsey & Company (120), Mercer (135), NERA Economic Consulting (120), Oliver Wyman (120), PwC Economics (125), Roland Berger (125), Willis Towers Watson (135)
- **Tech (Athey & Luca 2019):** Airbnb (107), Alibaba (120), Amazon (93), AppNexus (135), Apple (107), Block/Square (120), ByteDance/TikTok (120), CoreLogic (135), Coursera (120), Dstillery (135), Didichuxing (125), Digonex (135), DoorDash (107), eBay (107), ECONorthwest (125), Expedia (120), Forkcast (135), Glassdoor (120), Google (93), Granular (135), Groupon (125), Houzz (130), Huawei (120), IBM (120), Indeed (120), ING (130), Instacart (120), Intel (120), Kensho (125), Lending Club (135), LinkedIn (107), Lyft (120), Meta/Facebook (93), Microsoft (93), Netflix (93), Nuna (135), Nvidia (120), Pandora (135), PayPal (120), Pinterest (120), Prattle (135), Quantco (130), Quora (130), Redfin (120), Ripple (135), Roblox (120), Rover (135), Salesforce (120), Shopify (120), Spotify (120), Stripe (107), Trulia (125), Uber (107), Upwork (135), Visa (125), Walmart (125), Wealthfront (135), Yahoo (125), Yelp (125), Zillow (120)
- **Tech Labs:** Amazon Science (40), Apple Machine Learning Research (80), DeepMind (80), Google Research (40), Meta Research (80), Microsoft Research (40), Netflix Research (93), OpenAI Research (80), Uber Research (93)
- **Other:** AIG (135), Bank of America (125), Bloomberg (125), Boeing (135) Capital One (125), Exelon (135), ExxonMobil (135), Ford (135), General Electric (135), General Motors (135), Goldman Sachs (125), JPMorgan Chase (125), Liberty Mutual (135), Moody's Analytics (125), Morningstar (125), Nielsen (135), PNC (135), Progressive (135), S&P Global (125), Shell (135), Swiss Re (130), Toyota (130), Other (135), No Placement (150)

Data Description

A1: Dictionaries

- **Recommendation Letters:**
  - **Positive/Negative Words:** Young & Soroka (2012)'s Lexicoder Sentiment Dictionary
  - **Standout, Ability, Research, Grindstone, Communal Words:** Bai et al. (2022)
  - **Agentic Words:** assertive, confident, aggressive, ambitious, dominant, forceful, independent, daring, outspoken, intellectual (Madera et al. 2009); **AND** audacious, bold, command, compete, decisive, determine, drive, enterprise, fearless, influential, leader, pioneer, powerful, proactive, resilient, resourceful, self-assured, strategic, tenacious, vision
  - **Teaching Words:** academic, assess, clarity, coach, collaborate, communicate, cultivate, curriculum, demonstrate, develop, educate, encourage, engage, enlighten, evaluate, explain, facilitate, foster, guide, impart, inspire, insight, instruct, knowledge, learn, lesson, mentor, nurture, participate, pedagogy, present, scholar, support, teach, train, tutor, workshop
- **CVs and Essays:**
  - **Coding:** c/c++, eviews, fortran, java, jax, julia, latex, matlab, python, r, sas, stata, webscrape
  - **Publication:** Journal names from IDEAS/RePEc Aggregate Rankings for Journals
  - **Working Paper:** arxiv, center for economic policy research, cepr, econpapers, national bureau of economic research, nber, research papers in economics, repec, social science research network, ssrn, working paper
  - **Economics:** applied, asset, behavior, climate, computation, data, development, dynamic programming, economics, education, empirical, environment, experiment, finance, fiscal, game theory, health, immigration, industrial, inequality, international, io, labor, linear programming, macro, math, metrics, micro, monetary, policy, political, poverty, pricing, probability, public, sports, statistic, theory, trade, urban

⏩ Data Description

A1: Inference vs. Prediction

- Challenging to obtain consistent estimates due to omitted variables
  - **Example:** Don't observe effort (e.g. hours worked). Can bias coefficients upward if effort correlated with variables and with success irrespective of those variables
  - *Committee Score* proxies information the committee sees but I don't. Like BEMS (2022), proxy for effort with "grindstone terms" (e.g. depend*, diligen*)
- Prediction matters as much as inference
  - Mullainathan & Spiess (2017) distinguish $\hat{\beta}$ vs. $\hat{y}$ problems
  - If *Work Experience* predicts success, committee should use it even without knowing exact mechanism, provided relationship would still hold at this university

⏩ Data Description

A1: Sample Selection

- Determinants of success conditional on applying vs. matriculating
- BEMS (2022) prove that if $P(\text{Success}|\text{Skill})$ is an S-curve, then $\hat{\beta}$ will be smaller for more selected samples. Consistent with empirical results
- Outcomes observed only for matriculants (e.g. grades) $\Rightarrow$ Heckman correction
  - Let $y_n^* = w_n\beta + u_n$, $m_n = \mathbb{1}(q_n\gamma + v_n > 0)$, and $y_n = m_n y_n^*$, where $w = (x, z)$
  - **Step 1:** Probit of m on q for all applicants $\Rightarrow$ inverse Mills $\hat{\lambda}_n = \frac{\phi(q_n\hat{\gamma})}{\Phi(q_n\hat{\gamma})}$
  - **Step 2:** OLS of y on $(w, \hat{\lambda})$ for matriculants $\Rightarrow$ consistent $\hat{\beta}$
  - **Instruments:** Variables that affect matriculation but not the outcome

Data Description

A1: Acceptance, Matriculation, and Success AMEs

Dependent Variable	Accept	Matriculate	T100 Placement	T100 Placement
Age	-0.0001	-0.0223**	0.0037	0.0053*
Female	0.0163**	-0.0490	-0.0157	-0.0032
U.S.	-0.0013	0.1472*	0.0213	0.0192
International Asian	-0.0214***	0.1644**	0.0298	-0.0127
GRE Quantitative Percentile	0.0011**	-0.0147**	0.0028**	0.0006
Undergraduate GPA	0.0195	-0.1026	-0.0721	-0.1096**
TOEFL/IELTS	-0.0022	-0.0631	0.0219	0.0409*
Work Experience	0.0140**	0.1591**	0.0271	0.0358**
#Prolific Recommenders	0.0039	-0.0426	0.0137*	0.0076
Committee Score	0.0788***	-0.0371	0.0144**	0.0007
Advanced Math	0.0246	-0.1950**	-0.0317	-0.0287
Letters Terms (%) PC1	0.0055	0.0671**	0.0038	0.0159
CV Coding	-0.0044*	-0.0414**	0.0108	0.0125
Essay Economics Terms (%)	0.0001	0.0366**	-0.0175**	-0.0104
Essay Positive Terms (%)	-0.0013	-0.0072	0.0118*	0.0123*
Missing Committee Score	-0.0096		-0.0528**	-0.0277
Open House		0.2615***		
Program Rank				-0.0011***
Missing Program Rank				-0.3965***

Year Dummies
GRE Verbal Percentile
GRE Analytic Percentile
Elite U.S. University
Elite U.S. Liberal Arts College
Elite Foreign University
Other Foreign University
Graduate Degree
#Professor Recommenders
#Math Courses
Letters Length/100
Letters Terms (%) PC2
CV Length/100
CV Publication/Working Paper
Essay Length/100
Essay Specific Professor
Essay Research Terms (%)
Essay Negative Terms (%)
Missing Dummies

» Regression Results

A1: Diversity Objective

- **Demographics:** Squares vs. circles
- **Quality:** High vs. low

High Quality:



Low Quality:



⏩ Acceptance vs. Success AMEs

A1: Diversity Objective

- No preference for squares or circles

High Quality:



Low Quality:



Acceptance vs. Success AMEs

A1: Diversity Objective

- Small preference for squares $\Rightarrow$ consistent with maximizing quality

High Quality:



Low Quality:



Acceptance vs. Success AMEs

A1: Diversity Objective

- Large preference for squares $\Rightarrow$ inconsistent with maximizing quality

High Quality:



Low Quality:



⏩ Acceptance vs. Success AMEs

A1: Insufficiency of Comparing AMEs

- Cannot assess optimality by comparing AMEs across regressions. Suppose:
  - Work experience has large effect on success, graduate degree smaller effect
  - 25% of applicants have both, 25% experience only, 25% degree only, 25% neither
  - Committee can accept half of applicants, and other half attend different university
- Optimal committee accepts half with experience and ignores degree
  - **Acceptance Regression:** $\text{AME}_{\text{Experience}} = 1$, $\text{AME}_{\text{Degree}} = 0$
  - **Success Regression:** $1 > \text{AME}_{\text{Experience}} > \text{AME}_{\text{Degree}} > 0$
  - Despite optimality, acceptance and success AMEs not equal across regressions
- **Upshot:** Relationship between AMEs must be interpreted through a model

⏩ Acceptance vs. Success AMEs

A1: Wald Tests

- Test if acceptance and/or success AMEs are same across PhD programs
  - **Example:** Does elite undergraduate university predict acceptance and/or success more or less for economics program than for other programs?
  - **Sample:** Applicants to university's economics, government, history, linguistics, and psychology PhD programs from 2020–2023 (when multi-program data is available)
- Can I test if acceptance AMEs equal success AMEs within programs?
  - Predictors of acceptance only (success only) may be overweighted (underweighted), provided committee wants to select based on a "success" model
  - **Problem:** Not a test if admissions decisions are optimal
  - **Solution:** Need model of admissions to assess optimality

⏩ Acceptance vs. Success AMEs

A1: Logistic Wald Test Across Programs

Assume $\left(Y^E \perp\!\!\!\perp Y^{NE}\right)\big|X$. Let $P\left(Y_n^E = 1\middle|x_n^E;\theta^E\right) = \frac{\exp(x_{n,0}^E\theta_0^E + x_{n,1}^E\theta_1^E)}{1+\exp(x_{n,0}^E\theta_0^E + x_{n,1}^E\theta_1^E)}$, and $P\left(Y_n^{NE} = 1\middle|x_n^{NE};\theta^{NE}\right) = \frac{\exp(x_{n,0}^{NE}\theta_0^{NE} + x_{n,1}^{NE}\theta_1^{NE})}{1+\exp(x_{n,0}^{NE}\theta_0^{NE} + x_{n,1}^{NE}\theta_1^{NE})}$. Finally, let $N = N_E + N_{NE}$. Then:

$$\mathcal{L}_N^U\left(y\middle|x;\theta\right) = \prod_{n=1}^{N_E} P\left(Y_n^E = 1\middle|x_n^E;\theta^E\right)^{Y_n^E}\left[1 - P\left(Y_n^E = 1\middle|x_n^E;\theta^E\right)\right]^{1-Y_n^E}$$
$$\prod_{n=1}^{N_{NE}} P\left(Y_n^{NE} = 1\middle|x_n^{NE};\theta^{NE}\right)^{Y_n^{NE}}\left[1 - P\left(Y_n^{NE} = 1\middle|x_n^{NE};\theta^{NE}\right)\right]^{1-Y_n^{NE}}.$$

- $Y \in \{\mathbb{1}(\textit{Accept}), \mathbb{1}(\textit{Attend Program}), \mathbb{1}(\textit{Complete Program}), \mathbb{1}(\textit{Good Placement})\}$
- **Test:** Let $g(\hat{\theta}) = \frac{1}{N_E}\sum_{n=1}^{N_E}\hat{P}_n^E(1-\hat{P}_n^E)\hat{\theta}_1^E - \frac{1}{N_{NE}}\sum_{n=1}^{N_{NE}}\hat{P}_n^{NE}(1-\hat{P}_n^{NE})\hat{\theta}_1^{NE}$. Then: $W = g(\hat{\theta})'\left[\nabla_\theta g(\hat{\theta})\hat{V}(\hat{\theta})\nabla_\theta g(\hat{\theta})'\right]^{-1} g(\hat{\theta}) \sim \chi_Q^2$. Compare AMEs per Mood (2010)

Insufficiency of Comparing AMEs

A1: Linear Wald Test Across Programs

Assume $\left(Y^E \perp\!\!\!\perp Y^{NE}\right)\big|X$, i.e. $\left(\text{Economics} \perp\!\!\!\perp \text{Non-Economics}\right)\big|\text{Predictors}$.
Let $f\left(y_n^E \middle| x_n^E; \beta^E, \sigma_E^2\right) = \frac{1}{\sqrt{2\pi\sigma_E^2}} \exp\left[\frac{-1}{2\sigma_E^2}\left(y_n^E - x_{n,0}^E \beta_0^E - x_{n,1}^E \beta_1^E\right)^2\right]$, and
$f\left(y_n^{NE} \middle| x_n^{NE}; \beta^{NE}, \sigma_{NE}^2\right) = \frac{1}{\sqrt{2\pi\sigma_{NE}^2}} \exp\left[\frac{-1}{2\sigma_{NE}^2}\left(y_n^{NE} - x_{n,0}^{NE} \beta_0^{NE} - x_{n,1}^{NE} \beta_1^{NE}\right)^2\right]$. Then:

$$\mathcal{L}_N^U\left(y \middle| x; \beta, \sigma^2\right) = \prod_{n=1}^{N_E} f\left(y_n^E \middle| x_n^E; \beta^E, \sigma_E^2\right) \prod_{n=1}^{N_{NE}} f\left(y_n^{NE} \middle| x_n^{NE}; \beta^{NE}, \sigma_{NE}^2\right).$$

- $Y \in \{\textit{Standardized Committee Score}, \textit{Standardized Placement Rank}\}$
- **Test:** Let $g(\hat{\beta}_1) = \hat{\beta}_1^E - \hat{\beta}_1^{NE}$. (Can also do likelihood-ratio test.)
  Then $\mathrm{W} = g(\hat{\beta}_1)'\left[\nabla_{\beta,\sigma^2} g(\hat{\beta}_1) \hat{V}(\hat{\beta}, \hat{\sigma}^2) \nabla_{\beta,\sigma^2} g(\hat{\beta}_1)'\right]^{-1} g(\hat{\beta}_1) \sim \chi_Q^2$
- **Assumption:** Homoskedasticity (i.e. $\forall n,\ \sigma_n^2 = \sigma^2$)

⏩ Insufficiency of Comparing AMEs

A1: Additional Questions

- Are there different determinants of different types of success? (AKKLP 2007)
  - Need different skills at different stages in the program, such as grades vs. placement
  - For some programs, initial placement matters less than for economics
- Different determinants for international applicants? (KW 2000)
  - **Admission:** May have different preparation and/or interests
  - **Completion:** Depending on country, may have fewer outside options
  - **Placement:** May return to native country after graduation
- Different determinants over time? (2015 to 2023)
  - COVID-19 and other macro factors; changes in professions (e.g. more empirics)
  - For economics, increased importance of elite predocs and Fed RAships

⏩ Insufficiency of Comparing AMEs

A2: Models of Admissions

1. **Dynamic Model:** Committee ranks applicants based on conditional expectation in Round 1 about those who survive to Round 2, reads files of those above Cutoff 1, and rejects the rest. Favors those in Round 1 with objective variables positively correlated with subjective variables that predict success. Then reranks survivors based on all variables, accepts survivors above Cutoff 2, and rejects the rest
    1. **Static Model:** In Round 1, rank applicants based only on objective variables
    2. **Myopic Model:** Like dynamic, but distribution of z not conditional on x
2. **Matriculation-Dynamic Model:** In Round 1, rank applicants based on combination of success and matriculation probabilities (largely success probabilities)

⏩ Two-Round Selection

A2: Description of Models

- **Round 2:** If accept applicant, get success probability $P(y = S|x, z, t, \overline{k})$, where Success $= \mathbb{1}$(Complete Program) or $\mathbb{1}$(Good Placement). If reject applicant, get $P(y_2^* = S|t) = P$(Success) of year t's marginal accepted applicant

- **Round 1:** If reject applicant, get $L/P(y_1^* = S|t) = L/P$(Success) of year t's marginal transitioned applicant. Can be less or greater than $P(y_2^* = S|t)$ depending on value placed on committee's time vs. admitting a successful cohort
  - **Dynamic:** If transition applicant, get $\mathbb{E}[v_2(x, z, t, \overline{k}, \epsilon_2)|x, t, \overline{k}] =$ expected value of Round 2, where z is integrated out conditional on (x, t) based on $F(z|x, t)$
  - **Static:** If transition applicant, get $P(y = S|x, t, \overline{k})$, or P(Success) unconditional on z. It may differ from $P(y = S|x, z, t, \overline{k})$, though it still accounts for z's effect on y
  - **Myopic:** In $\mathbb{E}[v_2(x, z, t, \overline{k}, \epsilon_2)|x, t, \overline{k}]$, replace $F(z|x, t)$ with $F(z|t)$

Two-Round Selection

A2: Lagrangian (BEMS 2022)

Portfolio Choice Problem:

$$\max_{(d_1,\ldots,d_N)\in\{A,R\}^N} \sum_{n=1}^{N} P(y_n = S|x_n, z_n)\mathbb{1}(d_n = A), \text{ where: } \sum_{n=1}^{N} \mathbb{1}(d_n = A) \leq N_A$$

$$\mathcal{L}(d_1, \ldots, d_N; \lambda) = \sum_{n=1}^{N} \left[P(y_n = S|x_n, z_n)\mathbb{1}(d_n = A) + \lambda\mathbb{1}(d_n = R)\right] + \lambda(N_A - N)$$

Cutoff Rule Solution:

$$\forall n \in \{1, \ldots, N\},\ d_n(x_n, z_n|\lambda^*) = \begin{cases} A & \text{if } P(y_n = S|x_n, z_n) > \lambda^* \\ \{A, R\} & \text{if } P(y_n = S|x_n, z_n) = \lambda^* \\ R & \text{if } P(y_n = S|x_n, z_n) < \lambda^* \end{cases}$$

Upshot: λ^* is cutoff probability. Denote it hereafter as $P(y^* = S)$

- $P(y^* = S) \in \left[P(y_{N_A+1} = S|x_{N_A+1}, z_{N_A+1}), P(y_{N_A} = S|x_{N_A}, z_{N_A})\right]$

Portfolio Choice to Cutoff Rule

A2: Lagrangian with Shocks

Portfolio Choice Problem:

$$\max_{(d_1,\ldots,d_N)\in\{A,R\}^N} \sum_{n=1}^{N} \left\{ \left[P(y_n = S|x_n, z_n) + \epsilon_n(A)\right]\mathbb{1}(d_n = A) + \epsilon_n(R)\mathbb{1}(d_n = R)\right\},$$

$$\text{where: } \sum_{n=1}^{N} \mathbb{1}(d_n = A) \leq N_A. \quad \text{Then: } \mathcal{L}(d_1, \ldots, d_N; \lambda) =$$

$$\sum_{n=1}^{N} \left\{ \left[P(y_n = S|x_n, z_n) + \epsilon_n(A)\right]\mathbb{1}(d_n = A) + \left[\lambda + \epsilon_n(R)\right]\mathbb{1}(d_n = R)\right\} + \lambda(N_A - N)$$

Cutoff Rule Solution:

$$\forall n \in \{1, \ldots, N\}, d_n(x_n, z_n, \epsilon_n|\lambda^*) = \begin{cases} A & \text{if } P(y_n = S|x_n, z_n) + \epsilon_n(A) > \lambda^* + \epsilon_n(R) \\ \{A, R\} & \text{if } P(y_n = S|x_n, z_n) + \epsilon_n(A) = \lambda^* + \epsilon_n(R) \\ R & \text{if } P(y_n = S|x_n, z_n) + \epsilon_n(A) < \lambda^* + \epsilon_n(R) \end{cases}$$

Theorem 5

For large N, there is a cutoff rule solution with logistic CCPs. Portfolio Choice to Cutoff Rule

A2: Lagrangian, Round 2

Portfolio Choice Problem:

$$\max_{(d_{2,1},\ldots,d_{2,N_T})\in\{A,R\}^{N_T}} \sum_{n=1}^{N_T} P(y_n = S|x_n, z_n)\mathbb{1}(d_{2,n} = A), \text{ where: } \sum_{n=1}^{N_T} \mathbb{1}(d_{2,n} = A) \leq N_A$$

$$\mathcal{L}(d_{2,1}, \ldots, d_{2,N_T}; \lambda_2) = \sum_{n=1}^{N_T} \left[P(y_n = S|x_n, z_n)\mathbb{1}(d_{2,n} = A) + \lambda_2 \mathbb{1}(d_{2,n} = R)\right] + \lambda_2(N_A - N_T)$$

Cutoff Rule Solution:

$$\forall n \in \{1, \ldots, N_T\},\ d_{2,n}(x_n, z_n|\lambda_2^*) = \begin{cases} A & \text{if } P(y_n = S|x_n, z_n) > \lambda_2^* \\ \{A, R\} & \text{if } P(y_n = S|x_n, z_n) = \lambda_2^* \\ R & \text{if } P(y_n = S|x_n, z_n) < \lambda_2^* \end{cases}$$

Upshot: λ_2^* is cutoff probability. Denote it hereafter as $P(y_2^* = S)$

- $P(y_2^* = S) \in \left[P(y_{N_A+1} = S|x_{N_A+1}, z_{N_A+1}), P(y_{N_A} = S|x_{N_A}, z_{N_A})\right]$

Portfolio Choice to Cutoff Rule

A2: Lagrangian, Round 1 (Dynamic)

Portfolio Choice Problem:

$$\max_{(d_{1,1},\ldots,d_{1,N})\in\{T,R\}^N} \sum_{n=1}^{N} EV_{2,n}(x_n;\overline{\lambda_2^*})\mathbb{1}(d_{1,n}=T), \text{ where: } \sum_{n=1}^{N}\mathbb{1}(d_{1,n}=T)\leq N_T, \text{ where:}$$

$$EV_{2,n}(x_n)=\int_{\widetilde{z}_n}\max\{P(y_n=S|x_n,\widetilde{z}_n),\overline{\lambda_2^*}\}dF_{z|x}(\widetilde{z}_n|x_n)$$

$$\mathcal{L}(d_{1,1},\ldots,d_{1,N};\lambda_1)=\sum_{n=1}^{N}\left[EV_{2,n}(x_n)\mathbb{1}(d_{1,n}=T)+\lambda_1\mathbb{1}(d_{1,n}=R)\right]+\lambda_1(N_T-N)$$

Cutoff Rule Solution:

$$\forall n\in\{1,\ldots,N\},\ d_{1,n}(x_n|\lambda_1^*)=\begin{cases}T & \text{if } EV_{2,n}(x_n)>\lambda_1^*\\ \{T,R\} & \text{if } EV_{2,n}(x_n)=\lambda_1^*\\ R & \text{if } EV_{2,n}(x_n)<\lambda_1^*\end{cases}$$

Upshot: λ_1^* is cutoff level. Denote it hereafter as $L(y_1^*=S)\in\left[EV_{2,N_T+1},EV_{2,N_T}\right]$

Portfolio Choice to Cutoff Rule

A2: Lagrangian, Round 1 (Static)

Portfolio Choice Problem:

$$\max_{(d_{1,1},\ldots,d_{1,N})\in\{T,R\}^N} \sum_{n=1}^{N} P(y_n = S|x_n)\mathbb{1}(d_{1,n} = T), \text{ where: } \sum_{n=1}^{N} \mathbb{1}(d_{1,n} = T) \leq N_T$$

$$\mathcal{L}(d_{1,1},\ldots,d_{1,N};\lambda_1) = \sum_{n=1}^{N} \left[P(y_n = S|x_n)\mathbb{1}(d_{1,n} = T) + \lambda_1\mathbb{1}(d_{1,n} = R)\right] + \lambda_1(N_T - N)$$

Cutoff Rule Solution:

$$\forall n \in \{1,\ldots,N\},\ d_{1,n}(x_n|\lambda_1^*) = \begin{cases} T & \text{if } P(y_n = S|x_n) > \lambda_1^* \\ \{T,R\} & \text{if } P(y_n = S|x_n) = \lambda_1^* \\ R & \text{if } P(y_n = S|x_n) < \lambda_1^* \end{cases}$$

Upshot: λ_1^* is cutoff probability. Denote it hereafter as $P(y_1^* = S)$

- $P(y_1^* = S) \in \left[P(y_{N_T+1} = S|x_{N_T+1}), P(y_{N_T} = S|x_{N_T})\right]$

Portfolio Choice to Cutoff Rule

A2: Dynamic Parameterization 1

Unrestricted CCPs:

$$P_{d_1} = P\big(d_1 = T \big| x, t; \theta_T\big) = \frac{\exp(f_T(x,t)\theta_T)}{1 + \exp(f_T(x,t)\theta_T)}$$

$$P_{d_2} = P\big(d_2 = A \big| x, z, t; \theta_A\big) = \frac{\exp(f_A(x,z,t)\theta_A)}{1 + \exp(f_A(x,z,t)\theta_A)}$$

Success Probability:

$$P_y = P\big(y = S \big| x, z, t, k; \theta_S\big) = \frac{\exp(f_S(x,z,t,k)\theta_S)}{1 + \exp(f_S(x,z,t,k)\theta_S)}$$

Cutoff Level/Probability:

$$L_{y_1^*} = L\big(y_1^* = S \big| t; \theta_{S_1^*}\big) = \exp(f_{S_1^*}(t)\theta_{S_1^*})$$

$$P_{y_2^*} = P\big(y_2^* = S \big| t; \theta_{S_2^*}\big) = \frac{\exp(f_{S_2^*}(t)\theta_{S_2^*})}{1 + \exp(f_{S_2^*}(t)\theta_{S_2^*})}$$

Dynamic Model

A2: Dynamic Parameterization 2

$$\mathbb{E}[v_2(x,z,t,\overline{k},\epsilon_2)|x,t,\overline{k}] = \int_{\widetilde{z}} \sigma \log\left(\sum_{\widetilde{d_2}\in\{A,R\}} e^{u_2(x,\widetilde{z},t,\overline{k},\widetilde{d_2})/\sigma}\right) dF(\widetilde{z}|x,t)$$

$$= \int_{\widetilde{z_1}} \cdots \int_{\widetilde{z_J}} \left[\cdot\right] dF_1(\widetilde{z_1}|\widetilde{z_2},\ldots,\widetilde{z_J},x,t)\cdots dF_J(\widetilde{z_J}|x,t)$$

- **Binary:** Logistic regression on (x,t), so binary z's are independent given (x,t)
- **Continuous:** Jointly normal given (x,t) and binary z's
- **Computation:** Sum over configurations of binary z's, with 15-node Gauss–Hermite quadrature on $f_S(x,z,t,\overline{k})\theta_S$

Dynamic Model

A2: Dynamic Likelihood

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} \underbrace{f\left(z_n^c | z_n^{\neg c}, x_n, t_n; \theta_F\right)}_{\text{Density of } z^c | z^{\neg c}, x, t} \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional choice probabilities}}$$

$$\prod_{n=N_T+1}^{N} \underbrace{\left(1 - P_{d_{1,n}}(\theta_T)\right)}_{\text{Conditional choice probability}} \prod_{n=1}^{N} \underbrace{f\left(z_n^{\neg c} | x_n, t_n; \theta_F\right)}_{\text{Density of } z^{\neg c} | x, t} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}}$$

- Success mostly unobservable for applicants from fewer than six years ago

Maximum Likelihood Estimation

A2: Heckman (1979)-Corrected Dynamic Likelihood

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} \underbrace{\Phi(x_n, t_n; \theta_H)}_{\text{Selection}} \underbrace{f\left(z_n | x_n, t_n, \frac{\phi(x_n, t_n; \theta_H)}{\Phi(x_n, t_n; \theta_H)}; \theta_F\right)}_{\text{Density of } z|x,t} \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)}}_{\text{Conditional choice probabilities}}$$

$$\underbrace{\left(1 - P_{d_{2,n}}(\theta_A)\right)^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional choice probability}} \prod_{n=N_T+1}^{N} \underbrace{\left(1 - \Phi(x_n, t_n; \theta_H)\right)}_{\text{Selection}} \underbrace{f\left(z_n^{\neg c} | x_n, t_n, \frac{-\phi(x_n, t_n; \theta_H)}{1 - \Phi(x_n, t_n; \theta_H)}; \theta_F\right)}_{\text{Density of } z^{\neg c}|x,t}$$

$$\underbrace{\left(1 - P_{d_{1,n}}(\theta_T)\right)}_{\text{Conditional choice probability}} \prod_{n=1}^{N} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} \left(1 - P_{y_n}(\theta_S)\right)^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}}$$

- *Committee Score* unobservable for applicants who didn't survive Round 1

Maximum Likelihood Estimation

A2: Static Model

- **Motivation:** Interpretable cutoff probabilities in both rounds

Round 2:

$$v_2(x, z, t, \overline{k}, \epsilon_2) = \max_{d_2 \in \{A,R\}} u_2(x, z, t, \overline{k}, d_2) + \epsilon_2(d_2), \text{ where:}$$

$$u_2(x, z, t, \overline{k}, d_2) = \underbrace{P(y = S|x, z, t, \overline{k})\mathbb{1}(d_2 = A)}_{\text{If accept, get } P(\text{Success})} + \underbrace{P(y_2^* = S|t)\mathbb{1}(d_2 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

Round 1:

$$v_1(x, t, \overline{k}, \epsilon_1) = \max_{d_1 \in \{T,R\}} u_1(x, t, \overline{k}, d_1) + \epsilon_1(d_1), \text{ where:}$$

$$u_1(x, t, \overline{k}, d_1) = \underbrace{P(y = S|x, t, \overline{k})\mathbb{1}(d_1 = T)}_{\text{If transition, get } P(\text{Success})} + \underbrace{P(y_1^* = S|t)\mathbb{1}(d_1 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

- $P(y = S|x, t, \overline{k})$ estimates inconsistent, but fitted values account for z's effect on y

Dynamic Model

A2: Static Likelihood

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional choice probabilities}} \prod_{n=N_T+1}^{N} \underbrace{\left(1 - P_{d_{1,n}}(\theta_T)\right)}_{\text{Conditional choice probability}}$$

$$\prod_{n=1}^{N} \underbrace{\left[P_{y_n}(\theta_{S_1}) P_{y_n}(\theta_{S_2})\right]^{\mathbb{1}(y_n=S)} \left[(1 - P_{y_n}(\theta_{S_1}))(1 - P_{y_n}(\theta_{S_2}))\right]^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}}$$

- Success mostly unobservable for applicants from fewer than six years ago

⏩ Maximum Likelihood Estimation

A2: Lagrangian with Matriculation

Portfolio Choice Problem:

$$\max_{(d_1,\ldots,d_N)\in\{A,R\}^N} \sum_{n=1}^{N} P(y_n = S|w_n)P(m_n = M|w_n)\mathbb{1}(d_n = A),$$

$$\text{where: } \sum_{n=1}^{N} P(m_n = M|w_n)\mathbb{1}(d_n = A) \leq N_M, \text{ and } w = (x, z)$$

$$\mathcal{L}(d_1,\ldots,d_N;\lambda) = \sum_{n=1}^{N} \left[P(y_n = S|w_n)\mathbb{1}(d_n = A)+\lambda\mathbb{1}(d_n = R)\right]P(m_n = M|w_n)+\lambda(N_M-\overline{N_M})$$

Cutoff Rule Solution:

$$\forall n \in \{1,\ldots,N\},\ d_n(w_n|\lambda^*) = \begin{cases} A & \text{if } P(y_n = S|w_n) > \lambda^* \\ \{A, R\} & \text{if } P(y_n = S|w_n) = \lambda^* \\ R & \text{if } P(y_n = S|w_n) < \lambda^* \end{cases}$$

Cutoff Rule with Matriculation

A2: Lagrangian with Matriculation (Large N)

Portfolio Choice Problem:

$$\max_{\delta \in \{\delta | \delta : \mathbb{R}^K \to [0,1]\}} \int_{w \in \mathbb{R}^K} P(y = S|w) P(m = M|w) \delta(w) f(w) dw,$$

$$\text{where: } \int_{w \in \mathbb{R}^K} P(m = M|w) \delta(w) f(w) dw \leq N_M$$

$$\mathcal{L}(\delta; \lambda) = \int_{w \in \mathbb{R}^K} \left[P(y = S|w) \delta(w) + \lambda (1 - \delta(w)) \right] P(m = M|w) f(w) dw + \lambda (N_M - \overline{N_M})$$

Cutoff Rule Solution:

$$\forall w \in \mathbb{R}^K,\ \delta(w|\lambda^*) = \begin{cases} 1 & \text{if } P(y = S|w) > \lambda^* \\ [0,1] & \text{if } P(y = S|w) = \lambda^* \\ 0 & \text{if } P(y = S|w) < \lambda^* \end{cases}$$

Upshot: Point-identified cutoff probability λ^* is with respect to success probabilities

Cutoff Rule with Matriculation

A2: Lagrangian with Matriculation, Round 2

Portfolio Choice Problem:

$$\max_{(d_{2,1},\ldots,d_{2,N_T})\in\{A,R\}^{N_T}} \sum_{n=1}^{N_T} P(y_n = S|w_n)P(m_n = M|w_n)\mathbb{1}(d_{2,n} = A),$$

$$\text{where: } \sum_{n=1}^{N_T} P(m_n = M|w_n)\mathbb{1}(d_{2,n} = A) \leq N_M$$

$$\mathcal{L}(d_{2,1},\ldots,d_{2,N_T};\lambda_2) = \sum_{n=1}^{N_T} \left[P(y_n = S|w_n)\mathbb{1}(d_{2,n} = A) + \lambda_2 \mathbb{1}(d_{2,n} = R)\right] P(m_n = M|w_n) + \lambda_2(N_M - \overline{N_M})$$

Cutoff Rule Solution:

$$\forall n \in \{1,\ldots,N_T\},\ d_{2,n}(w_n|\lambda_2^*) = \begin{cases} A & \text{if } P(y_n = S|w_n) > \lambda_2^* \\ \{A,R\} & \text{if } P(y_n = S|w_n) = \lambda_2^* \\ R & \text{if } P(y_n = S|w_n) < \lambda_2^* \end{cases}$$

Cutoff Rule with Matriculation

A2: Lagrangian with Matriculation, Round 1

Portfolio Choice Problem:

$$\max_{(d_{1,1},\ldots,d_{1,N})\in\{T,R\}^N} \sum_{n=1}^{N} EV_{2,n}(x_n;\overline{\lambda_2^*})\mathbb{1}(d_{1,n}=T), \text{ where: } \sum_{n=1}^{N}\mathbb{1}(d_{1,n}=T)\leq N_T, \text{ where:}$$

$$EV_{2,n}(x_n)=\int_{\widetilde{z}_n}\max\{P(y_n=S|x_n,\widetilde{z}_n)-\overline{\lambda_2^*},0\}P(m_n=M|x_n,\widetilde{z}_n)dF_{z|x}(\widetilde{z}_n|x_n)$$

$$\mathcal{L}(d_{1,1},\ldots,d_{1,N};\lambda_1)=\sum_{n=1}^{N}\left[EV_{2,n}(x_n)\mathbb{1}(d_{1,n}=T)+\lambda_1\mathbb{1}(d_{1,n}=R)\right]+\lambda_1(N_T-N)$$

Cutoff Rule Solution:

$$\forall n\in\{1,\ldots,N\},\ d_{1,n}(x_n|\lambda_1^*)=\begin{cases}T & \text{if } EV_{2,n}(x_n)>\lambda_1^*\\ \{T,R\} & \text{if } EV_{2,n}(x_n)=\lambda_1^*\\ R & \text{if } EV_{2,n}(x_n)<\lambda_1^*\end{cases}$$

Cutoff Rule with Matriculation

A2: Matriculation-Dynamic Likelihood

Matriculation Probability:

$$P_m = P\big(m = M \big| x, z, t; \theta_M\big) = \frac{\exp(f_M(x, z, t)\theta_M)}{1 + \exp(f_M(x, z, t)\theta_M)}$$

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_A} \underbrace{P_{m_n}(\theta_M)^{\mathbb{1}(m_n = M)}(1 - P_{m_n}(\theta_M))^{\mathbb{1}(m_n = D)}}_{\text{Matriculation probabilities}}$$

$$\prod_{n=1}^{N_T} \underbrace{f\left(z_n^c | z_n^{\neg c}, x_n, t_n; \theta_F\right)}_{\text{Density of } z^c | z^{\neg c}, x, t} \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n} = A)}(1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n} = R)}}_{\text{Conditional choice probabilities}}$$

$$\prod_{n=N_T+1}^{N} \underbrace{\left(1 - P_{d_{1,n}}(\theta_T)\right)}_{\text{Conditional choice probability}} \prod_{n=1}^{N} \underbrace{f\left(z_n^{\neg c} | x_n, t_n; \theta_F\right)}_{\text{Density of } z^{\neg c} | x, t} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n = S)}(1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n = F)}}_{\text{Success probabilities}}$$

Maximum Likelihood Estimation

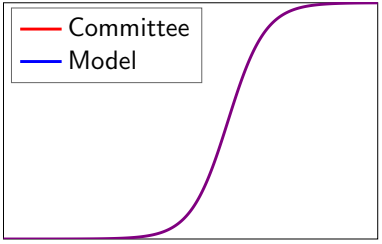
A2: Empirical Considerations

- **Selection Effects:** Track success for all applicants, including non-matriculants
  - In selected samples, success parameters are biased down (e.g. quantitative GRE)
- **Potential Outcomes:** Track which graduate program all applicants attended, and add *Program Rank* as a control variable in success blocks of likelihood
  - In decision blocks containing success, fix *Program Rank* at this university's rank
  - If a variable only predicts success by helping applicant get into elite graduate program, then committee should not use that variable to rank applicants
- **Unobservables:** I may not observe all information that committee observes
  - **Example:** Recommendation letter has nuances that algorithms can't detect
  - Include subjective committee score in z as a proxy for this information

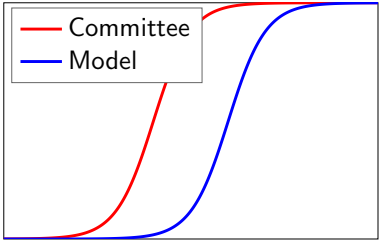
⏩ Maximum Likelihood Estimation

A2: Deviation Gains Procedure

1. Write Bellman equations for committee's optimization problem
2. Estimate parameters. Rationality assumption distorts estimates
3. Re-estimate parameters without rationality, and use to solve model
4. Compare selections from model's decision rule to committee's decision rule
5. For this comparison, evaluate selections at perturbations of estimates



Optimal Committee



Suboptimal Committee

Deviation Gains

A2: Deviation Gains Algorithm

Dynamic/Static/Myopic:

1. Rank each t's applicants by $\mathbb{E}[v_2(x, z, t, \overline{k}, \epsilon_2)|x, t, \overline{k}]$ with $\sigma = 0$ for dynamic and myopic, or $P(y = S|x, t, \overline{k})$ for static. Transition top N_T, and reject the rest
2. Rank surviving applicants by $P(y = S|x, z, t, \overline{k})$, and accept top N_A survivors. Compute $\overline{P(y = S|x, z, t, \overline{k})}$ of acceptance set, and compare to committee's set

Matriculation-Dynamic/Static:

1. Rank each t's applicants by $\mathbb{E}[v_2(x, z, t, \overline{k}, \epsilon_2)|x, t, \overline{k}]$ with $\sigma = 0$ for dynamic, or $P(y = S|x, t, \overline{k})$ for static (best analogue). Transition top N_T, and reject the rest
2. Rank surviving applicants by $P(y = S|x, z, t, \overline{k})$ and accept them in descending order, each matriculating with probability $P(m = M|x, z, t)$, until N_M matriculate. Compute $\overline{P(y = S|x, z, t, \overline{k})}$ of matriculation set, and compare to committee's set

Deviation Gains

A2: Deviation Gains Interpretation

- Under meaningfully-sized deviation gains, committee may have incorrect beliefs about which types of applicants are most likely to succeed
  - Too few observations for a true out-of-sample test, and cannot account for changes in the determinants of success over time, such as writing essays with AI
- Committee may use success-predicting variables that I do not observe
  - Because the success probability measure used to evaluate gains cannot detect such variables, any edge the committee's private information provides goes unmeasured
  - Comprehensive data, especially committee score in z, protects against this limitation

⏩ Deviation Gains Correction

A2: Extensions

- **Extension 1:** Uncover additional variables that predict matriculation but not failure
  - **Example:** Applicants with work experience more likely to succeed and matriculate
  - Can help committee find "people that we can afford" (*Moneyball*)
- **Extension 2:** Relax assumption that committee maximizes $P(\text{Success})$
  - Instead maximize combination of $P(\text{Success})$ and other objectives
  - **Example:** Committee may value diverse applicants (e.g. across fields): $\alpha P(\text{Success}) + (1 - \alpha)\text{Diversity}$, where $\alpha \in [0, 1]$ can be estimated
- **Extension 3:** Waitlist and multiyear models (see online appendix)
  - Three-round models, and two-round models that repeat over infinite horizon
  - States contain info about past rounds/years, and committee values future rounds/years (memory includes matriculation probability and diversity)

Conclusion

A3: Summary Statistics

Objective Variables

	Full Sample		Transition = 1		Accept = 1		Matriculate = 1	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Transition	0.4204	0.4937	1.0000	0.0000	1.0000	0.0000	1.0000	0.0000
Waitlist	0.3324	0.4712	0.7686	0.4220	0.0743	0.2628	0.0896	0.2877
Accept	0.1103	0.3134	0.2625	0.4402	1.0000	0.0000	1.0000	0.0000
Matriculate	0.0339	0.1811	0.0807	0.2725	0.3074	0.4623	1.0000	0.0000
Open House	0.0353	0.1847	0.0817	0.2741	0.3267	0.4702	0.5522	0.5010
Attend Program	0.6024	0.4895	0.6752	0.4685	0.7704	0.4214	1.0000	0.0000
Complete Program	0.5578	0.4968	0.6210	0.4854	0.6654	0.4728	0.6709	0.4729
Top 100 Placement	0.1971	0.3979	0.2472	0.4316	0.3074	0.4623	0.3418	0.4773
Program Rank	88.9133	57.4823	78.5844	57.1361	63.3828	54.1845	37.0000	0.0000
Missing Program Rank	0.4199	0.4937	0.3422	0.4747	0.2490	0.4333	0.0000	0.0000
Placement Rank	127.7505	34.9306	123.0222	37.2570	117.2696	40.5746	117.4430	37.5728
Missing Placement Rank	0.4444	0.4970	0.3830	0.4864	0.3346	0.4728	0.3291	0.4729
2016	0.2052	0.4040	0.1696	0.3754	0.1790	0.3841	0.2025	0.4045
2017	0.2014	0.4011	0.1788	0.3833	0.1829	0.3873	0.1772	0.3843
2018	0.1928	0.3946	0.2686	0.4435	0.2062	0.4054	0.2532	0.4376
2019	0.2027	0.4021	0.2084	0.4064	0.2179	0.4136	0.2152	0.4136
Age	25.9910	3.1479	26.2635	3.1823	26.0817	3.0652	26.3291	3.2806
Female	0.3912	0.4881	0.4147	0.4929	0.4397	0.4973	0.4177	0.4963
U.S.	0.2357	0.4245	0.2380	0.4261	0.2996	0.4590	0.3038	0.4628
International Asian	0.4809	0.4997	0.5230	0.4997	0.3424	0.4754	0.4304	0.4983
GRE Verbal Percentile	68.0129	22.5869	76.8421	18.3703	79.5703	18.3047	76.9744	19.4041
GRE Quantitative Percentile	88.4550	11.6330	93.1362	5.4135	93.5391	4.4614	92.3077	5.7554
GRE Analytic Percentile	50.0613	27.5744	56.3653	26.5466	60.3398	26.0880	57.4615	26.6481
Undergraduate GPA	3.6307	0.3026	3.6645	0.3003	3.7227	0.2610	3.7068	0.2880
TOEFL/IELTS	7.4376	0.5587	7.5641	0.5834	7.7011	0.6667	7.4464	0.6137
Elite U.S. University	0.1267	0.3327	0.1736	0.3790	0.2257	0.4188	0.1772	0.3843
Elite U.S. Liberal Arts College	0.0296	0.1696	0.0347	0.1832	0.0428	0.2028	0.0633	0.2450
Elite Foreign University	0.0386	0.1928	0.0490	0.2160	0.0350	0.1842	0.0253	0.1581
Other Foreign University	0.4049	0.4910	0.4188	0.4936	0.3619	0.4815	0.4430	0.4999
Missing Undergraduate Institution	0.2203	0.4145	0.1910	0.3933	0.1984	0.3996	0.1266	0.3346
Graduate Degree	0.5067	0.5001	0.5730	0.4949	0.5136	0.5008	0.5190	0.5028
Work Experience	0.5247	0.4995	0.5894	0.4922	0.6770	0.4685	0.7468	0.4376
#Professor Recommenders	2.6230	0.7371	2.6302	0.7165	2.5992	0.7118	2.5696	0.7104
#Prolific Recommenders	1.2602	1.0686	1.3912	1.0840	1.4825	1.0424	1.3038	1.0664
Missing GRE	0.0335	0.1800	0.0102	0.1006	0.0039	0.0624	0.0127	0.1125
Missing Undergraduate GPA	0.7853	0.4107	0.7487	0.4340	0.6770	0.4685	0.6835	0.4681
Missing TOEFL/IELTS	0.6076	0.4884	0.5935	0.4914	0.6615	0.4741	0.6456	0.4814
Missing Age	0.0009	0.0293	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Observations	2329		979		257		79	

Subjective Variables

	Full Sample		Transition = 1		Accept = 1		Matriculate = 1	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Committee Score	6.4070	1.8701	6.4070	1.8701	8.5003	0.8151	8.4854	0.7157
#Math Courses	5.7637	1.8154	5.8173	1.7904	5.6992	1.8640	5.6585	1.6219
Advanced Math	0.4860	0.5004	0.4909	0.5007	0.5680	0.4973	0.4651	0.5047
Letters Length/100	8.3474	3.5749	8.5333	3.8263	10.0514	4.2864	9.7088	3.4876
Letters Positive Terms (%)	11.9770	2.2993	11.9976	2.2077	12.2167	1.9255	12.5110	2.0904
Letters Negative Terms (%)	5.7286	1.3427	5.7216	1.3318	5.7446	1.2313	5.8938	1.2871
Letters Standout Terms (%)	0.6241	0.4020	0.6241	0.4029	0.7151	0.4509	0.7077	0.5109
Letters Ability Terms (%)	1.3326	0.6239	1.3457	0.5964	1.3796	0.5620	1.4334	0.6037
Letters Research Terms (%)	3.4837	1.3319	3.5208	1.3423	3.8548	1.1984	3.8395	1.2121
Letters Grindstone Terms (%)	1.8189	0.7734	1.8674	0.7808	1.9975	0.7588	2.1600	0.8093
Letters Teaching Terms (%)	1.7379	0.7710	1.7489	0.7759	1.7275	0.7536	1.8976	0.7577
Letters Communal Terms (%)	0.5946	0.3890	0.6019	0.3954	0.5494	0.3234	0.5630	0.3875
Letters Agentic Terms (%)	0.4768	0.3150	0.4842	0.3229	0.5039	0.3222	0.5597	0.3314
Letters Terms (%) PC1	-0.0579	1.6183	0.0000	1.5768	0.2176	1.3817	0.5697	1.5386
Letters Terms (%) PC2	-0.0271	1.1177	0.0000	1.1446	0.1002	1.1774	0.1699	1.2823
CV Length/100	4.3532	2.4237	4.3982	2.4758	4.9641	2.7175	4.8261	2.3473
CV Coding	2.8321	2.1135	2.7931	2.1000	2.8618	2.1129	2.5122	1.6450
CV Publication/Working Paper	0.2518	0.4346	0.2571	0.4377	0.3659	0.4836	0.3902	0.4939
Essay Length/100	4.5099	1.5741	4.5173	1.5805	4.7488	1.6373	4.9816	1.7932
Essay Specific Professor	0.4822	0.5003	0.5219	0.5003	0.5691	0.4972	0.6512	0.4822
Essay Economics Terms (%)	9.7284	2.6219	9.7216	2.6635	9.5303	2.7324	9.8941	3.2608
Essay Research Terms (%)	5.7975	2.0501	5.7402	2.0168	5.7313	1.8797	5.5231	1.8432
Essay Positive Terms (%)	11.8293	2.6518	11.7324	2.6650	11.4734	2.4592	11.3169	2.4455
Essay Negative Terms (%)	6.2844	1.9056	6.3233	1.9712	6.2889	2.0860	6.2651	1.9436
Missing Committee Score	0.5844	0.4929	0.0112	0.1055	0.0195	0.1384	0.0633	0.2450
Missing Application Pdf	0.8154	0.3881	0.6650	0.4722	0.5136	0.5008	0.4557	0.5012
Missing #Math Courses	0.8201	0.3842	0.6701	0.4704	0.5214	0.5005	0.4810	0.5028
Missing Letters	0.8175	0.3863	0.6691	0.4708	0.5136	0.5008	0.4557	0.5012
Missing CV	0.8210	0.3835	0.6742	0.4689	0.5214	0.5005	0.4810	0.5028
Missing Essay	0.8192	0.3849	0.6731	0.4693	0.5214	0.5005	0.4557	0.5012
Observations	2329		979		257		79	

Regression Results

A3: Correlation Matrix

	Trans	Vaxx	Anx	Mar	GHI	Art	Case	TSIR	PtoR	NPreR	PtoR	NPreR	2014	2017	2018	2019	Age	Fem	US	Inc3pc	ICRSV	GRSQ	GSES	DPS	TJ	81to	BLA	8Tor	20Tor	90Sq	Grad	Work	Prof	Pov	MSNR	BSDPS	MT.D	MAge	Soon	NMark	AMark	Ineff	LPos	1Reg	1Sta	1Abs	1Dis	1Occ	1Tea	1Com	1Dge	LP21	LP22	CVN	CVCod	CVPay	RadN	RadPred	RadDem	RadPos	RadSq	MSLow	MPSP	MRMMark	ML	MCV	MRes
Transition		

A3: Recommendation Letters Principal Components

	PC1	PC2
Positive	0.5044	-0.1769
Negative	0.1479	0.3759
Standout	0.2115	-0.5082
Ability	0.4667	-0.0177
Research	0.1214	0.6418
Grindstone	0.3879	0.3075
Teaching	0.3708	0.0743
Communal	0.1284	-0.1983
Agentic	0.3768	-0.1324
Proportion of Variance	0.2763	0.1456

Regression Results

A3: Logistic Regression Results

Dependent Variable	Accept	Accept	Matriculate	7100 Placement	7200 Placement	7200 Placement	Placement Rank	Placement Rank	Placement Rank
2016	-0.1626	-0.1609	-0.2573	0.0230	0.0302	0.0295	1.0028	0.7607	-0.0419
	(0.1581)	(0.1555)	(0.1631)	(0.1706)	(0.1711)	(0.1707)	(0.1597)	(0.1561)	(0.2056)
2017	-0.0954	-0.1047	0.1317	0.0629	0.0721	0.1468	2.7736	0.6235	0.7673
	(0.1599)	(0.1574)	(0.1652)	(0.1727)	(0.1732)	(0.1729)	(0.1617)	(0.1581)	(0.2082)
2018	-0.2206	-0.7689***	0.0474	0.0424	-0.0350	0.0142	2.7295	0.3685	0.9773
	(0.2648)	(0.2453)	(0.2439)	(0.1761)	(0.1887)	(0.2011)	(0.1631)	(0.1611)	(0.2786)
2019	-0.2660	-0.8139***	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.2653)	(0.2114)	(0.2455)	(0.1821)	(0.1938)	(0.2052)	(0.1658)	(0.1636)	(0.2423)
Age	-0.0216	-0.0200	-0.0997**	0.0160	0.0162	0.0230*	0.1006	0.0632	-0.1277
	(0.0230)	(0.0227)	(0.0396)	(0.0248)	(0.0248)	(0.0248)	(0.0233)	(0.0227)	(0.0299)
Female	0.1750**	0.1753**	-0.1645	-0.0882	-0.1024	-0.0256	1.6895	1.2651	-1.6375
	(0.1497)	(0.1611)	(0.0736)	(0.1126)	(0.1118)	(0.1252)	(0.0774)	(0.0755)	(2.5117)
U.S.	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000
	(0.1562)	(0.1616)	(0.1658)	(0.1081)	(0.1084)	(0.1216)	(0.1766)	(0.1761)	(0.1844)
International Asian	-1.1568***	-1.1567***	-1.1999**	0.1495	0.1568	-0.1006	7.0212**	6.3569**	5.3887
	(0.1124)	(0.4146)	(0.4956)	(0.1376)	(0.1391)	(0.4568)	(3.0116)	(3.0706)	(3.5577)
GRE Verbal Percentile	0.0264***	0.0075	-0.0167	0.0040	0.0035	0.0087	-0.0769**	-0.0729*	-0.0155
	(0.0097)	(0.0107)	(0.0107)	(0.0064)	(0.0065)	(0.0067)	(0.0038)	(0.0038)	(0.0053)
GRE Quantative Percentile	0.1319***	0.0938***	0.1031**	0.0247***	0.0187***	0.0167	-0.0567**	-0.0467**	-0.1543***
	(0.0202)	(0.0275)	(0.0444)	(0.0077)	(0.0075)	(0.0084)	(0.0347)	(0.0370)	(0.0460)
GRE Analytic Percentile	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Undergraduate GPA	1.2736*	-1.0909	-0.7259	-0.3976	-0.4722	-0.6687*	6.6560	6.6511	17.2907**
	(0.6855)	(0.7139)	(0.7139)	(0.3855)	(0.3855)	(0.3855)	(4.4000)	(4.4000)	(7.0000)
TOEFL/IELTS	0.4236	-0.1185	0.4416	0.1778	0.1437	0.3287	-0.2954	-0.2968	-7.6857*
	(0.2824)	(0.5076)	(0.9529)	(0.1671)	(0.1666)	(0.1767)	(4.5852)	(4.5431)	(3.9269)
Elite U.S. University	0.1000**	0.0999**	0.0999**	0.0999**	0.0999**	0.0999**	0.0999**	0.0999**	0.0999**
	(0.1000)	(0.1000)	(0.0799)	(0.1000)	(0.1000)	(0.1000)	(0.1000)	(0.1000)	(0.0800)
Elite U.S. Liberal Arts College	0.0167	0.5666	0.9907	0.0731	0.0883	-0.1050	-0.6670	-0.6567	-0.6806
	(0.1558)	(0.4896)	(0.6437)	(0.1070)	(0.1061)	(0.3936)	(0.6071)	(0.7041)	(0.7222)
Elite Foreign University	-0.2166	-0.1607	-0.0673	-0.1216	0.0751	0.0909	-0.7726	-0.5414	-2.2142
	(0.1558)	(0.1558)	(0.1558)	(0.1558)	(0.1558)	(0.1558)	(0.1558)	(0.1558)	(0.1558)
Other Foreign University	0.6119*	0.6119	0.7273	0.1693	0.1693	0.1693	-0.3567	-0.7556	0.7268**
	(0.3222)	(0.4065)	(0.6167)	(0.3941)	(0.3975)	(0.2216)	(0.9079)	(0.9622)	(0.3486)
Missing Undergraduate Institution	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.3211)	(0.4041)	(0.7240)	(0.3557)	(0.3522)	(0.2160)	(0.1101)	(0.1235)	(0.5809)
Graduate Degree	0.1726	-0.3058	-0.3954	0.0494	-0.0921	-0.1835	-0.2283	-0.2242	-2.6253
	(0.1558)	(0.1558)	(0.1558)	(0.1558)	(0.1558)	(0.1558)	(0.1558)	(0.1558)	(0.1558)
Work Experience	0.1709***	0.7019***	0.1923**	0.1558*	0.1572	0.2629***	0.0452	-2.2304	-4.4358**
	(0.1631)	(0.1660)	(0.9291)	(0.1251)	(0.1270)	(0.1422)	(3.2006)	(3.2181)	(2.0893)
#Professor Recommenders	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.1237)	(0.1245)	(0.1213)	(0.0745)	(0.0754)	(0.0842)	(0.0492)	(0.0495)	(0.1746)
#Possible Recommenders	0.2268***	0.2681	0.2993	0.1013*	0.0899*	0.0668	-1.4623	-1.1967	0.1728
	(0.0811)	(0.1721)	(0.2048)	(0.0592)	(0.0513)	(0.0595)	(1.5040)	(1.3975)	(0.1756)
Missing GRE	-0.1099	-1.0141		-0.4515	-0.5354	0.6176	13.5282	9.9163	-3.7108
	(0.3211)	(0.4041)		(0.3557)	(0.3522)	(0.2160)	(0.1101)	(0.1235)	(0.5809)
Missing Undergraduate GPA	-0.1422	-0.1799	-0.0768	-0.1018	-0.0639	-0.0512	-0.3296	-0.3294	-0.8723
	(0.2216)	(0.4413)	(0.8543)	(0.1560)	(0.1598)	(0.1756)	(3.6808)	(3.6662)	(3.3972)
Missing TOEFL/IELTS	0.1000**	0.1000**	0.1000**	0.1000**	0.1000**	0.1000**	0.1000**	0.1000**	0.1000**
	(0.2662)	(0.4372)	(0.5955)	(0.1347)	(0.1342)	(0.1453)	(0.1208)	(0.1216)	(2.9967)
Committee Score		4.3624***	-0.2608		0.0942***	0.0899		2.5589**	0.4884
		(0.1558)	(0.1558)		(0.1558)	(0.1558)		(0.1558)	(0.1558)
Grade Course	-0.0949	-0.1556	0.0255	0.0626	0.0642	0.0858	-2.1833	-2.4958	0.5124
	(0.1077)	(0.1629)	(0.1648)	(0.0795)	(0.0764)	(0.0895)	(1.6031)	(1.6414)	(1.5902)
Advanced Math	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.1025)	(0.1625)	(0.1621)	(0.1065)	(0.1028)	(0.1203)	(0.2606)	(0.4227)	(0.5824)
Letters Length, LSE	0.1358***	0.1084	0.0910	0.0067	-0.0018	-0.0156	-0.2591	-0.6078	0.7356
	(0.0511)	(0.0742)	(0.1193)	(0.0331)	(0.0339)	(0.0358)	(0.2254)	(0.2372)	(0.7000)
Letters Terms (TL) PCU	0.0107	0.2930	0.4734***	0.0362	0.0251	0.1268	1.4082	1.8374	-0.9412
	(0.1042)	(0.2930)	(0.1792)	(0.1242)	(0.1242)	(0.1242)	(0.1242)	(0.1242)	(0.1242)
Letters Terms (TL) PCU	-0.0013	-0.0000	0.0756	-0.1238	-0.1199	-0.1199	-0.1199	0.5908	-0.1199
	(0.1044)	(0.4163)	(0.2616)	(0.1162)	(0.1177)	(0.1272)	(2.7992)	(2.8081)	(2.5856)
CV Length, LSE	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.0417)	(0.1184)	(0.1372)	(0.0942)	(0.0941)	(0.0955)	(1.2232)	(1.1130)	(1.0354)
CV Coding	-0.2439	-0.2366*	-0.2913**	0.0734	0.0710	0.0943	-1.1601	-1.0941	-2.0142
	(0.1558)	(0.1558)	(0.1558)	(0.1558)	(0.1558)	(0.1558)	(0.1558)	(0.1558)	(0.1558)
Publication/Working Paper	0.4163*	0.4163*	0.4163*	0.2982	0.3014	0.3118	-0.1344	-0.1344	-4.4205
	(0.1621)	(0.1623)	(0.1574)	(0.3998)	(0.3034)	(0.1302)	(7.2871)	(7.1131)	(6.7855)
Essay Length, LSE	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.1621)	(0.1623)	(0.1574)	(0.0751)	(0.0764)	(0.0821)	(0.4642)	(0.5721)	(0.4264)
Essay Specific Professor	0.1662	0.4170	0.6655	0.2164	0.1774	0.2672	-0.4668	-0.9204	-7.5268
	(0.1558)	(0.7014)	(0.5118)	(0.1607)	(0.2102)	(0.2758)	(0.2215)	(0.9963)	(0.3716)
Essay Economics Terms (%)	-0.0160	0.0670	0.2976**	-0.1134**	-0.1163**	-0.1822	0.6420***	0.7277***	2.1702*
	(0.1558)	(0.1558)	(0.1558)	(0.1558)	(0.1558)	(0.1558)	(0.1558)	(0.1558)	(0.1558)
Essay Research Terms (%)	-0.0310	-0.1459	-0.1868	0.0697	0.0642	0.0191	-1.7354**	-1.6799**	-1.1975
	(0.0736)	(0.1495)	(0.1151)	(0.0631)	(0.0635)	(0.0674)	(1.4220)	(1.4209)	(1.2855)
Essay Positive Terms (%)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.0512)	(0.1641)	(0.1151)	(0.0493)	(0.0498)	(0.0542)	(1.5416)	(1.1496)	(0.9752)
Essay Negative Terms (%)	-0.0151	-0.0899	0.0019	-0.0396	-0.0454	-0.0472	0.0164	0.9436	0.9937
	(0.0779)	(0.1641)	(0.1071)	(0.0693)	(0.0698)	(0.0742)	(1.4221)	(1.4221)	(1.2855)
Missing Committee Score		-0.4303			-0.3606**	-0.2196		0.2254***	1.2669
		(0.4303)			(0.1597)	(0.1519)		(3.4582)	(3.6256)
Missing Application Pref	-3.0227***	0.1926	0.1463		0.2686	0.3316	-0.1611	6.4397	-6.8456
	(0.1558)	(0.4686)	(0.4562)		(0.2562)	(0.2476)	(0.2192)	(4.0218)	(4.1416)
Open House			1.8884***						
			(0.4316)						
Program Rank							-0.6667***		0.1611***
							(0.0257)		(0.0257)
Missing Program Rank							-1.1107***		16.4717***
							(0.4077)		(0.8947)
Sample	All	All	Accept = 1	All	All	All	All	All	All
Observations	2506	2506	864	2506	2506	2506	2506	2506	2506
Pseudo-R^2	0.0997	0.1035	0.2977	0.0506	0.0513	0.1048	0.0896	0.0897	0.1845

*Robust standard errors in parentheses *p<0.05, **p<0.01, ***p<0.001*

- **Accept ($p < .1$):** 2017 (-), 2018 (-), 2019 (-), Female (+), International Asian (-), GRE Quantitative Percentile (+), Work Experience (+), Committee Score (+), CV Coding (-)
- **Success ($p < .1$):** Age (+), Undergraduate GPA (-), TOEFL/IELTS (+), Work Experience (+), Essay Positive Terms (+), Program Rank (-), Missing Program Rank (-)

Regression Results

A3: Logistic Regression AMEs

Dependent Variable	Accept	Accept	Matriculate	T100 Placement	T100 Placement	T100 Placement	Placement Rank	Placement Rank	Placement Rank
2016	-0.0271	-0.0091	0.0167	0.0035	0.0046	0.0112	0.0650	0.0690	-0.1639
	(0.0171)	(0.0180)	(0.0974)	(0.0242)	(0.0241)	(0.0218)	(0.0615)	(0.0622)	(0.1469)
2017	-0.0060	-0.0304***	0.0161	0.0127	0.0169	0.0166	1.1574	1.4571	6.1631
	(0.0166)	(0.0116)	(0.0945)	(0.0235)	(0.0234)	(0.0212)	(0.5969)	(0.6035)	(1.4251)
2018	-0.0159	-0.0697***	0.1328	0.0096	0.0166	0.0216	1.3167	1.7091	7.2907
	(0.0180)	(0.0186)	(0.0957)	(0.0270)	(0.0270)	(0.0254)	(1.1439)	(1.1466)	(1.7791)
2019	-0.0192	-0.0766***	0.1328	0.0096	0.0166	0.0216	1.3167	1.7091	7.2907
	(0.0192)	(0.0168)	(0.0975)	(0.0275)	(0.0275)	(0.0259)	(2.2356)	(2.2361)	(1.8977)
Age	-0.0004	-0.0001	0.0035**	0.0007	0.0007	0.0004*	0.1756	0.1751	-0.0086
	(0.0004)	(0.0003)	(0.0016)	(0.0005)	(0.0005)	(0.0002)	(0.0772)	(0.0772)	(0.0177)
Female	0.0268**	0.0243**	0.0095	0.0216	0.0217	0.0202	0.3717	0.3822	0.7117
	(0.0116)	(0.0096)	(0.0532)	(0.0172)	(0.0172)	(0.0159)	(1.3967)	(1.3951)	(1.1316)
U.S.	-0.0358*	-0.0013	0.1077*	0.0279	0.0311	0.0192	-0.321	-0.4701	-0.3241
	(0.0196)	(0.0184)	(0.0564)	(0.0292)	(0.0291)	(0.0192)	(1.3132)	(1.3093)	(1.0491)
International Asian	-0.0146	-0.0178*	0.0845	0.0172	0.0172	0.0178	(1.7114)	(1.7199)	(1.1496)
	(0.0146)	(0.0097)	(0.0845)	(0.0212)	(0.0211)	(0.0178)	(1.7114)	(1.7199)	(1.1496)
GRE Verbal Percentile	0.0220***	0.0005	0.0015	0.0008	0.0005	0.0005	-0.0949**	-0.0358*	-0.0351
	(0.0042)	(0.0033)	(0.0183)	(0.0059)	(0.0059)	(0.0055)	(0.0373)	(0.0206)	(0.0221)
GRE Quantitative Percentile	0.0205***	0.0151***	0.0277*	0.0207***	0.0207***	0.0004	-0.0987**	-0.0418**	-0.1114*
	(0.0041)	(0.0035)	(0.0164)	(0.0052)	(0.0051)	(0.0011)	(0.0372)	(0.0192)	(0.0670)
GRE Analytic Percentile	0.0052	0.0030	0.0011	0.0014	0.0014	0.0014	-0.0121	-0.0121	-0.0192
	(0.0052)	(0.0032)	(0.0181)	(0.0064)	(0.0064)	(0.0064)	(0.0222)	(0.0222)	(0.0202)
Undergraduate GPA	0.0918*	0.0195	0.1026	0.0610	0.0721	-0.1078**	2.3266	5.1666	7.9332**
	(0.0499)	(0.0729)	(0.1003)	(0.0660)	(0.0567)	(0.0516)	(1.9511)	(3.1699)	(3.5364)
TOEFL/IELTS	0.0332	-0.0022	-0.0431	0.0272	0.0219	0.0439**	-0.1847	-0.4266	-1.6277*
	(0.0211)	(0.0211)	(0.1063)	(0.0277)	(0.0274)	(0.0218)	(0.6997)	(0.7000)	(0.9222)
Elite U.S. University	0.0503***	0.0116	-0.0708	-0.0034	0.0116	-0.0132	-1.167	-0.7717	-1.1561
	(0.0196)	(0.0111)	(0.0834)	(0.0227)	(0.0226)	(0.0196)	(2.9414)	(2.9479)	(2.0402)
Elite U.S. Liberal Arts College	0.0503***	0.0116	-0.0708	-0.0034	0.0116	-0.0132	-1.167	-0.7717	-1.1561
	(0.0197)	(0.0116)	(0.1097)	(0.0317)	(0.0311)	(0.0269)	(4.3365)	(4.3366)	(2.7491)
Elite Foreign University	0.0389	-0.0102	-0.0996	0.0186	0.0115	0.0110	-0.3354	-0.7426	-0.0344
	(0.0242)	(0.0127)	(0.0844)	(0.0228)	(0.0227)	(0.0198)	(2.9414)	(2.9479)	(2.0402)
Other Foreign University	0.0192*	-0.0118	-0.1008	0.0258	0.0167	0.0208	-0.3376	-2.7964	-4.1994**
	(0.0104)	(0.0110)	(0.0886)	(0.0181)	(0.0181)	(0.0279)	(2.3566)	(2.3912)	(1.9486)
Missing Undergraduate Institution	0.0192*	-0.0118	-0.1008	0.0258	0.0167	0.0208	-0.3376	-2.7964	-4.1994**
	(0.0102)	(0.0111)	(0.1002)	(0.0283)	(0.0288)	(0.0256)	(2.4439)	(2.4439)	(2.0496)
Graduate Degree	0.0123	-0.0058	0.0429	0.0062	-0.0142	-0.0151	-1.5821	-0.6795	1.2056
	(0.0116)	(0.0077)	(0.0712)	(0.0169)	(0.0164)	(0.0172)	(1.5257)	(1.5401)	(1.2666)
Work Experience	0.0307***	0.0140**	0.0997**	0.0354*	0.0271	0.0358**	-0.7125	-1.0872	-2.9916**
	(0.0114)	(0.0069)	(0.0501)	(0.0191)	(0.0189)	(0.0179)	(0.8966)	(0.8966)	(1.3966)
cPosNone Recommenders	-0.0027	-0.0011	0.0016	-0.0057	-0.0078	-0.0069	0.0229	0.0006	0.0196
	(0.0077)	(0.0065)	(0.0427)	(0.0113)	(0.0113)	(0.0108)	(0.0691)	(0.0692)	(0.0705)
cPosMin Recommenders	-0.0027	-0.0011	0.0016	-0.0057	-0.0078	-0.0069	0.0229	0.0006	0.0196
	(0.0056)	(0.0031)	(0.0295)	(0.0081)	(0.0081)	(0.0073)	(0.4995)	(0.4995)	(0.5172)
Missing GRE	-0.1302	0.0168		-0.0693	-0.0497	0.0222	0.5829	4.1215	-1.7054
	(0.0796)	(0.0222)		(0.0454)	(0.0454)	(0.0429)	(1.1757)	(2.7252)	(3.0596)
Missing Undergraduate GPA	-0.1256	-0.0022	0.0014	-0.0206	-0.0081	0.0005	0.1757	0.7252	-0.0596
	(0.0156)	(0.0083)	(0.0703)	(0.0260)	(0.0262)	(0.0231)	(1.1547)	(1.1050)	(1.5432)
Missing TOEFL/IELTS	0.0077	0.0075	-0.0204	-0.0294	-0.023	-0.0058	0.1006	0.2661	-0.0017
	(0.0167)	(0.0075)	(0.0836)	(0.0196)	(0.0195)	(0.0182)	(1.4097)	(1.4025)	(1.1913)
Committee Score		0.0000			0.0000	0.0000		0.0000	0.0000
		(0.0000)			(0.0000)	(0.0000)		(0.0000)	(0.0000)
cMath Courses	-0.0066	-0.0065	0.0107	0.0094	0.0101	0.0065	-0.1495	0.1022	0.4195
	(0.0066)	(0.0045)	(0.0337)	(0.0096)	(0.0095)	(0.0092)	(0.5970)	(0.5970)	(0.5970)
Returned Math	0.0307***	0.0144	0.0929**	0.0296	0.0317	0.0187	-0.1483	1.9466	0.5011
	(0.0104)	(0.0104)	(0.0741)	(0.0199)	(0.0202)	(0.0272)	(1.1561)	(1.1022)	(2.5421)
Letters Length/100	0.0004	0.0001	0.0004	0.0001	0.0001	0.0001	0.0019	0.0019	0.0019
	(0.0007)	(0.0006)	(0.0031)	(0.0010)	(0.0010)	(0.0008)	(0.1970)	(0.1812)	(0.1970)
Letters Terms (%) PC1	0.0108	0.0056	0.0672**	0.0056	0.0018	0.0139	0.7752	0.6855	-0.2976
	(0.0086)	(0.0055)	(0.0335)	(0.0098)	(0.0098)	(0.0085)	(0.5055)	(0.5055)	(0.5055)
Letters Terms (%) PC2	0.0084	0.0084	0.0252	0.0168	0.0168	0.0135	0.7256	0.7256	0.6556
	(0.0105)	(0.0077)	(0.0372)	(0.0170)	(0.0170)	(0.0160)	(1.1695)	(1.1939)	(1.0706)
CV Length/100	0.0026	0.0026	0.0250	0.0068	0.0062	0.0056	-0.0468	-0.1877	-0.3706
	(0.0042)	(0.0037)	(0.0190)	(0.0066)	(0.0066)	(0.0075)	(0.4370)	(0.4329)	(0.4796)
CV Publication	0.0107**	0.0107**	0.0425**	0.0107	0.0107	0.0107	0.0000	0.0000	0.0000
	(0.0054)	(0.0053)	(0.0213)	(0.0082)	(0.0082)	(0.0082)	(0.7083)	(0.7083)	(0.6139)
CV Publication/Working Paper	0.0407	0.0156	-0.0237	0.0491	0.0421	0.0145	-0.3511	-0.2409	-0.1150
	(0.0257)	(0.0154)	(0.1107)	(0.0307)	(0.0307)	(0.0284)	(1.1695)	(1.1939)	(1.0706)
Essay Length/100	0.0139*	0.0045	0.0233	-0.0061	-0.0081	-0.0098	-0.1293	-0.2670	0.6818
	(0.0067)	(0.0072)	(0.0316)	(0.0113)	(0.0117)	(0.0102)	(0.9988)	(0.9022)	(0.7161)
Essay Specific Professor	0.0206	0.0135	0.0743	0.0189	0.0189	0.0189	(2.5564)	(2.5564)	(2.1437)
	(0.0206)	(0.0135)	(0.0743)	(0.0189)	(0.0189)	(0.0189)	(2.5564)	(2.5564)	(2.1437)
Essay Economic Terms (%)	-0.0022	0.0065	0.0846**	-0.0272**	-0.0276**	-0.0156	1.7410***	1.7949***	0.9944*
	(0.0059)	(0.0054)	(0.0342)	(0.0085)	(0.0081)	(0.0073)	(0.5566)	(0.4837)	(0.5116)
Essay Research Terms (%)	-0.0061	-0.0069	-0.0266	0.0167	0.0086	0.0039	-1.7076**	-1.7725**	-0.4895
	(0.0061)	(0.0059)	(0.0344)	(0.0107)	(0.0107)	(0.0095)	(0.7997)	(0.7997)	(0.6183)
Essay Positive Terms (%)	-0.0014	0.0011	0.0057	0.0014	0.0014	0.0013*	0.0127	-0.1664	-0.4785
	(0.0037)	(0.0033)	(0.0180)	(0.0064)	(0.0064)	(0.0075)	(0.0064)	(0.5609)	(0.5642)
Essay Negative Terms (%)	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
	(0.0001)	(0.0000)	(0.0002)	(0.0002)	(0.0002)	(0.0002)	(0.0004)	(0.0004)	(0.0004)
Missing Committee Score		-0.0056			-0.0029**	-0.0277		6.6892***	1.3226
		(0.0035)			(0.0014)	(0.0172)		(1.7783)	(1.6420)
Missing Application PDF	-0.0773***	0.0035	-0.1056	0.0224	0.0162	0.0106	-2.4277	-4.2072	-1.1012
	(0.0227)	(0.0168)	(0.0956)	(0.0375)	(0.0375)	(0.0318)	(2.9066)	(2.9641)	(2.6462)
Open House			0.3630***						
			(0.0912)						
Program Rank							-0.6611***		0.0833***
							(0.0002)		(0.0181)
Missing Program Rank							-0.5667***		41.0729***
							(0.0612)		(0.8832)
Sample	All	All	Accept = 1	All	All	All	All	All	All
Observations	2026	2026	367	2026	2026	2026	2026	2026	2026

Robust standard errors in parentheses *p<0.10, **p<0.05, ***p<0.01

- **Accept ($p < .1$):** 2017 (-), 2018 (-), 2019 (-), Female (+), International Asian (-), GRE Quantitative Percentile (+), Work Experience (+), Committee Score (+), CV Coding (-)
- **Success ($p < .1$):** Age (+), Undergraduate GPA (-), TOEFL/IELTS (+), Work Experience (+), Essay Positive Terms (+), Program Rank (-), Missing Program Rank (-)

Regression Results

A3: Logistic Regression Results (Synthetic)

Dependent Variable	Accept	Accept	Matriculate	T100 Placement	T100 Placement	T100 Placement	Placement Rank	Placement Rank	Placement Rank
2016	-0.1681	-0.1571	0.1256	0.0997	0.0997	0.0728	-0.6542	-0.5631	-1.5907
	(0.1596)	(0.2017)	(0.1479)	(0.1472)	(0.1614)	(0.1973)	(0.5724)	(0.5702)	(1.1436)
2017	-0.1591	-0.1607*	0.0747	0.1527	0.2901	0.1545	-12.1499***	-12.2024***	-9.3630
	(0.1599)	(0.1967)	(0.1482)	(0.1475)	(0.1772)	(0.2000)	(0.5737)	(0.5722)	(1.1452)
2018	0.2889	0.2951	0.2947	0.3437	0.3787	0.3787	0.2272	-0.7259**	0.7459**	-0.6068
	(0.2094)	(0.2212)	(0.1917)	(0.1906)	(0.1827)	(0.1872)	(0.2272)	(0.4278)	(0.4180)	(0.9875)
2019	0.3111	0.3111	0.3111	0.3111	0.3111	0.3111	0.3111	0.3111	0.3111	0.3111
	(0.2227)	(0.2480)	(0.4049)	(0.1927)	(0.1910)	(0.2061)	(0.6027)	(0.5058)	(1.1452)	(1.9171)
Age	-0.0168**	-0.0414*	0.0045	-0.0141	-0.0150	-0.0094	0.6484	0.4381	0.2496	
	(0.0082)	(0.0215)	(0.0076)	(0.0090)	(0.0092)	(0.0098)	(0.4004)	(0.3992)	(0.8004)	
Female	-0.1624	-0.1535	0.0845	0.1361	0.1379	0.2109*	-0.1697	-1.4358	-0.7134*	
	(0.1466)	(0.1568)	(0.4036)	(0.1167)	(0.1168)	(0.1259)	(2.6557)	(2.6532)	(2.6709)	
U.S.	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	
	(0.2282)	(0.2464)	(0.6453)	(0.1795)	(0.1821)	(0.1957)	(0.4491)	(0.4469)	(0.8966)	
International Asian	-0.1150	-0.0648	-0.1255	0.2014	0.1972	0.1708	-0.7475	-0.8423	0.1783	
	(0.1668)	(0.1835)	(0.5028)	(0.1394)	(0.1392)	(0.1502)	(0.4541)	(0.3754)	(0.3721)	(0.7231)
GRE Verbal Percentile	-0.0025	-0.0081	0.0004	0.0004	0.0001	-0.0023	-0.0948	-0.0623	0.0181	
	(0.0035)	(0.0055)	(0.0028)	(0.0028)	(0.0028)	(0.0029)	(0.0966)	(0.0963)	(0.1931)	
GRE Quantitative Percentile	-0.0039	-0.0005	-0.0019	0.0149**	0.0143**	-0.0002	-0.6943***	-0.6564***	-0.0164	
	(0.0075)	(0.0086)	(0.0139)	(0.0064)	(0.0066)	(0.0069)	(0.2370)	(0.2340)	(0.1276)	
GRE Analytic Percentile	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
	(0.0031)	(0.0031)	(0.0076)	(0.0024)	(0.0024)	(0.0026)	(0.0347)	(0.0347)	(0.0693)	
Undergraduate GPA	-0.4230	-0.3257	0.8077	0.4075	0.3986	0.4637	-2.7870	-2.7490	-5.6113	
	(0.3284)	(0.3777)	(0.5973)	(0.2752)	(0.2766)	(0.2977)	(0.8855)	(0.8824)	(1.7689)	
TOEFL/IELTS	0.1329	0.1776	0.2673	-0.0662	-0.0635	-0.0142	0.4257	0.2464	-0.4374	
	(0.3164)	(0.3302)	(0.4438)	(0.1939)	(0.1927)	(0.1970)	(0.7072)	(0.7021)	(1.1666)	
Elite U.S. University	-0.1230	-0.1230	-0.1230	-0.1230	-0.1230	-0.1230	-0.1230	-0.1230	-0.1230	
	(0.1287)	(0.1364)	(0.7142)	(0.1094)	(0.1062)	(0.1158)	(0.3795)	(0.3875)	(0.7855)	
Elite U.S. Liberal Arts College	-0.4442	-0.4893		0.0999	-0.0068	-0.0956	-0.2641	-0.2265	0.3486	
	(0.3168)	(0.3764)		(0.3894)	(0.3691)	(0.3964)	(0.9356)	(0.8721)	(1.7472)	
Elite Foreign University	-0.1797	-0.1798	0.0194	-0.0018	-0.0042	-0.0298	-0.4302	-0.9315	-0.2745	
	(0.1654)	(0.1778)	(0.3126)	(0.1358)	(0.1345)	(0.1396)	(0.4705)	(0.4686)	(0.9408)	
Elite Foreign University	-0.6494	-0.1699	0.6613	0.1099	0.1129	0.0656	-0.0213	-0.2658	-0.6009	
	(0.3259)	(0.2342)	(0.9139)	(0.1668)	(0.1689)	(0.1868)	(0.3122)	(0.2870)	(0.6627)	
Missing Undergraduate Institution	-0.1217	-0.1217	-0.1217	-0.1217	-0.1217	-0.1217	-0.1217	-0.1217	-0.1217	
	(0.2317)	(0.2411)	(0.9278)	(0.1916)	(0.1842)	(0.1977)	(0.5567)	(0.5885)	(1.1622)	
Graduate Degree	-0.0366	-0.1354	-0.0330	0.1010	0.1611	0.1906	-1.4794	-1.6161	-2.0725	
	(0.2284)	(0.2468)	(0.3218)	(0.1956)	(0.1966)	(0.2116)	(0.6284)	(0.6258)	(1.2539)	
Work Experience	0.0427*	0.0149	-0.2103	0.0765	0.0695	0.1043	0.6884	0.3542	-2.7342	
	(0.1460)	(0.1668)	(0.3573)	(0.1341)	(0.1340)	(0.1292)	(2.8462)	(2.8451)	(2.8260)	
#Professor Recommenders	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	
	(0.0956)	(0.1021)	(0.2719)	(0.0709)	(0.0701)	(0.0756)	(0.1982)	(0.1916)	(0.7884)	
#Possible Recommenders	-0.0168	-0.0072	0.0434	0.0027	-0.0000	0.0105	0.1027	0.5170	-1.1079	
	(0.0977)	(0.1072)	(0.1695)	(0.0721)	(0.0722)	(0.0775)	(1.1242)	(1.1251)	(2.2351)	
Missing GPA	0.1662	0.0736		0.3136	0.3314	0.2676	-2.0515	-1.9975	-3.1660	
	(0.3282)	(0.3778)		(0.2752)	(0.2766)	(0.2977)	(0.8855)	(0.8824)	(1.7689)	
Missing Undergraduate GPA	0.1747	0.1720	0.2910	-0.1533	-0.1369	-0.1814	0.7205	0.7474	1.4181	
	(0.1861)	(0.2034)	(0.5422)	(0.1391)	(0.1389)	(0.1516)	(1.5497)	(1.5423)	(3.1894)	
Missing TOEFL/IELTS	0.1329	0.1776	0.2673	-0.0662	-0.0635	-0.0142	0.4257	0.2464	-0.4374	
	(0.3164)	(0.3302)	(0.4438)	(0.1939)	(0.1927)	(0.1970)	(0.7072)	(0.7021)	(1.1666)	
Committee Score		0.0377	-0.1004		0.0681	0.0330		-1.6407	-0.4906	
		(0.1554)	(0.2744)		(0.1134)	(0.1158)		(1.1452)	(2.2904)	
Objective Courses	0.0432	0.0706	0.1603	-0.1269	-0.1587*	-0.1554**	1.1826	1.2071	1.8699*	
	(0.0687)	(0.0733)	(0.1601)	(0.0786)	(0.0760)	(0.0802)	(1.7397)	(1.7208)	(1.1610)	
Advanced Math	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
	(0.1956)	(0.2017)	(0.7299)	(0.1664)	(0.1613)	(0.1784)	(0.4233)	(0.4257)	(0.8515)	
Letters Length, 100	-0.0354	-0.0128	0.0679	-0.1049***	-0.1057***	-0.1133***	1.4082**	1.4686**	2.1816***	
	(0.0344)	(0.0418)	(0.0874)	(0.0397)	(0.0376)	(0.0426)	(0.7071)	(0.7065)	(0.6994)	
Letters Terms (%) PC1	0.0230	-0.0073	0.0361	0.0574	0.0643	0.0998	1.1561	1.1016	1.0356	
	(0.0703)	(0.0777)	(0.1556)	(0.0574)	(0.0577)	(0.0618)	(0.9166)	(0.9166)	(1.8332)	
Letters Terms (%) PC2	-0.2128*	-0.2128*	-0.4913	0.0802	0.0802	0.0802	0.1318	-0.6599	-1.7884	
	(0.1234)	(0.1295)	(0.3026)	(0.1330)	(0.1211)	(0.1250)	(2.6915)	(2.7093)	(2.1872)	
CV Length, 100	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	
	(0.0302)	(0.0368)	(0.1142)	(0.0548)	(0.0551)	(0.0611)	(0.1268)	(0.1268)	(0.2536)	
CV Coding	-0.0168	-0.0473	-0.0816	0.0668	0.0720*	0.0787	-1.4029	-1.4770	-1.0259	
	(0.0703)	(0.0777)	(0.1556)	(0.0574)	(0.0577)	(0.0618)	(0.9166)	(0.9166)	(1.8332)	
CV Publication Working Paper	0.3579	0.3695	1.6897*	0.1111	0.1163	0.1111	0.7524	1.1016	5.9956	
	(0.2852)	(0.2942)	(0.9175)	(0.3264)	(0.3272)	(0.3402)	(7.0936)	(7.0748)	(6.2211)	
Essay Length/100	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	
	(0.0273)	(0.0331)	(0.1499)	(0.0785)	(0.0794)	(0.0871)	(0.1764)	(0.1722)	(0.3471)	
Essay Specific Professor	-0.1784	-0.2499	0.1223	-0.0947	-0.0924	0.0899	1.2439	1.2259	-0.1567	
	(0.2564)	(0.2714)	(0.6502)	(0.2774)	(0.2796)	(0.3023)	(0.3568)	(0.2682)	(0.5757)	
Essay Economic Terms (%)	-0.0123	-0.0099	-0.0404	-0.0091	-0.0082	-0.0222	0.2378	0.2960	0.6769	
	(0.0213)	(0.0258)	(0.0606)	(0.0196)	(0.0197)	(0.0212)	(0.2370)	(0.2340)	(0.1276)	
Essay Research Terms (%)	0.0310	0.0304	0.0844	0.1042	0.1004	0.1076	-1.8879	-1.8699	-1.8172	
	(0.0464)	(0.0702)	(0.1578)	(0.0703)	(0.0705)	(0.0723)	(1.5182)	(1.5183)	(3.2696)	
Essay Positive Terms (%)	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	
	(0.0253)	(0.0464)	(0.1213)	(0.0541)	(0.0517)	(0.0605)	(1.1516)	(1.1568)	(1.1858)	
Essay Negative Terms (%)	0.0333	0.0322	-0.0711	-0.0364	-0.0200	-0.0168	0.5011	0.4716	0.2611	
	(0.0354)	(0.0377)	(0.0748)	(0.0359)	(0.0367)	(0.0384)	(0.8855)	(0.8824)	(1.7689)	
Missing Committee Score		-0.5337**	-0.0749		-0.3639**	-0.4239***		-0.8682	-4.1957**	
		(0.4111)			(0.1547)	(0.1588)			(2.0764)	
Missing Application Pdf	-0.6021***	-0.3668	0.6473	0.0974	-0.0946	-0.1530	0.6138	4.0299	5.8616	
	(0.2016)	(0.2743)	(0.4743)	(0.2616)	(0.2619)	(0.2402)	(5.6728)	(4.0592)	(5.4166)	
Open House			1.0943							
			(0.3294)							
Program Rank						-0.0587**				0.1176***
						(0.0239)				(0.0436)
Missing Program Rank						-2.6337***				54.3253***
						(0.4372)				(5.6181)
Sample	All	All	Accept = 1	All	All	All	All	All	All	All
Observations	2506	2506	1164	2506	2506	2506	2506	2506	2506	2506
Pseudo R^2	0.0052	0.0083	0.0402	0.0015	0.0045	0.0195	0.0066	0.0542	0.0953	

*Robust standard errors in parentheses *p<0.10, **p<0.05, ***p<0.01*

- **Accept ($p < .1$):** 2017 (-), Age (-), Letters Terms (%) PC2 (-), Missing Committee Score (-)
- **Success ($p < .1$):** Female (+), Elite U.S. University (+), Missing TOEFL/IELTS (+), #Math Courses (-), Letters Length/100 (-), CV Coding (+), Missing Committee Score (+), Program Rank (-), Missing Program Rank (-)

Regression Results

A3: Logistic Regression AMEs (Synthetic)

Dependent Variable	Accept	Accept	Matriculate	T10D Placement	T10D Placement	T10E Placement	Placement Rank	Placement Rank	Placement Rank
2016	-0.0185	-0.0185	-0.0201	0.0136	0.0137	0.0133	-0.3764	-0.3729	-0.4113
	(0.0202)	(0.0197)	(0.0205)	(0.0343)	(0.0336)	(0.0335)	(0.6715)	(1.1401)	(1.1382)	(1.0189)
2017	-0.0297	-0.0417**	0.1262	0.0387	0.0396	0.0185	-5.3264***	-5.7097***	-5.7939***	-2.7939
	(0.0202)	(0.0197)	(0.0205)	(0.0343)	(0.0336)	(0.0335)	(0.6715)	(1.1401)	(1.1382)	(1.0189)
2018	0.0256	0.0229	0.0308	0.0253	0.0259	0.0150	-4.3977***	-4.3379***	-4.3379***	-2.4375
	(0.0182)	(0.0168)	(0.0894)	(0.0247)	(0.0250)	(0.0233)	(0.6722)	(1.0972)	(1.0968)	(1.7765)
2019	0.0257	0.0257	0.0257	0.0257	0.0257	0.0257	-4.0000***	-4.0000***	-4.0000***	-2.0000
	(0.0197)	(0.0183)	(0.0942)	(0.0264)	(0.0264)	(0.0243)	(0.6722)	(1.1011)	(1.1061)	(1.7765)
Age	-0.0697**	-0.0832**	0.0833	-0.0233	-0.0233	-0.0211	0.0000	0.0000	0.2051	0.1095
	(0.0336)	(0.0327)	(0.0341)	(0.0571)	(0.0559)	(0.0557)	(0.0000)	(0.0000)	(0.1896)	(0.1819)
Female	-0.0322	-0.0117	0.0297	-0.0206	-0.0191	0.0267	-0.7964	-0.7963	-0.7963	-0.3997
	(0.0139)	(0.0121)	(0.0602)	(0.0240)	(0.0239)	(0.0238)	(0.6148)	(1.1705)	(1.1725)	(1.1767)
U.S.	0.0030	-0.0027	0.0295	0.0027	0.0013	-0.0015	1.1597	1.2035	1.3635	1.9003
	(0.0238)	(0.0188)	(0.1235)	(0.0284)	(0.0284)	(0.0236)	(0.6139)	(1.0914)	(1.0905)	(1.6373)
International Asian	-0.0192	-0.0192	-0.0192	-0.0192	-0.0192	-0.0192	-0.0000	-0.0000	-0.0000	-0.0000
	(0.0192)	(0.0164)	(0.0762)	(0.0241)	(0.0241)	(0.0192)	(0.6142)	(1.1706)	(1.1709)	(1.1706)
GRE Verbal Percentile	-0.0002	-0.0000	0.0001	0.0001	0.0000	-0.0002	-0.0011	-0.0291	0.0289	
	(0.0002)	(0.0002)	(0.0011)	(0.0003)	(0.0003)	(0.0003)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
GRE Quantitative Percentile	-0.0002	-0.0000	-0.0019	0.0029**	0.0029**	-0.0000	-0.0000***	-0.0000***	-0.0000***	-0.0000
	(0.0007)	(0.0006)	(0.0030)	(0.0009)	(0.0009)	(0.0008)	(0.0000)	(0.0000)	(0.0734)	(0.0000)
GRE Analytic Percentile	-0.0002	-0.0000	-0.0019	0.0029**	0.0029**	-0.0000	-0.0000***	-0.0000***	-0.0000***	-0.0000
	(0.0002)	(0.0002)	(0.0011)	(0.0003)	(0.0003)	(0.0002)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Undergraduate GPA	-0.0375	-0.0361	0.1421	0.0908	0.0944	0.0948	-0.3043	-0.2824	-0.5904	
	(0.0342)	(0.0304)	(0.2879)	(0.0443)	(0.0430)	(0.0418)	(0.6135)	(1.0272)	(1.0250)	(1.5902)
TOEFL/IELTS	-0.0296	-0.0212	-0.0282	-0.0090	-0.0095	-0.0107	0.1161	0.1699	-0.1831	
	(0.0242)	(0.0222)	(0.1000)	(0.0313)	(0.0308)	(0.0303)	(0.6135)	(1.0272)	(1.0250)	(1.5902)
Elite U.S. University	-0.0158	-0.0158	-0.0158	0.0243	0.0243	0.0243*	-0.3158	-0.3158	-0.3158	-0.3158
	(0.0232)	(0.0225)	(0.1108)	(0.0261)	(0.0262)	(0.0241)	(0.6139)	(1.0220)	(1.0219)	(1.5902)
Elite U.S. Liberal Arts College	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000
	(0.0000)	(0.0000)		(0.0000)	(0.0000)	(0.0000)	(0.6139)	(1.1317)	(1.1317)	(1.1317)
Elite Foreign University	-0.0159	-0.0137	0.0691	-0.0070	-0.0095	-0.0155	-0.4822	-0.4160	-0.1724	
	(0.0232)	(0.0225)	(0.1108)	(0.0261)	(0.0262)	(0.0241)	(0.6139)	(1.0220)	(1.0219)	(1.5902)
Other Foreign University	-0.0036	-0.0001	0.1205	0.0164	0.0154	0.0095	-0.6329	-0.7813	-0.3811	
	(0.0192)	(0.0173)	(0.0801)	(0.0251)	(0.0251)	(0.0237)	(0.6722)	(1.1682)	(1.1689)	
Missing Undergraduate Institution	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000
	(0.0196)	(0.0184)	(0.0882)	(0.0262)	(0.0261)	(0.0218)	(2.1699)	(2.1684)	(2.1684)	(3.8264)
Graduate Degree	-0.0075	-0.0103	0.0924	0.0297	0.0239	0.0226	-0.4896	-0.7648	-0.9099	
	(0.0139)	(0.0124)	(0.0996)	(0.0240)	(0.0239)	(0.0237)	(0.5494)	(1.0395)	(1.1391)	
Work Experience	0.0236*	0.0244	-0.0159	0.0133	0.0132	0.0125	0.1911	0.1666	-0.2884	
	(0.0139)	(0.0124)	(0.0996)	(0.0240)	(0.0239)	(0.0237)	(0.5494)	(1.0395)	(1.1391)	
LPorNone Recommenders	-0.0027	-0.0041	0.0990	-0.0009	0.0041	-0.0036	-0.1644	-0.1816	-0.1021	
	(0.0094)	(0.0079)	(0.0423)	(0.0104)	(0.0104)	(0.0096)	(0.6933)	(0.8961)	(0.7340)	
LProfile Recommenders	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.6933)	(0.8961)	(0.7340)	
Missing GRE	0.0266	0.0004		0.0429	0.0463	0.0159	-0.4177	-0.5961	-0.5830	
	(0.0139)	(0.0124)		(0.0240)	(0.0239)	(0.0237)	(0.5494)	(1.0395)	(1.1391)	
Missing Undergraduate GPA	0.0177	0.0131	0.0852	-0.0351	-0.0376	-0.0386	0.1979	0.2324	0.7831	
	(0.0144)	(0.0154)	(0.0967)	(0.0260)	(0.0259)	(0.0232)	(1.4452)	(1.4455)	(1.4526)	
Missing TOEFL/IELTS	0.0154	-0.0068	0.0984	0.0367	0.0336	0.0273*	-0.7367	-0.6600	-1.3605	
	(0.0102)	(0.0132)	(0.0690)	(0.0175)	(0.0175)	(0.0142)	(1.4995)	(1.4692)	(1.2122)	
Committee Score		0.0000	0.0000		0.0000	0.0000		0.0000	0.0000	
		(0.0000)	(0.0000)		(0.0000)	(0.0000)		(0.0000)	(0.0000)	
#Math Courses	0.0056	0.0059	-0.0262	-0.0170	-0.0170*	-0.0208**	0.0035	0.0650	0.1398**	
	(0.0035)	(0.0036)	(0.0162)	(0.0043)	(0.0043)	(0.0041)	(0.1202)	(0.1202)	(0.0699)	
Advanced Math	0.0020	0.0016	0.1004	-0.0256	-0.0251	-0.0191	0.1600	0.0752	0.7994	
	(0.0229)	(0.0210)	(0.1222)	(0.0301)	(0.0301)	(0.0266)	(0.6139)	(1.0362)	(1.0720)	
Letters Length/100	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000
	(0.0235)	(0.0210)	(0.1222)	(0.0301)	(0.0301)	(0.0266)	(0.6139)	(1.0362)	(1.0720)	(1.5902)
Letters Terms (%) PC1	-0.0025	-0.0006	0.0899	0.0079	0.0068	0.0047	0.5318	0.5530	0.6165	
	(0.0139)	(0.0124)	(0.0996)	(0.0240)	(0.0239)	(0.0237)	(0.5494)	(1.0395)	(1.1391)	
Letters Terms (%) PC2	-0.0367	-0.0367	-0.0183	-0.0110	-0.0110	-0.0146	-0.1588	-0.1911	-0.2247	
	(0.0139)	(0.0099)	(0.0462)	(0.0240)	(0.0239)	(0.0147)	(1.2609)	(1.2951)	(0.9467)	
CV Length/100	0.0009**	0.0001	0.0009	-0.0008	-0.0008	-0.0011	1.1607**	1.1707**	0.1350	
	(0.0001)	(0.0009)	(0.0010)	(0.0075)	(0.0075)	(0.0076)	(0.5002)	(0.5025)	(0.6468)	
CV Coding	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000
	(0.0053)	(0.0034)	(0.0247)	(0.0095)	(0.0095)	(0.0094)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
CV Publication/Working Paper	0.0271	0.0230	0.3997*	0.0152	0.0162	0.0133	1.2881	1.1160	2.4357	
	(0.0139)	(0.0124)	(0.0996)	(0.0240)	(0.0239)	(0.0237)	(0.5494)	(1.0395)	(1.1391)	
Essay Length/100	0.0000	0.0000	-0.0007	-0.0001	-0.0007	-0.0002	0.0000	0.0000	0.0000	0.1000
	(0.0068)	(0.0060)	(0.0231)	(0.0107)	(0.0106)	(0.0103)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Essay Specific Professor	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.0237)	(0.0237)	(0.0870)	(0.0380)	(0.0382)	(0.0337)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Essay Economic Terms (%)	-0.0021	-0.0007	0.0060	-0.0052	-0.0021	-0.0036	0.1680	0.1395	0.2971	
	(0.0068)	(0.0064)	(0.0257)	(0.0074)	(0.0075)	(0.0074)	(0.6148)	(1.0415)	(0.6177)	
Essay Research Terms (%)	0.0027	0.0005	-0.0072	0.0141	0.0143	0.0139	-0.0747	-0.0825	-0.7975	
	(0.0068)	(0.0064)	(0.0257)	(0.0074)	(0.0075)	(0.0074)	(0.6148)	(1.0415)	(0.6177)	
Essay Positive Terms (%)	-0.0014	0.0006	0.0670***	0.0159*	0.0133*	0.0112	-0.7274	-0.6931	-0.5156	
	(0.0062)	(0.0057)	(0.0182)	(0.0070)	(0.0071)	(0.0072)	(0.5356)	(0.5419)	(0.6022)	
Essay Negative Terms (%)	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000
	(0.0035)	(0.0027)	(0.0125)	(0.0085)	(0.0085)	(0.0082)	(0.6535)	(0.4229)	(0.4031)	(0.5117)
Missing Committee Score		-0.0367**			-0.0367**	-0.0437***		-0.4410	-0.7331**	
		(0.0139)			(0.0240)	(0.0237)		(0.5494)	(1.0395)	(1.1391)
Missing Application Ref	-0.0354***	-0.0606	0.1472	0.0079	0.0079	0.0130	2.1131	2.1643	2.1643	1.4366
	(0.0229)	(0.0210)	(0.1994)	(0.0366)	(0.0366)	(0.0367)	(0.7720)	(1.8437)	(1.8437)	(1.5706)
Open House			0.1664							
			(0.1229)							
Program Rank						-0.0088**				0.0614***
						(0.0002)				(0.0160)
Missing Program Rank						-0.1667***				41.4667***
						(0.0282)				(0.3858)
Sample	All	All	Accept = 1	All	All	All	All	All	All	All
Observations	2626	2626	317	2626	2626	2626	2626	2626	2626	2626

Robust standard errors in parentheses *p<0.05, **p<0.01, ***p<0.01

- **Accept ($p < .1$):** 2017 (-), Age (-), Letters Terms (%) PC2 (-), Missing Committee Score (-)
- **Success ($p < .1$):** Female (+), Elite U.S. University (+), Missing TOEFL/IELTS (+), #Math Courses (-), Letters Length/100 (-), CV Coding (+), Missing Committee Score (+), Program Rank (-), Missing Program Rank (-)

Regression Results

A3: Lasso Logistic Regression Results

Dependent Variable	Accept	Accept	Matriculate	T100 Placement	T100 Placement	T100 Placement	Placement Rank	Placement Rank	Placement Rank
2016	-0.1107								
2017		-0.4928							
2018		-2.4456	0.0186						
2019		-1.7752							
Age				0.0084	0.0047	0.0184			
Female	0.2674	0.5589							
U.S.	-0.1959							0.0477	
International Asian	-0.8686	-0.8651	0.1271				-0.5295	-0.7927	
GRE Verbal Percentile	0.0227	0.0047	-0.0015	0.0015			-0.0085		
GRE Quantitative Percentile	0.1027	0.0174	-0.0273	0.0179	0.0125		-0.2706	-0.2243	-0.0070
GRE Analytic Percentile	0.0056	0.0035							
Undergraduate GPA	0.9812	0.7135				-0.1316			0.2561
TOEFL/IELTS	0.3055		-0.2943	0.0300		0.0934	-0.6903	-0.5850	-2.2692
Elite U.S. University	0.4024								
Elite U.S. Liberal Arts College		0.1112							
Elite Foreign University		-0.4025							
Other Foreign University	0.0790	0.0940	0.1998						-0.2265
Missing Undergraduate Institution			-0.0036						
Graduate Degree	0.1178								
Work Experience	0.6006	0.4068	0.0404	0.0961	0.0470	0.1976			-2.0809
#Professor Recommenders									0.0922
#Prolific Recommenders	0.1534	0.0301	-0.0870	0.0526	0.0363		-0.0789	-0.0358	
Missing GRE	-0.6534			-0.0697					
Missing Undergraduate GPA	-0.2348	-0.1148							
Missing TOEFL/IELTS	0.1447			-0.1243	-0.1295		1.8402	2.2548	
Committee Score		3.4243			0.0353			-0.4899	
#Math Courses	-0.0553	-0.0354							
Advanced Math	0.4302	0.7352							
Letters Length/100	0.1196	0.0587							
Letters Terms (%) PC1	0.0746	0.1485	0.1258			0.0102			
Letters Terms (%) PC2	-0.0162	-0.2601							
CV Length/100	0.0219	0.0298		0.0014			-0.0286	-0.1179	
CV Coding	-0.0038	-0.1754		0.0165	0.0076		-0.2540	-0.4154	-0.0998
CV Publication/Working Paper	0.5563	0.6436		0.0892			-0.0687		
Essay Length/100	0.1202	0.0295							
Essay Specific Professor	0.3403	0.3958	0.0674						
Essay Economics Terms (%)	-0.0016			-0.0405	-0.0325	-0.0104	0.4473	0.6570	0.2117
Essay Research Terms (%)		-0.0278						-0.1506	
Essay Positive Terms (%)	-0.0049	-0.0182		0.0192	0.0140	0.0267			
Essay Negative Terms (%)		-0.0417							
Missing Committee Score		-0.8416			-0.2615	-0.0714		4.0146	0.7971
Missing Application Pdf	-1.0365								
Open House			1.0982						
Program Rank						-0.0082			0.1385
Missing Program Rank						-2.2557			19.4273
Sample	All	All	Accept = 1	All	All	All	All	All	All
Observations	2329	2329	257	2329	2329	2329	2329	2329	2329
Pseudo-R^2	0.3841	0.8007	0.1281	0.0196	0.0213	0.2265	0.0206	0.0278	0.2693
CV Pseudo-R^2	0.2441	0.7605	0.0596	0.0082	0.0129	0.2199	0.0121	0.0183	0.2638

- **Accept:** 2017 (-), 2018 (-), 2019 (-), Female (+), International Asian (-), GRE Verbal Percentile (+), GRE Quantitative Percentile (+), GRE Analytic Percentile (+), Undergraduate GPA (+), Elite U.S. Liberal Arts College (+), Elite Foreign University (-), Other Foreign University (+), Work Experience (+), # Prolific Recommenders (+), Missing Undergraduate GPA (-), Committee Score (+), #Math Courses (-), Advanced Math (+), Letters Length/100 (+), Letters Terms (%) PC1 (+), Letters Terms (%) PC2 (-), CV Length/100 (+), CV Coding (-), CV Publication/Working Paper (+), Essay Length/100 (+), Essay Specific Professor (+), Essay Research Terms (%) (-), Essay Positive Terms (%) (-), Essay Negative Terms (%) (-), Missing Committee Score (-)
- **Matriculate:** 2018 (+), International Asian (+), GRE Verbal Percentile (-), GRE Quantitative Percentile (-), TOEFL/IELTS (-), Other Foreign University (+), Missing Undergraduate Institution (-), Work Experience (+), #Prolific Recommenders (-), Letters Terms (%) PC1 (+), Essay Specific Professor (+), Open House (+)
- **T100 Placement:** Age (+), Undergraduate GPA (-), TOEFL/IELTS (+), Work Experience (+), Letters Terms (%) PC1 (+), Essay Economics Terms (%) (-), Essay Positive Terms (%) (+), Missing Committee Score (-), Program Rank (-), Missing Program Rank (-)

Regression Results

A3: Dynamic Estimates

Unrestricted

	Committee Score	Transition		Average		Scarcity	
	Coefficients	Coefficients	AME	Coefficients	AME	Coefficients	AME
Intercept	-2.7856	-11.0041***	-1.4696***	-16.5370***	-1.2876***	-0.8447**	-0.5635**
	(2.1741)	(1.5491)	(0.2121)	(4.1661)	(0.2913)	(0.3951)	(0.2702)
2016	0.0615	-0.2899	-0.0438	-0.5294	-0.0237	0.0767	0.0299
	(0.1894)	(0.1764)	(0.0266)	(0.5264)	(0.0216)	(0.1653)	(0.0218)
2017	0.2354	-0.1552	-0.0233	-1.5364**	-0.0998**	0.1266	0.0340
	(0.1894)	(0.1764)	(0.0266)	(0.5264)	(0.0216)	(0.1653)	(0.0218)
2018	0.3898***	-1.4420***	-0.2116***	-7.6814***	-0.4498***	0.2009	0.0512
	(0.1777)	(0.1868)	(0.0291)	(0.4291)	(0.0205)	(0.2011)	(0.0256)
2019	0.4118***	0.2905	0.0437	1.0769***	0.1137***	0.3514	0.0816
	(0.1867)	(0.1855)	(0.0275)	(0.4699)	(0.0257)	(0.1975)	(0.0250)
Age	-0.0009	-0.0181	-0.0027	-0.0417	-0.0017	-0.0009	-0.0002
	(0.0191)	(0.0181)	(0.0062)	(0.0612)	(0.0037)	(0.0193)	(0.0026)
Female	-0.0562	0.3918***	0.0589***	0.8939***	0.0398***	-0.0422	-0.0263
	(0.1178)	(0.1195)	(0.0184)	(0.3672)	(0.0216)	(0.1236)	(0.0158)
U.S.	-0.9257**	-0.1218	-0.0189	0.0291	0.0011	0.1097	0.0282
	(0.4211)	(0.3966)	(0.0606)	(1.1264)	(0.0662)	(0.4256)	(0.0550)
International Asian	-0.7533***	0.2887	0.0414	1.1404***	0.0747***	-0.2501	-0.0316
	(0.1952)	(0.1996)	(0.0310)	(0.4183)	(0.0172)	(0.1940)	(0.0239)
GRE Verbal Percentile	-0.0001	-0.0004	-0.0001	-0.0008	-0.0000	0.0000	0.0000
	(0.0056)	(0.0050)	(0.0008)	(0.0131)	(0.0006)	(0.0054)	(0.0007)
GRE Quantitative Percentile	0.0027***	0.0029***	0.0275***	0.0032	0.0016	0.0029	0.0004
	(0.0011)	(0.0041)	(0.0006)	(0.0036)	(0.0015)	(0.0060)	(0.0010)
GRE Analytic Percentile	0.0026	0.0052***	0.0003***	0.0078	0.0004	-0.0003	-0.0003
	(0.0020)	(0.0020)	(0.0003)	(0.0048)	(0.0002)	(0.0020)	(0.0002)
TOEFL IELTS	0.0027***	0.0000	-0.0003	-0.1286	-0.0068	0.0257	0.0037
	(0.3096)	(0.1602)	(0.0196)	(0.9631)	(0.0250)	(0.1752)	(0.0231)
Elite U.S. University	0.3037***	0.9894***	0.1387***	-0.1617	-0.0208	-0.0961	-0.0119
	(0.1994)	(0.2165)	(0.0312)	(0.4618)	(0.0291)	(0.2219)	(0.0291)
Elite U.S. Liberal Arts College	0.1911	0.3151	0.0453	0.4630	0.0243	-0.0000	-0.0000
	(0.1661)	(0.1661)	(0.0162)	(0.4014)	(0.0264)	(0.2022)	(0.0269)
Elite Foreign University	0.2662	0.0618	0.0078	0.9137	0.0458	0.0958	0.0121
	(0.2991)	(0.2965)	(0.0268)	(0.5824)	(0.0219)	(0.3019)	(0.0370)
Other Foreign University	0.4797**	0.1161	0.0174	0.2299	0.0103	0.1627	0.0209
	(0.2178)	(0.2178)	(0.0329)	(0.5263)	(0.0243)	(0.2243)	(0.0289)
Missing Undergraduate Institution	0.3097**	-0.4107**	-0.0610**	0.0506	0.0023	0.1768	0.0224
	(0.1557)	(0.1996)	(0.0308)	(0.4668)	(0.0299)	(0.2221)	(0.0296)
Graduate Degree	-0.1757	1.0237***	0.1507***	-0.1619	-0.0183	-0.0213	-0.0027
	(0.1279)	(0.1262)	(0.0177)	(0.3993)	(0.0171)	(0.1397)	(0.0172)
Work Experience	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.1585)	(0.1296)	(0.0178)	(0.3012)	(0.0154)	(0.1419)	(0.0179)
pProfessor Recommendation	0.3618	-0.0231	-0.0032	0.1896	-0.0085	-0.0822	-0.0079
	(0.1966)	(0.1966)	(0.0306)	(0.4618)	(0.0291)	(0.2219)	(0.0291)
pPhotNo Recommendations	0.3667***	-0.1329***	-0.0186***	0.1063	0.0061	0.0006	0.0000
	(0.0648)	(0.0563)	(0.0097)	(0.1797)	(0.0077)	(0.0662)	(0.0073)
Missing GRE	1.4087*	1.8143***	0.2723***	1.4616	0.0790	0.0247	0.0031
	(0.9622)	(0.5268)	(0.0772)	(1.8092)	(0.0911)	(0.3772)	(0.0477)
Missing TOEFL IELTS	0.1097**	0.0967	0.0162	0.2000	0.0107	0.0000	0.0000
	(0.1519)	(0.1295)	(0.0194)	(0.4218)	(0.0238)	(0.1635)	(0.0211)
Committee Score				0.7599***	0.0407***	-0.0094	-0.0091
				(0.3944)	(0.0186)	(0.0982)	(0.0124)
pMath Courses	0.0780			0.2894	0.0112	0.0287	0.0063
	(0.1599)			(0.3944)	(0.0186)	(0.0982)	(0.0124)
Advanced Math	0.2679			1.1001	0.0482	-0.2230	-0.0301
	(0.1899)			(0.7916)	(0.0347)	(0.2926)	(0.0370)
Letters Length/ 100	0.3858***			0.2078	0.0034	-0.0105	-0.0017
	(0.0245)			(0.1003)	(0.0045)	(0.0381)	(0.0049)
Letters Tems (%) PC1	0.0141			0.0000	0.0000	0.0000	0.0000
	(0.3941)			(0.2689)	(0.0112)	(0.0951)	(0.0123)
Letters Tems (%) PC2	0.0208			0.3767	0.0169	-0.1364	-0.0172
	(0.0833)			(0.3055)	(0.0153)	(0.1357)	(0.0200)
CV Length/ 100	0.0513			0.0991	0.0057	0.0197	0.0068
	(0.0245)			(0.1003)	(0.0045)	(0.0381)	(0.0049)
CV Coding	0.1591			-0.1238**	-0.0187	0.1324	0.0320
	(0.0624)			(0.1051)	(0.0966)	(0.0951)	(0.0082)
CV Publication/Working Paper	0.5650***			0.7911	0.0161	0.1683	0.0182
	(0.1912)			(0.8128)	(0.0364)	(0.3293)	(0.0425)
Essay Length/ 100	0.0997			0.0000	0.0000	0.0000	0.0000
	(0.0997)			(0.3081)	(0.0137)	(0.0910)	(0.0202)
Essay Specific Professor	0.1411			0.4135	0.0185	0.2894	0.0368
	(0.1911)			(0.8128)	(0.0364)	(0.3293)	(0.0425)
Essay Economics Terms (%)	-0.2297			-0.0134	0.0007	-0.0864	-0.0102
	(0.3312)			(0.9014)	(0.0356)	(0.3643)	(0.0477)
Essay Research Terms (%)	0.0096			-0.0761	-0.0056	0.0146	0.0018
	(0.3626)			(0.1652)	(0.0067)	(0.0672)	(0.0085)
Essay Positive Terms (%)	0.1911			(0.3081)	(0.0137)	(0.0910)	(0.0202)
	(0.3626)			(0.2964)	(0.0161)	(0.2539)	(0.0368)
Essay Negative Terms (%)	0.0681			0.0863	0.0039	0.0461	-0.0056
	(0.3643)			(0.1968)	(0.0361)	(0.0793)	(0.0094)
Missing Committee Score						-0.1178	-0.0178
						(0.1178)	(0.0178)
Missing Application Pdf	-0.5375***			0.2614	0.0117	0.2699	0.0316
	(0.1641)			(0.7055)	(0.0311)	(0.2756)	(0.0338)
Program Rank						-0.0006***	-0.0001***
						(0.0021)	(0.0001)
Missing Program Rank						-0.0000	-0.0000
						(0.4376)	(0.0541)
$\hat{\rho}$	1.6667***						
	(0.1976)						
Obs / Param / LL / AIC	2129	562	-1013.6	2918.1			

Robust standard errors in parentheses. *p<0.10, **p<0.05, ***p<0.01.

Restricted

	Committee Score	Cutoff 1	#(Cutoff)	Cutoff 2	#(Cutoff)	Success	Sigma
Intercept	-0.4749**	-0.9660***	-0.3731	-0.5937***	-36.58%	-0.7183*	
	(0.1661)	(0.1092)	(0.3067, 0.4439)	(0.1941)	(29.61%, 43.92%)	(0.3791)	
2016	0.0555	0.0171	0.3796	0.0594	37.31%	0.1033	
	(0.1527)	(0.1286)	(0.3625, 0.4791)	(0.2016)	(29.28%, 45.99%)	(0.2169)	
2017	0.1266	0.0340	0.4291	0.0594	37.31%	0.1033	
	(0.1744)	(0.1085)	(0.3561, 0.4776)	(0.1784)	(34.02%, 46.10%)	(0.1791)	
2018	0.5029***	0.0237	0.3834	0.0672	37.68%	-0.0864	
	(0.1777)	(0.1868)	(0.3625, 0.4897)	(0.1941)	(31.72%, 43.97%)	(0.1791)	
2019	0.4276**	0.4161	0.3997	0.0947	38.18%	0.3232	
	(0.1961)	(0.1142)	(0.3268, 0.4426)	(0.1852)	(33.89%, 43.81%)	(0.1661)	
Age	-0.0009					(0.0009)	
	(0.0206)					(0.0009)	
Female	-0.0326					0.0114	
	(0.1281)					(0.0009)	
U.S.	-0.3132					0.0255	
	(0.4581)					(0.0009)	
International Asian	-0.3029**					-0.0851	
	(0.1577)					(0.0070)	
GRE Verbal Percentile	0.0000					0.0000	
	(0.0061)					(0.0009)	
GRE Quantitative Percentile	0.0069***					0.0007	
	(0.0037)					(0.0009)	
GRE Analytic Percentile	0.0026					0.0001	
	(0.0021)					(0.0009)	
TOEFL IELTS	0.1047					0.0007	
	(0.1703)					(0.0073)	
Elite U.S. University	0.1000					0.0000	
	(0.1775)					(0.0009)	
Elite U.S. Liberal Arts College	0.0000					0.0007	
	(0.1233)					(0.0051)	
Elite Foreign University	0.2109					-0.0086	
	(0.2224)					(0.0009)	
Other Foreign University	0.0749**					0.0004	
	(0.1502)					(0.0112)	
Missing Undergraduate Institution	0.1000					0.0000	
	(0.2282)					(0.0009)	
Graduate Degree	0.0001					-0.0022	
	(0.1000)					(0.0009)	
Work Experience	0.1000					0.0000	
	(0.1432)					(0.0079)	
pProfessor Recommendation	0.1035					-0.0005	
	(0.0764)					(0.0057)	
pPhotNo Recommendations	0.0000					0.0000	
	(0.0099)					(0.0035)	
Missing GRE	1.2274					0.0154	
	(0.3000)					(0.0009)	
Missing TOEFL IELTS	0.0000					0.0006	
	(0.1292)					(0.0041)	
Committee Score						0.0013	
						(0.0095)	
pMath Courses	0.1040					0.0000	
	(0.1497)					(0.0047)	
Advanced Math	0.4000					-0.0022	
	(0.1000)					(0.0009)	
Letters Length/ 100	0.3936**					0.0005	
	(0.0436)					(0.0033)	
Letters Tems (%) PC1	0.2137*					0.0013	
	(0.0962)					(0.0056)	
Letters Tems (%) PC2	0.1000					0.0000	
	(0.1961)					(0.0047)	
CV Length/ 100	0.0181					0.0013	
	(0.0436)					(0.0033)	
CV Coding	0.0356					-0.0044	
	(0.0981)					(0.0040)	
CV Publication/Working Paper	1.0000**					0.0009	
	(0.4362)					(0.0133)	
Essay Length/ 100	0.0000					0.0000	
	(0.1564)					(0.0095)	
Essay Specific Professor	0.1231					-0.0060	
	(0.1564)					(0.0095)	
Essay Economics Terms (%)	0.3966					0.0004	
	(0.0592)					(0.0041)	
Essay Research Terms (%)	0.0000					0.0000	
	(0.1564)					(0.0095)	
Essay Positive Terms (%)	0.0422					-0.0004	
	(0.0742)					(0.0074)	
Essay Negative Terms (%)	0.1000					0.0008	
	(0.0766)					(0.0041)	
Missing Committee Score						-0.0003	
						(0.1292)	
Missing Application Pdf	-0.7040**					0.0000	
	(0.2764)					(0.0095)	
Program Rank						-0.0000***	
						(0.0003)	
Missing Program Rank						0.0777***	
						(0.4841)	
$\hat{\rho}$	1.7333***						
	(0.0475)						
Sigma							0.0000
							(0.0001)
Obs / Param / LL / AIC	2129	529	-1013.6	2918.9			

Robust standard errors in parentheses. *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Restricted Dynamic Estimates

A3: Restricted Static Estimates

	Round 1 Cutoff	Round 2 Cutoff	Sigma
2015	29.36% (23.20%, 36.38%)	39.20% (32.75%, 46.05%)	
⋮	⋮	⋮	
2019	34.04% (28.67%, 39.85%)	39.62% (32.54%, 47.16%)	
Sigma			0.0064* (0.0035)

- Static model has interpretable Round 1 cutoff probabilities
- Round 2 cutoff probabilities always greater than those in Round 1

Restricted Dynamic Estimates

A3: Static Estimates

Unrestricted

	Transition		Accept		Scores 1		Scores 2	
	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-21.0041***	-1.4454***	-21.5773***	-1.1942***	-4.4547***	-0.5996***	-4.6315***	-0.5423***
	(2.6491)	(0.2471)	(4.9091)	(0.2932)	(1.4190)	(0.2054)	(1.1831)	(0.1752)
2016	0.1792	0.0123	0.2744	0.0152	0.0599	0.0081	0.0599	0.0070
	(0.1756)	(0.0164)	(0.5244)	(0.0316)	(0.1001)	(0.0137)	(0.1063)	(0.0128)
2017	-0.1533	-0.0155	-1.3384**	-0.0938**	0.1186	0.0169	0.1266	0.0160
	(0.1726)	(0.0160)	(0.5312)	(0.0336)	(0.1077)	(0.0160)	(0.1076)	(0.0154)
2018	1.5429***	0.2018***	-1.6804***	-0.1407***	0.0122	0.0014	0.0095	0.0012
	(0.1825)	(0.0247)	(0.5284)	(0.0349)	(0.1102)	(0.0151)	(0.1102)	(0.0142)
2019	0.2806	0.0403	-1.0294**	-0.1177***	0.0177	0.0023	0.0214	0.0016
	(0.1855)	(0.0271)	(0.4699)	(0.0291)	(0.1061)	(0.0160)	(0.1074)	(0.0156)
Age	-0.0000	-0.0000	0.0000	0.0000	-0.0000	-0.0000	-0.0000	-0.0000
	(0.0183)	(0.0012)	(0.0612)	(0.0037)	(0.0129)	(0.0019)	(0.0129)	(0.0018)
Female	0.3919***	0.0589***	0.8899**	0.0599***	-0.0162	-0.0044	-0.0422	-0.0055
	(0.1190)	(0.0164)	(0.3672)	(0.0156)	(0.1720)	(0.0358)	(0.1599)	(0.0218)
U.S.	-0.1124	-0.0169	0.0741	0.0053	0.3436	0.0269	0.3197	0.0232
	(0.1196)	(0.0164)	(0.3672)	(0.0156)	(0.1720)	(0.0358)	(0.1599)	(0.0218)
International Asian	0.0007	0.0001	-1.1997***	-0.0914***	-0.0391	-0.0114	-0.0521	-0.0134
	(0.1196)	(0.0210)	(0.4133)	(0.0172)	(0.1489)	(0.0301)	(0.1502)	(0.0199)
GRE Verbal Percentile	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.0035)	(0.0003)	(0.0131)	(0.0006)	(0.0034)	(0.0004)	(0.0034)	(0.0003)
GRE Quantitative Percentile	0.0029***	0.0276***	0.0662	0.0016	0.0006	0.0008	0.0004	0.0006
	(0.0035)	(0.0003)	(0.0131)	(0.0006)	(0.0034)	(0.0004)	(0.0034)	(0.0003)
GRE Analytic Percentile	0.0029***	0.0277***	0.0670	0.0016	0.0001	0.0001	0.0002	0.0000
	(0.0035)	(0.0004)	(0.0090)	(0.0004)	(0.0020)	(0.0004)	(0.0020)	(0.0004)
TOEFL/IELTS	-0.0097	-0.0013	-0.1284	-0.0098	0.0167*	0.0023*	0.0170*	0.0021*
	(0.0422)	(0.0045)	(0.1621)	(0.0092)	(0.0751)	(0.0103)	(0.0752)	(0.0101)
Elite U.S. University	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.2102)	(0.0312)	(0.4436)	(0.0250)	(0.2180)	(0.0353)	(0.2193)	(0.0293)
Elite U.S. Liberal Arts College	0.3248	0.0481	0.3263	0.0186	-0.0022	-0.0004	-0.0066	-0.0010
	(0.2102)	(0.0312)	(0.4436)	(0.0250)	(0.2180)	(0.0353)	(0.2193)	(0.0293)
Elite Foreign University	0.3632	0.0559	0.6107	0.0366	0.0661	0.0117	0.0854	0.0125
	(0.2999)	(0.0483)	(0.9288)	(0.0424)	(0.5413)	(0.0427)	(0.5029)	(0.0419)
Other Foreign University	0.1031	0.0175	0.2264	0.0165	0.0503	0.0084	0.0437	0.0059
	(0.1948)	(0.0294)	(0.4698)	(0.0196)	(0.2166)	(0.0370)	(0.2048)	(0.0276)
Missing Undergraduate Institution	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.1996)	(0.0302)	(0.4668)	(0.0299)	(0.2182)	(0.0364)	(0.2201)	(0.0297)
Graduate Degree	1.0229***	0.1637***	0.1618	0.0151	-0.1727	-0.0176	-0.1648	-0.0215
	(0.1196)	(0.0164)	(0.3672)	(0.0156)	(0.1720)	(0.0358)	(0.1599)	(0.0218)
Work Experience	0.7870***	0.1187***	0.9518*	0.0749*	0.1418**	0.0185**	0.1358*	0.0167*
	(0.1196)	(0.0270)	(0.5612)	(0.0316)	(0.1361)	(0.0272)	(0.1424)	(0.0179)
JD/Professor Recommenders	-0.0221	-0.0032	-0.1896	-0.0165	-0.0670	-0.0071	-0.0624	-0.0079
	(0.0409)	(0.0124)	(0.2576)	(0.0113)	(0.0686)	(0.0100)	(0.0697)	(0.0109)
JD/Public Recommenders	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.0161)	(0.0017)	(0.1757)	(0.0077)	(0.0191)	(0.0072)	(0.0082)	(0.0073)
Missing GRE	-1.8142***	-0.2727***	1.4013	0.0793	-0.0087	-0.0011	0.0257	0.0033
	(0.1196)	(0.0164)	(0.3672)	(0.0156)	(0.1720)	(0.0358)	(0.1599)	(0.0218)
Missing TOEFL/IELTS	0.0947	0.0142	0.1811	0.0252	-0.0998	-0.0018	-0.0578	-0.0095
	(0.1196)	(0.0164)	(0.4236)	(0.0189)	(0.1419)	(0.0182)	(0.1405)	(0.0181)
Committee Score			0.7268***	0.1663***			-0.0036	-0.0001
			(0.3564)	(0.0391)			(0.0286)	(0.0062)
JMark Courses			0.0000	0.0000			0.0000	0.0000
			(0.1396)	(0.0158)			(0.0079)	(0.0121)
Advanced Math			0.1003	0.0160			-0.2132	-0.0291
			(0.1396)	(0.0158)			(0.0079)	(0.0121)
Letters Length/100			0.0770	0.0204			-0.0119	-0.0017
			(0.1696)	(0.0366)			(0.0189)	(0.0289)
Letters Terms (%) PC1			0.0000	0.0000			0.0000	0.0000
			(0.1696)	(0.0112)			(0.0012)	(0.0191)
Letters Terms (%) PC2			-0.1747	-0.0349			-0.1566	-0.0172
			(0.1696)	(0.0191)			(0.1297)	(0.0199)
CV Length/100			0.0091	0.0027			0.0347	0.0068
			(0.1696)	(0.0191)			(0.1297)	(0.0199)
CV Coding			-0.2238*	-0.0387*			0.0219	0.0133
			(0.1331)	(0.0366)			(0.0841)	(0.0382)
CV Publication/Working Paper			0.0000	0.0000			0.0000	0.0000
			(0.3126)	(0.0366)			(0.1006)	(0.0425)
Essay Length/100			0.2567	0.0392			-0.0716	-0.0091
			(0.1693)	(0.0197)			(0.0818)	(0.0325)
Essay Specific Professor			0.0126	0.0195			0.3056	0.0366
			(0.1696)	(0.0191)			(0.1297)	(0.0199)
Essay Economics Terms (%)			0.0156	0.0307			-0.0359	-0.0158
			(0.3914)	(0.0356)			(0.0414)	(0.0077)
Essay Research Terms (%)			0.0000	0.0000			0.0000	0.0000
			(0.1693)	(0.0367)			(0.0672)	(0.0085)
Essay Positive Terms (%)			0.0649	0.0151			0.0977*	0.0126*
			(0.0966)	(0.0350)			(0.0539)	(0.0066)
Essay Negative Terms (%)			-0.0342	-0.0039			-0.0421	-0.0056
			(0.1346)	(0.0361)			(0.0539)	(0.0066)
Missing Committee Score							-0.2179	-0.0376
							(0.3116)	(0.0191)
Missing Application Pdf			0.2614	0.0117			0.0000	0.0000
			(0.7026)	(0.0131)			(0.1756)	(0.0168)
Program Rank					-0.0998***	-0.0151***	-0.0994***	-0.0171***
					(0.0025)	(0.0853)	(0.0031)	(0.0001)
Missing Program Rank					-0.1387***	-0.0210***	-0.1379***	-0.0237***
					(0.4021)	(0.0853)	(0.4078)	(0.0001)
Obs / Params / LL / BIC	2226	131	-2964.5	6166.0				

Robust standard errors in parentheses *p<0.10, **p<0.05, ***p<0.01

Restricted

	Cutoff 1	P(Cutoff)1	Cutoff 2	P(Cutoff)2	Success 1	Success 2	Sigma
Intercept	-0.8751***	(8.38%)	-0.1999***	(9.32%)	-1.3317***	-0.7052**	
	(0.1429)	(23.28%, 36.18%)	(0.1412)	(32.76%, 49.06%)	(0.1484)	(0.4442)	
2016	0.0000	(8.38%)	0.0000	(9.32%)	0.0000	0.0000	
	(0.1984)	(22.21%, 37.17%)	(0.1767)	(31.96%, 47.56%)	(0.1977)	(0.1771)	
2017	0.3832*	(7.59%)	0.3235	(41.66%,	0.3167*	0.0469	
	(0.1963)	(32.18%, 42.24%)	(0.1727)	(34.93%, 48.96%)	(0.1956)	(0.1705)	
2018	0.8536**	(8.12%)	0.8042	(9.36%,	0.4288**	-0.0912	
	(0.2078)	(33.96%, 44.64%)	(0.1829)	(32.76%, 47.56%)	(0.2068)	(0.1772)	
2019	0.2163	(41.04%,	0.3179	(9.42%,	0.2035	-0.0617	
	(0.2017)	(28.67%, 39.85%)	(0.1791)	(32.56%, 47.34%)	(0.2016)	(0.1775)	
Age					0.0000	0.0000	
					(0.0013)	(0.0026)	
Female					0.0116	0.0239	
					(0.0077)	(0.0146)	
U.S.					-0.0034	0.0032	
					(0.0077)	(0.0146)	
International Asian					-0.0056	-0.0162	
					(0.0081)	(0.0187)	
GRE Verbal Percentile					0.0010*	0.0001	
					(0.0005)	(0.0002)	
GRE Quantitative Percentile					0.0000	0.0000	
					(0.0003)	(0.0001)	
GRE Analytic Percentile					0.0009*	0.0002	
					(0.0004)	(0.0002)	
TOEFL/IELTS					-0.0008	-0.0016	
					(0.0087)	(0.0146)	
Elite U.S. University					0.0292	0.0013	
					(0.0191)	(0.0178)	
Elite U.S. Liberal Arts College					0.0000	0.0000	
					(0.0212)	(0.0216)	
Elite Foreign University					0.0013	-0.0239	
					(0.0289)	(0.0272)	
Other Foreign University					0.0030	0.0073	
					(0.0212)	(0.0216)	
Missing Undergraduate Institution					-0.0119	0.0000	
					(0.0366)	(0.0172)	
Graduate Degree					0.0000	0.0000	
					(0.0175)	(0.0172)	
Work Experience					0.0239*	0.0148	
					(0.0141)	(0.0130)	
JD/Professor Recommenders					-0.0084	-0.0097	
					(0.0062)	(0.0059)	
JD/Public Recommenders					0.0039	0.0042	
					(0.0026)	(0.0050)	
Missing GRE					0.0000	0.0000	
					(0.0361)	(0.0410)	
Missing TOEFL/IELTS					0.0229	0.0133	
					(0.0041)	(0.0077)	
Committee Score						0.0941*	
						(0.0054)	
JMark Courses						0.0000	
						(0.0048)	
Advanced Math						0.0297	
						(0.0048)	
Letters Length/100						0.0000	
						(0.0029)	
Letters Terms (%) PC1						0.0063	
						(0.0029)	
Letters Terms (%) PC2						0.0000	
						(0.0156)	
CV Length/100						0.0010	
						(0.0037)	
CV Coding						0.0000	
						(0.0041)	
CV Publication/Working Paper						0.0297	
						(0.0056)	
Essay Length/100						0.0000	
						(0.0029)	
Essay Specific Professor						0.0156	
						(0.0156)	
Essay Economics Terms (%)						0.0000	
						(0.0021)	
Essay Research Terms (%)						-0.0019	
						(0.0035)	
Essay Positive Terms (%)						-0.0014	
						(0.0035)	
Essay Negative Terms (%)						-0.0023	
						(0.0036)	
Missing Committee Score						0.0000	
						(0.1347)	
Missing Application Pdf						0.0079	
						(0.0145)	
Program Rank					-0.0995***	-0.0075***	
					(0.0031)	(0.0001)	
Missing Program Rank					-0.8239***	-0.1171***	
					(0.4085)	(0.4041)	
Sigma							0.0094*
							(0.0005)
Obs / Params / LL / BIC	2226	60	-2992.0	6168.1			

Robust standard errors in parentheses *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Restricted Dynamic Estimates

A3: Myopic Estimates

Unrestricted

	Committee Score Coefficients	Transition Coefficients	Transition AME	Accept Coefficients	Accept AME	Scalarize Coefficients	Scalarize AME
Intercept	-8.6650***	-11.0412***	-1.4006***	-16.5237***	-1.2876***	-0.8167**	-0.5635**
	(0.9962)	(1.5491)	(0.1212)	(4.581)	(0.3913)	(2.1651)	(0.2762)
2016	0.1751	-0.2899	-0.0438	-0.5294	-0.0237	0.3767	0.0299
	(0.1966)	(0.1794)	(0.0266)	(0.5294)	(0.0233)	(0.1835)	(0.0218)
2017	0.4393**	-0.1552	-0.0233	-1.7394**	-0.0998**	0.1266	0.0160
	(0.1966)	(0.1794)	(0.0266)	(0.5294)	(0.0233)	(0.1835)	(0.0218)
2018	0.9018***	-1.5420***	-0.2319***	-7.6814***	-0.4498***	0.2009	0.0212
	(0.1873)	(0.1856)	(0.0291)	(0.4291)	(0.0205)	(0.2011)	(0.0254)
2019	0.7997***	0.2905	0.0437	-3.0597***	-0.1377***	0.3114	0.0214
	(0.1987)	(0.1855)	(0.0275)	(0.8099)	(0.0257)	(0.1975)	(0.0250)
Age		0.0181	0.0027	0.0611	0.0027	0.0116	0.0009
		(0.0218)	(0.0032)	(0.0612)	(0.0027)	(0.0218)	(0.0016)
Female		0.3918***	0.0589***	0.8997**	0.0399**	-0.0422	-0.0263
		(0.1195)	(0.0184)	(0.3671)	(0.0164)	(0.1256)	(0.0159)
U.S.		-0.1116	-0.0168	0.0241	0.0011	0.1097	0.0282
		(0.1195)	(0.0184)	(0.3671)	(0.0164)	(0.1256)	(0.0159)
International Asian		0.0387	0.0058	1.1404***	0.0506***	-0.0501	-0.0216
		(0.1396)	(0.0218)	(0.4133)	(0.0172)	(0.1545)	(0.0195)
GRE Verbal Percentile		-0.0107**	-0.0016**	-0.0222**	-0.0010**	0.0016	0.0004
		(0.0054)	(0.0008)	(0.0111)	(0.0004)	(0.0054)	(0.0005)
GRE Quantitative Percentile		0.0297***	0.0045***	0.0402**	0.0018**	0.0024	0.0004
		(0.0042)	(0.0016)	(0.0156)	(0.0007)	(0.0040)	(0.0003)
GRE Analytic Percentile		0.0152***	0.0023***	0.0078	0.0004	-0.0022	-0.0003
		(0.0042)	(0.0006)	(0.0156)	(0.0007)	(0.0040)	(0.0003)
TOEFL/IELTS		-0.0060	-0.0010	-0.1286	-0.0058	0.1057*	0.0107*
		(0.1622)	(0.0196)	(0.9621)	(0.0290)	(0.1752)	(0.0231)
Elite U.S. University		0.9994***	0.1567***	-0.1017	-0.0238	-0.0561	-0.0119
		(0.2335)	(0.0312)	(0.6618)	(0.0291)	(0.2319)	(0.0291)
Elite U.S. Liberal Arts College		0.1911	0.0274	0.2193	0.0103	0.1627	0.0359
		(0.1601)	(0.0162)	(0.4012)	(0.0184)	(0.1622)	(0.0189)
Elite Foreign University		0.0918	0.0078	0.9137	0.0408	0.0958	0.0121
		(0.2965)	(0.0268)	(0.9268)	(0.0424)	(0.3019)	(0.0670)
Other Foreign University		0.1161	0.0174	0.2293	0.0103	0.1627	0.0359
		(0.1396)	(0.0218)	(0.4133)	(0.0172)	(0.1545)	(0.0195)
Missing Undergraduate Institution		-0.4103**	-0.0643**	0.0936	0.0023	0.1768	0.0224
		(0.1996)	(0.0188)	(0.6668)	(0.0299)	(0.2221)	(0.0256)
Graduate Degree		1.0237***	0.1607***	-0.1819	-0.0353	-0.1883	-0.0211
		(0.1262)	(0.0177)	(0.3903)	(0.0171)	(0.1397)	(0.0172)
Work Experience		-0.0139	-0.0022	0.0178	0.0008	0.0124	0.0028
		(0.1396)	(0.0176)	(0.3012)	(0.0154)	(0.1423)	(0.0179)
pFirstmax Recommendations		-0.0211	-0.0032	-0.1896	-0.0085	-0.0822	-0.0079
		(0.1396)	(0.0176)	(0.3012)	(0.0154)	(0.1423)	(0.0179)
pPrintMin Recommendations		-0.1125***	-0.0166***	0.2483	0.0091	0.0847	0.0094
		(0.0563)	(0.0057)	(0.1757)	(0.0077)	(0.0562)	(0.0073)
Missing GRE		1.8143***	0.2723***	1.4616	0.0740	0.0247	0.0021
		(0.5268)	(0.0772)	(1.8062)	(0.0811)	(0.3772)	(0.0477)
Missing TOEFL/IELTS		0.0967	0.0142	0.2293	0.0103	0.1627	0.0359
		(0.1396)	(0.0194)	(0.4133)	(0.0189)	(0.1435)	(0.0181)
Committee Score				0.7004***	0.0407***	-0.0094	-0.0091
				(0.1994)	(0.0086)	(0.0086)	(0.0082)
pMath Courses	0.0834			0.2494	0.0112	0.0287	0.0053
	(0.0242)			(0.1003)	(0.0045)	(0.0241)	(0.0029)
Advanced Math	0.2068*			1.1001	0.0492	-0.2250	-0.0391
	(0.1699)			(0.7916)	(0.0347)	(0.2926)	(0.0370)
Letters Length/ 100	0.1107***			0.0276	0.0034	0.0105	0.0017
	(0.0242)			(0.1003)	(0.0045)	(0.0391)	(0.0049)
Letters Tems (%) PC1	0.0116			0.2689	0.0122	0.0113	0.0023
	(0.0616)			(0.2689)	(0.0122)	(0.0113)	(0.0023)
Letters Tems (%) PC2	0.0826			0.3767	0.0169	0.1364	0.0172
	(0.0242)			(0.1003)	(0.0152)	(0.1357)	(0.0169)
CV Length/ 100	0.0587			0.0391	0.0017	0.0167	0.0068
	(0.0242)			(0.1003)	(0.0045)	(0.0241)	(0.0029)
CV Coding	0.3951			-0.1728**	-0.0187	0.1324	0.0159
	(0.3617)			(0.1051)	(0.0966)	(0.0951)	(0.0082)
CV Publication/Working Paper	0.5107***			0.7911	0.0351	0.1665	0.0182
	(0.1975)			(0.8128)	(0.0364)	(0.1293)	(0.0425)
Essay Length/ 100	0.0167			0.0081	0.0007	0.0010	0.0001
	(0.0947)			(0.3983)	(0.0197)	(0.0953)	(0.0202)
Essay Specific Professor	0.1442			0.4135	0.0185	0.2994	0.0368
	(0.0242)			(0.1003)	(0.0045)	(0.0241)	(0.0029)
Essay Economics Terms (%)	0.0512			0.0134	0.0007	-0.0064	-0.0006
	(0.0314)			(0.0814)	(0.0036)	(0.0644)	(0.0077)
Essay Research Terms (%)	0.0304			-0.0761	-0.0034	0.0146	0.0016
	(0.0646)			(0.1692)	(0.0067)	(0.0672)	(0.0089)
Essay Positive Terms (%)	0.0112			0.0134	0.0007	-0.0064	-0.0006
	(0.0314)			(0.0964)	(0.0041)	(0.0509)	(0.0068)
Essay Negative Terms (%)	0.0628			0.0863	0.0039	0.0461	-0.0096
	(0.0960)			(0.1968)	(0.0084)	(0.0793)	(0.0094)
Missing Committee Score						-0.1178	-0.0376
						(0.1178)	(0.0147)
Missing Application Pdf	-0.6004**			0.2614	0.0117	0.2699	0.0316
	(0.1892)			(0.7055)	(0.0311)	(0.2756)	(0.0338)
Program Rank						-0.0006***	-0.0001***
						(0.0021)	(0.0001)
Missing Program Rank						-0.0006**	-0.0001**
						(0.0276)	(0.0041)
$\sqrt{\text{age}}$	1.7520***						
	(0.3853)						
Obs / Param / LL / AIC	2129	297	-1643.7	3919.4			

Robust standard errors in parentheses. *p<0.10, **p<0.05, ***p<0.01.

Restricted

	Committee Score	Cutoff 1	1/Cutoffs	Cutoff 2	2*Cutoff2	Income	Sigma
Intercept	5.2798***	-0.9372***	0.2594	0.9222***	99.58%	-1.2864**	
	(1.2648)	(0.0966)	(0.5116, 0.4644)	(0.1954)	(79.68%, 43.66%)	(0.5656)	
2016	0.3976	-0.0309	0.1762	-0.0546	99.47%	-0.516	
	(0.1986)	(0.1329)	(0.2992, 0.4711)	(0.2133)	(59.37%, 49.91%)	(0.3139)	
2017	0.5981***	-0.0146	0.1762	-0.0546	99.47%	-0.516	
	(0.2332)	(0.1299)	(0.3701, 0.4703)	(0.1764)	(59.05%, 49.95%)	(0.1771)	
2018	1.0719***	0.0146	0.3832	0.0296	97.39%	-0.1309	
	(0.2332)	(0.1299)	(0.3701, 0.4703)	(0.1764)	(59.05%, 49.95%)	(0.1771)	
2019	0.8995***	0.0035	0.3784	0.0004	96.94%	-0.3494	
	(0.1982)	(0.1256)	(0.3086, 0.4994)	(0.2002)	(79.43%, 44.58%)	(0.2645)	
Age						0.0003	
						(0.0087)	
Female						0.0552	
						(0.0362)	
U.S.						0.0053	
						(0.0362)	
International Asian						-0.0075	
						(0.0501)	
GRE Verbal Percentile						0.0007	
						(0.0007)	
GRE Quantitative Percentile						0.0039**	
						(0.0012)	
GRE Analytic Percentile						0.0004**	
						(0.0012)	
TOEFL/IELTS						-0.0069	
						(0.0502)	
Elite U.S. University						0.0072	
						(0.0371)	
Elite U.S. Liberal Arts College						0.0084	
						(0.0358)	
Elite Foreign University						-0.0022	
						(0.0358)	
Other Foreign University						0.0084	
						(0.0358)	
Missing Undergraduate Institution						0.0001	
						(0.0501)	
Graduate Degree						0.0156**	
						(0.0358)	
Work Experience						0.0148**	
						(0.0358)	
pFirstmax Recommendations						-0.0014	
						(0.0202)	
pPrintMin Recommendations						-0.0014	
						(0.0358)	
Missing GRE						-0.0449	
						(0.0358)	
Missing TOEFL/IELTS						0.0084	
						(0.0358)	
Committee Score						0.0063**	
						(0.0282)	
pMath Courses	0.0637					0.0001	
	(0.0885)					(0.0371)	
Advanced Math	0.1298					0.0023	
	(0.0885)					(0.0371)	
Letters Length/ 100	0.1107***					0.0021	
	(0.0317)					(0.0029)	
Letters Tems (%) PC1	0.1135					0.0024	
	(0.0772)					(0.0087)	
Letters Tems (%) PC2	0.0261					0.0021	
	(0.1152)					(0.0189)	
CV Length/100	0.0628					0.0022	
	(0.0317)					(0.0029)	
CV Coding	0.1866					-0.0091	
	(0.0982)					(0.0039)	
CV Publication/Working Paper	0.1665					0.0182	
	(0.1965)					(0.0169)	
Essay Length/ 100	0.1186					0.0083	
	(0.1886)					(0.0121)	
Essay Specific Professor	0.1367					0.0021	
	(0.0317)					(0.0029)	
Essay Economics Terms (%)	-0.0024					-0.0019	
	(0.0396)					(0.0021)	
Essay Research Terms (%)	0.0146					0.0016	
	(0.0396)					(0.0046)	
Essay Positive Terms (%)	0.0401					0.0011	
	(0.0476)					(0.0002)	
Essay Negative Terms (%)	0.0249					-0.0089	
	(0.0476)					(0.0089)	
Missing Committee Score						-0.1177	
						(0.1482)	
Missing Application Pdf	-0.6539***					0.0316	
	(0.2254)					(0.0197)	
Program Rank						-0.0039***	
						(0.0002)	
Missing Program Rank						-0.0039**	
						(0.0458)	
$\sqrt{\text{age}}$	1.3670***						
	(0.3436)						
Sigma						0.0027**	
						(0.0011)	
Obs / Param / LL / AIC	2129	246	-1638.3	3568.5			

Robust standard errors in parentheses. *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Restricted Dynamic Estimates

A3: Matriculation-Dynamic Estimates

Unrestricted

	Committee Score	Transition	Accept	Success	Matriculation
	Cutoff	Cutoff × ME	Cutoff × ME	Cutoff × ME	Cutoff × ME
Intercept	-0.7564	23.5842***	-0.6647**	23.5729***	-0.4953**	23.6547***	-0.6347**	23.5814***	-0.5373**
	(-2.1791)	(7.6491)	(-2.0271)	(9.9682)	(-2.2933)	(7.6511)	(-2.2752)	(9.9668)	(-2.5045)
2016	0.0015	-0.2286	-0.0225	-0.3296	-0.2237	0.0762	0.0095	0.3196	0.0507
	(0.1856)	(-0.1776)	(-0.0264)	(-0.5230)	(-0.4235)	(0.1463)	(0.0230)	(0.6120)	(0.1032)
2017	0.0004	-0.2286	-0.0225	-0.3296	-0.2237	0.0762	0.0095	0.3196	0.0507
	(0.0418)	(-0.1736)	(-0.0264)	(-0.5212)	(-0.4239)	(0.1459)	(0.0264)	(0.6116)	(0.1050)
2018	0.0998***	1.5429***	0.1519***	1.6382***	0.1649***	0.0895	0.0022	1.6854***	0.2610***
	(3.1107)	(3.1107)	(3.1107)	(3.1107)	(3.1107)	(1.4999)	(0.0264)	(3.1107)	(3.1107)
2019	0.0123***	0.2930	0.0427	2.0787***	0.2177***	0.0114	0.0004	1.9547	0.1727*
	(0.1987)	(1.1631)	(0.6271)	(3.4499)	(3.6197)	(0.1974)	(0.0292)	(3.6106)	(0.0971)
Age	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.8230)	(0.6218)	(0.8822)	(0.3612)	(0.4227)	(0.6218)	(0.8822)	(0.4736)	(0.6213)
Female	-0.0562	6.3918***	0.0589***	6.3989***	0.0594***	-0.0422	-0.0053	-0.0562	-0.0722
	(-0.4126)	(3.1139)	(0.6542)	(3.3672)	(3.6154)	(-0.1293)	(-0.0193)	(0.3056)	(-0.4056)
U.S.	-0.1627**	-0.1128	-0.0169	0.0266	0.0311	0.1597	0.0202	0.9160	0.1609
	(-2.1179)	(-0.1128)	(-0.0169)	(0.0266)	(0.0311)	(0.1597)	(0.0202)	(0.9160)	(0.1609)
International Asian	-0.7537***	0.0047	0.0011	-0.1854***	-0.0277	-0.0011	-0.0134	-1.1854***	-0.1564***
	(-3.1852)	(0.1196)	(0.0253)	(-3.4115)	(-3.6172)	(-0.3563)	(-0.5295)	(-3.8531)	(-3.8680)
GRE Verbal Percentile	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.8230)	(0.6218)	(0.8822)	(0.3612)	(0.4227)	(0.6218)	(0.8822)	(0.4736)	(0.6213)
GRE Quantitative Percentile	0.0007***	0.1629***	0.0275***	0.1662	0.0018	0.0020	0.0004	-0.1358***	-0.0233***
	(3.1852)	(3.1852)	(3.1852)	(3.1852)	(3.1852)	(3.1852)	(3.1852)	(3.1852)	(3.1852)
GRE Analytic Percentile	0.0004	0.0253***	0.0075	0.0078	0.0004	-0.0022	-0.0003	-0.0297**	-0.0041
	(0.0027)	(0.0026)	(0.0004)	(0.0096)	(0.0006)	(0.0029)	(0.0006)	(0.0097)	(0.0022)
TOEFL/IELTS	0.0007***	-0.0007	-0.0015	-0.1196	-0.0098	0.1575*	0.0027	-0.1140	-0.0011
	(0.1496)	(0.1612)	(0.0265)	(0.5621)	(0.6292)	(0.1752)	(0.0221)	(0.5175)	(0.0687)
Elite U.S. University	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.1956)	(0.2120)	(0.0312)	(0.6458)	(0.6293)	(0.2114)	(0.0293)	(0.6479)	(0.0892)
Elite U.S. Liberal Arts College	-0.0711	0.1268	0.0495	0.1261	0.0138	-0.0984	-0.0133	0.0830	0.1599
	(-0.4126)	(0.1268)	(0.0495)	(0.1261)	(0.0138)	(-0.0984)	(-0.0133)	(0.0830)	(0.1599)
Elite Foreign University	0.3562	0.0918	0.0076	-0.5037	-0.0388	0.0854	0.0231	-0.2799	-0.0237
	(3.2991)	(1.2999)	(0.6565)	(0.9280)	(0.6424)	(0.3439)	(0.0429)	(0.9777)	(0.1560)
Other Foreign University	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.3021)	(0.1566)	(0.0256)	(0.4369)	(0.4184)	(0.1544)	(0.0209)	(0.4193)	(0.0602)
Missing Undergraduate Institution	0.5809***	-0.4337**	-0.0663**	0.0506	0.0035	0.1769	0.0256	-0.0258	-0.1299
	(0.2157)	(0.2156)	(0.0355)	(0.4468)	(0.2199)	(0.2721)	(0.2256)	(0.6059)	(0.1050)
Graduate Degree	-0.0757	1.0229***	0.1507***	0.1629	-0.0191	-0.1664	-0.0213	-0.0043	-0.0101
	(-0.4126)	(3.1139)	(0.6542)	(0.3672)	(-0.1293)	(-0.0984)	(-0.0133)	(0.0830)	(0.1599)
Work Experience	0.0000**	0.1923***	0.1163***	0.0914*	0.0000	0.1758*	0.0227*	0.0434**	0.0057
	(0.1165)	(0.1256)	(0.0179)	(0.3412)	(0.0164)	(0.1424)	(0.0179)	(0.0411)	(0.0967)
#Professor Recommenders	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.4795)	(0.6838)	(0.0124)	(0.2976)	(0.4112)	(0.8993)	(0.0288)	(0.2728)	(0.0032)
#Postdoc Recommenders	0.1667***	0.1235***	0.0159***	0.1903	0.0089	0.0667	0.0086	-0.2648	-0.0393
	(3.1852)	(3.1852)	(3.1852)	(3.1852)	(3.1852)	(3.1852)	(3.1852)	(3.1852)	(3.1852)
Missing GRE	0.4685	0.4317**	0.0557**	1.1453	0.1477	0.0527	0.0053	0.3971	0.0511
	(0.9022)	(0.0246)	(0.0772)	(1.8092)	(0.6911)	(0.3772)	(0.0477)	(1.2435)	(0.1602)
Missing TOEFL/IELTS	0.1039**	0.0547	0.0107	0.4511	0.0232	-0.0036	-0.0005	-0.0285	-0.0004
	(0.1026)	(0.1299)	(0.0064)	(0.4230)	(0.4195)	(0.1433)	(0.0182)	(0.5931)	(0.0885)
Committee Score	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
				(0.3964)	(0.6984)	(0.0000)	(0.0842)	(0.3399)	(0.0529)
#Math Courses	-0.0795			-0.2484	-0.0112	0.0007	0.0001	-0.0182	-0.0029
	(-0.4126)			(-0.3964)	(-0.6984)	(0.0000)	(0.0842)	(0.3399)	(0.0529)
Advanced Math	0.3979			1.1555	0.0482	-0.2230	-0.0284	-1.1143**	-0.1762**
	(0.1680)			(0.7934)	(0.1047)	(0.2934)	(0.0370)	(0.9155)	(0.0877)
Letters Length/100	0.0000			0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.5287)			(0.1555)	(0.6163)	(0.0000)	(0.0842)	(0.0711)	(0.0113)
Letters Temp (%) PC1	0.1037*			0.2691	0.0132	0.1279	0.0162	0.9997**	0.0610**
	(0.4551)			(0.1555)	(0.5112)	(0.0951)	(0.0113)	(0.1663)	(0.0267)
Letters Temp (%) PC2	-0.0208			0.1797	0.0149	0.1964	0.0173	0.2997	0.0458
	(0.1680)			(0.1555)	(0.5112)	(0.0951)	(0.0113)	(0.1663)	(0.0267)
CV Length/100	0.0011			0.0391	0.0027	0.0366	0.0042	-0.1189	-0.0175
	(0.1529)			(0.1680)	(0.0047)	(0.0962)	(0.0075)	(0.1335)	(0.0175)
CV Coding	0.0000			0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.8230)			(0.1383)	(0.5589)	(0.0951)	(0.0842)	(0.1179)	(0.0181)
CV Publication/Working Paper	0.6825***			0.7911	0.0341	0.1440	0.0162	-0.5451	-0.0569
	(3.1852)			(0.1555)	(0.5112)	(0.0951)	(0.0113)	(0.1663)	(0.0267)
Essay Length/100	0.0002			0.2647	0.0189	0.0718	0.0095	0.0000	0.0000
	(0.0997)			(0.1662)	(0.0167)	(0.0814)	(0.0103)	(0.1396)	(0.0209)
Essay Specific Professor	0.1611			0.4125	0.0276	0.2856	0.0366	0.6012	0.1078
	(0.1756)			(0.4130)	(0.0275)	(0.1773)	(0.0392)	(0.4295)	(0.0773)
Essay Economics Terms (%)	0.0000			0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.1512)			(0.0814)	(0.0096)	(0.0814)	(0.0077)	(0.1075)	(0.0163)
Essay Research Terms (%)	0.0396			-0.0751	-0.0034	0.0164	0.0019	-0.1984	-0.0301
	(0.1680)			(0.1555)	(0.5112)	(0.0951)	(0.0113)	(0.1663)	(0.0267)
Essay Positive Terms (%)	0.0171			-0.0899	-0.0055	0.0977*	0.0130*	-0.0170	-0.0027
	(0.3165)			(0.3966)	(0.0041)	(0.0559)	(0.0066)	(0.1151)	(0.0162)
Essay Negative Terms (%)	0.0000			-0.0841	-0.0059	0.0000	0.0000	-0.0176	-0.0026
	(0.8230)			(0.1383)	(0.5589)	(0.0951)	(0.0842)	(0.1179)	(0.0181)
Missing Committee Score	0.0000					(0.1383)	(0.5589)	(0.0951)	(0.0842)
								(0.1179)	(0.0181)
Missing Application Pdf	-0.5378***			0.3616	0.0127	0.1669	0.0164	-0.7160	-0.1177
	(0.1661)			(0.7084)	(0.0211)	(0.1662)	(0.0167)	(0.1396)	(0.0209)
Program Rank						-0.0000***	-0.0001***		
						(0.0021)	(0.0000)		
Missing Program Rank						-0.0000***	-0.0001***		
						(0.4078)	(0.0016)		
$\sqrt{\frac{\text{GPA}}{\text{max}}}$	1.5687***								
	(0.1956)								
Sigma									
Obs / Param / LL / AIC	2329	565	-14356.1	28746.1					

Robust standard errors in parentheses *p<0.10, **p<0.05, ***p<0.01

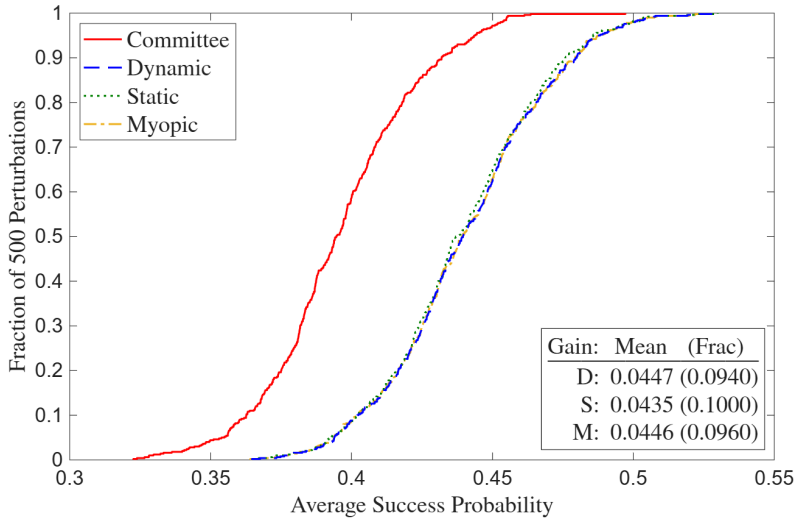
Restricted

	Committee Score	Cutoff × 1	I(Cutoff)	Cutoff × 2	PC1×Cutoff	Success	Matriculate	Sigma
Intercept	-0.7615***	-0.7611***	-0.0015	-0.7605***	34.76%	-0.6174***	-0.5445**	1.5484
	(-2.5331)	(-2.5302)	(-0.0024, 0.0054)	(-2.5264)	(-0.475, 0.5185)	(-3.3643)	(-2.5457)	(1.5457)
2016	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.2975)	(0.1305)	(0.0018, 0.0035)	(0.1304)	(-0.0195, 0.0195)	(0.1304)	(0.1304)	(0.7294)
2017	0.6424***	0.3092**	0.0108	0.1616	69.96%	0.1646	-0.3684	0.0000
	(3.2382)	(2.1786)	(-0.0041, 0.0257)	(0.1615)	(39.15%, 69.96%)	(0.1615)	(0.1672)	(0.0000)
2018	0.8914***	0.1956	0.0045	0.0197	39.97%	-0.3018	1.3369*	0.0000
	(3.2382)	(1.2999)	(-0.0041, 0.0257)	(0.1615)	(39.15%, 69.96%)	(0.1615)	(0.1672)	(0.0000)
2019	0.9997***	0.2882	0.0059	0.1616	57.41%	-0.2696	0.7667	0.0000
	(3.2382)	(0.2102)	(0.0035, 0.0259)	(0.1960)	(10.84%, 64.36%)	(0.1975)	(0.0962)	(0.0000)
Age	0.0000					0.0000	0.0000	0.0000
	(0.8230)					(0.4351)	(0.9841)	(0.0841)
Female	-0.0776					0.0116	0.0637	0.0000
	(0.1162)					(0.0662)	(0.0662)	(0.0000)
U.S.	0.1707*					-0.0233	0.8496	0.0000
	(1.9511)					(0.0662)	(0.0662)	(0.0000)
International Asian	-0.0824***					-0.0112	0.0938*	0.0000
	(-3.1852)					(0.0662)	(0.5833)	(0.5833)
GRE Verbal Percentile	0.0000**					0.0000	0.0000	0.0000
	(0.8230)					(0.4351)	(0.9841)	(0.0841)
GRE Quantitative Percentile	0.0007***					0.0000	0.0000	0.0000
	(3.1852)					(0.0662)	(0.0662)	(0.0000)
GRE Analytic Percentile	0.0004					0.0000	0.0021	0.0000
	(0.0027)					(0.0662)	(0.0662)	(0.0000)
TOEFL/IELTS	0.2068					-0.0012	-0.0645	0.0000
	(0.1756)					(0.0161)	(0.2753)	0.0000
Elite U.S. University	0.0000***					-0.0098	-0.0637	0.0000
	(0.1082)					(0.0162)	(0.3925)	0.0000
Elite U.S. Liberal Arts College	0.0000					0.0000	0.0000	0.0000
	(0.1956)					(0.0162)	(0.3925)	0.0000
Elite Foreign University	0.1956					-0.0197	0.4559	0.0000
	(0.2680)					(0.0162)	(0.3925)	0.0000
Other Foreign University	0.0000***					0.0038	0.2911	0.0000
	(0.2680)					(0.0162)	(0.3764)	0.0000
Missing Undergraduate Institution	0.5809***					0.0000	0.0000	0.0000
	(0.2130)					(0.0162)	(0.4967)	0.0000
Graduate Degree	-0.0757					-0.0043	-0.0101	0.0000
	(-0.4126)					(0.0162)	(0.3925)	0.0000
Work Experience	0.0765					0.0088	0.9662	0.0000
	(0.1165)					(0.0162)	(0.3925)	0.0000
#Professor Recommenders	0.0000					0.0000	0.0000	0.0000
	(0.0022)					(0.0694)	(0.1692)	0.0000
#Postdoc Recommenders	0.0000					-0.0002	-0.1797	0.0000
	(0.0995)					(0.0012)	(0.1641)	0.0000
Missing GRE	0.4685					0.0053	0.0511	0.0000
	(0.3761)					(0.0012)	(0.1641)	0.0000
Missing TOEFL/IELTS	0.2637**					0.0023	0.0051	0.0000
	(0.1161)					(0.0012)	(0.1641)	0.0000
Committee Score	0.0000					0.0000***	0.5117***	0.0000
						(0.0299)	(0.1697)	0.0000
#Math Courses	-0.1660					0.0000	0.0000	0.0000
	(0.1531)					(0.0297)	(0.1619)	0.0000
Advanced Math	0.3979					0.0000	0.0000	0.0000
	(0.1161)					(0.0158)	(0.1557)	0.0000
Letters Length/100	0.1099**					0.0013	-0.0132	0.0000
	(0.5287)					(0.0158)	(0.1557)	0.0000
Letters Temp (%) PC1	0.3697***					0.0000	0.0000	0.0000
	(0.0988)					(0.0071)	(0.1642)	0.0000
Letters Temp (%) PC2	-0.3836**					-0.0158	-0.3648	0.0000
	(0.1756)					(0.0071)	(0.1642)	0.0000
CV Length/100	0.0011					-0.0013	-0.0648	0.0000
	(0.1529)					(0.0071)	(0.1642)	0.0000
CV Coding	0.0167					-0.0049	-0.9843	0.0000
	(0.8230)					(0.0071)	(0.1642)	0.0000
CV Publication/Working Paper	1.3617***					0.0000	-0.1974	0.0000
	(0.1832)					(0.0162)	(0.5220)	0.0000
Essay Length/100	0.2717*					-0.0000	-0.0000	0.0000
	(0.1675)					(0.0162)	(0.1622)	0.0000
Essay Specific Professor	0.5000					(0.0162)	(0.1622)	0.0000
	(0.1988)					(0.0162)	(0.1622)	0.0000
Essay Economics Terms (%)	0.0081					0.0030	0.0869	0.0000
	(0.1512)					(0.0162)	(0.1622)	0.0000
Essay Research Terms (%)	0.1330					0.0000	-0.1729	0.0000
	(0.0662)					(0.0830)	(0.1293)	0.0000
Essay Positive Terms (%)	0.0000					-0.0000	0.0000	0.0000
	(0.0761)					(0.0829)	(0.0946)	0.0000
Essay Negative Terms (%)	0.0000					-0.0000	0.0000	0.0000
	(0.1082)					(0.0829)	(0.1032)	0.0000
Missing Committee Score						-0.1430		
						(0.0829)		
Missing Application Pdf	-0.6098					0.0116	-1.9307***	
	(0.2666)					(0.0129)	(0.2216)	
Program Rank						-0.0000***		
						(0.0026)		
Missing Program Rank						-0.0000***		
						(0.4041)		
$\sqrt{\frac{\text{GPA}}{\text{max}}}$	1.7622***							
	(0.2466)							
Sigma								0.0030**
								(0.0002)
Obs / Param / LL / AIC	2329	568	-1526.9	2963.8				

Robust standard errors in parentheses *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

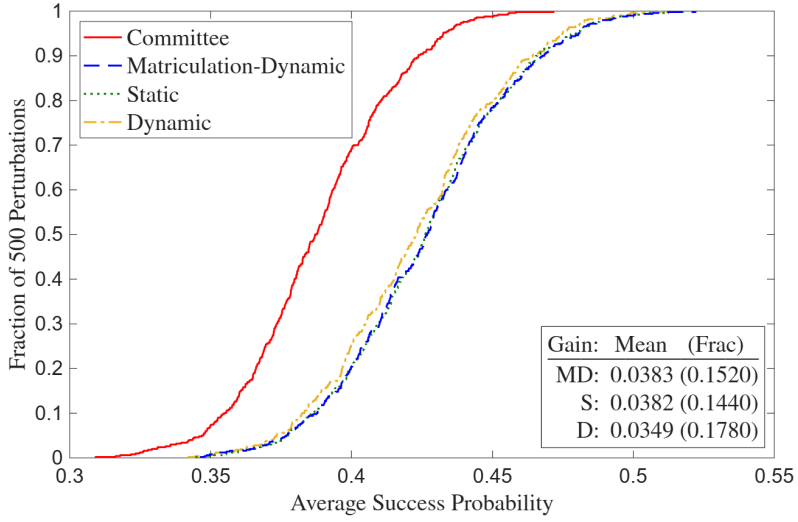
Restricted Matriculation-Dynamic Estimates

A3: Full Deviation Gains (Acceptance Set)



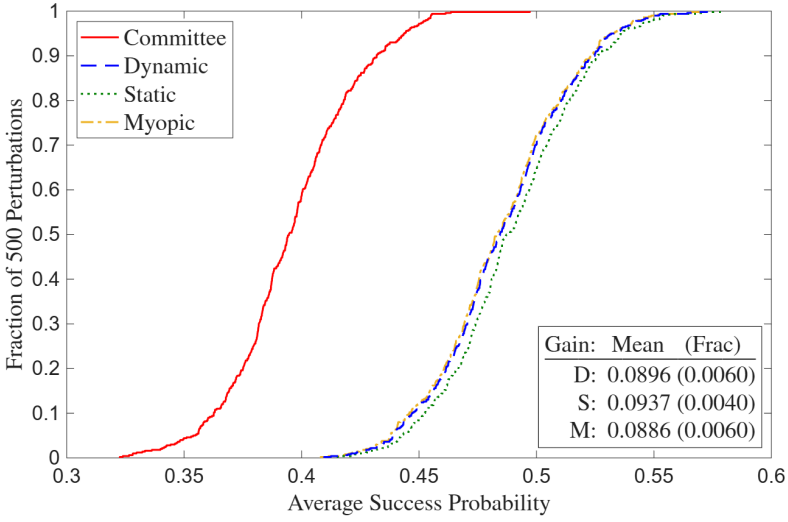
Deviation Gains (Acceptance Set)

A3: Full Deviation Gains (Matriculation Set)



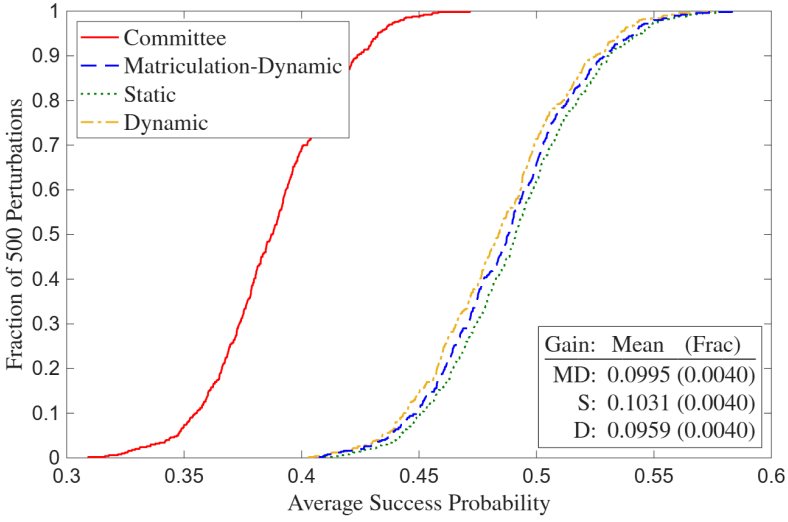
Deviation Gains (Matriculation Set)

A3: Uncorrected Deviation Gains (Acceptance Set)



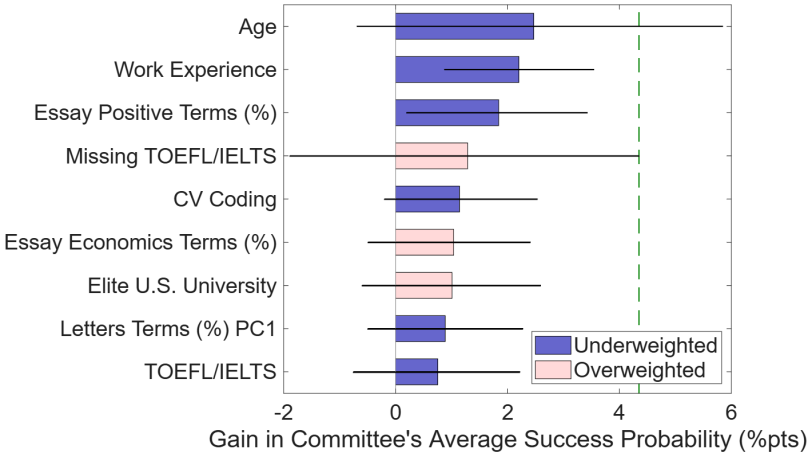
Deviation Gains (Acceptance Set)

A3: Uncorrected Deviation Gains (Matriculation Set)



Deviation Gains (Matriculation Set)

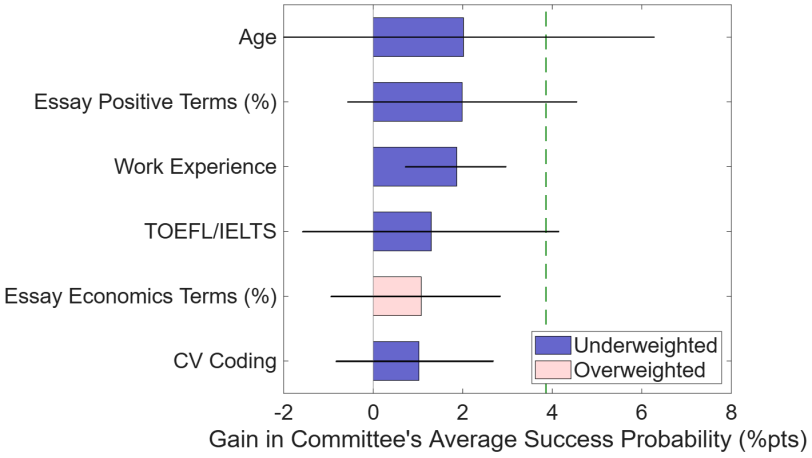
A3: Full Deviation Gains Attribution (Acceptance Set)



1. Experience
2. English Proficiency
3. Essay/Letters Subtext
4. Industry Background
5. Undergraduate

Deviation Gains Attribution (Acceptance Set)

A3: Full Deviation Gains Attribution (Matriculation Set)



1. Experience
2. English Proficiency
3. Essay Subtext
4. Industry Background

Deviation Gains Attribution (Matriculation Set)

A3: Static Model Attribution (Acceptance Set)

	Gain (%pt)	Sample Mean	Committee Mean	1-Variable Optimal Mean	Joint Optimal Mean	1-Variable Difference (SD)	Joint Difference (SD)
Age	2.4692	25.9910	26.2840	29.9572	29.5331	1.1674	1.0326
Work Experience	2.2037	0.5247	0.7276	0.9844	0.9650	0.5141	0.4752
Essay Positive Terms (%)	1.8431	11.7499	11.6521	12.5376	12.1183	0.7858	0.4137
Missing TOEFL/IELTS	1.2872	0.6076	0.6654	0.0739	0.4086	-1.2110	-0.5258
CV Coding	1.1426	2.8001	2.7820	3.3399	2.9415	0.6244	0.1785
Essay Economics Terms (%)	1.0382	9.7228	9.7423	9.0284	9.5388	-0.6411	-0.1828
Elite U.S. University	1.0086	0.1267	0.2335	0.0078	0.0389	-0.6784	-0.5848
Letters Terms (%) PC1	0.8868	-0.0106	0.1374	0.4969	0.1387	0.5202	0.0018
TOEFL/IELTS	0.7541	7.4376	7.5675	7.7759	7.6827	0.5957	0.3294

⏩ Deviation Gains Attribution (Acceptance Set)

A3: Static Model Attribution (Matriculation Set)

	Gain (%pt)	Sample Mean	Committee Mean	1-Variable Optimal Mean	Joint Optimal Mean	1-Variable Difference (SD)	Joint Difference (SD)
Age	2.0176	25.9910	26.3284	30.9159	29.7737	1.4580	1.0950
Essay Positive Terms (%)	1.9874	11.7499	11.6107	12.9136	12.2763	1.1534	0.5905
Work Experience	1.8659	0.5247	0.7938	0.9949	0.9799	0.4077	0.3725
TOEFL/IELTS	1.2973	7.4376	7.4939	7.8501	7.6128	1.0238	0.3397
Essay Economics Terms (%)	1.0726	9.7228	9.9994	9.0856	9.5738	-0.8271	-0.3821
CV Coding	1.0219	2.8001	2.6661	3.3395	2.8782	0.7618	0.2374

⏩ Deviation Gains Attribution (Matriculation Set)

A3: Deviation Gains Robustness

	Acceptance Set	Matriculation Set
Observed Gain, Uncorrected	0.1015	0.1124
Observed Gain, Corrected	0.0514	0.0475
Perturbed Gain, Uncorrected	0.0937	0.1031
Perturbed Gain, Corrected	0.0435	0.0382
Likelihood-Ratio p-Value	0.1166	0.1166
Calibrated Signal Scale	0.4197	0.4197
Calibrated In-Sample Gain	0.1242	0.1481
Calibrated True Gain	0.0491	0.0560

⏩ Deviation Gains (Acceptance) ⏩ Deviation Gains (Matriculation)

1. Scale down (x, z) parameters with values of scaling factor κ from 0 to 1
2. Simulate *Success*, and find specific κ yielding observed joint LR test statistic
3. Calculate true deviation gain for that κ

A4: Graduate School Means by Program

	Economics	Government	History	Linguistics	Psychology
Accept	0.1121	0.0430	0.0986	0.0678	0.0535
Covid Year	0.2478	0.2792	0.3029	0.3460	0.3004
Age	26.2782	27.9633	26.8478	28.0692	25.8230
Female	0.4201	0.4182	0.4232	0.6133	0.7490
U.S. African/Hispanic/Native	0.0289	0.0766	0.1159	0.0665	0.1728
U.S. Asian	0.0313	0.0421	0.0304	0.0353	0.0576
International Asian	0.5322	0.2510	0.1058	0.2714	0.1523
International Other	0.2517	0.2967	0.2203	0.3256	0.1584
GRE Verbal Percentile	71.8677	81.3486	82.6054	73.5451	75.6192
GRE Quantitative Percentile	86.2163	63.3428	49.3295	57.8455	58.0000
GRE Analytic Percentile	52.6351	72.1905	75.0460	63.6824	69.2077
Undergraduate GPA	3.6534	3.5675	3.6531	3.6567	3.5695
TOEFL/IELTS	7.5232	7.5422	7.4252	7.5444	7.3158
Elite U.S. University	0.1516	0.1847	0.2565	0.1954	0.2840
Elite U.S. Liberal Arts College	0.0423	0.0672	0.0696	0.0231	0.0535
Elite Foreign University	0.0717	0.0578	0.0478	0.0488	0.0206
Other Foreign University	0.5736	0.3953	0.2333	0.5102	0.1852
Graduate Degree	0.7478	0.7553	0.6652	0.7544	0.4259
Work Experience	0.5938	0.7463	0.6638	0.7544	0.7593
#Professor Recommenders	2.5173	2.3469	2.6261	2.4355	2.1831
#Prolific Recommenders	1.0760	0.6912	0.5043	0.7069	0.3128
#Math Courses	6.6396				
Advanced Math	0.4827				
Missing GRE	0.0255	0.4612	0.6217	0.6839	0.4650
Missing Undergraduate GPA	0.6607	0.4827	0.3072	0.5726	0.2490
Missing TOEFL/IELTS	0.5958	0.7714	0.8449	0.7707	0.8827
Observations	2078	2231	690	737	486

Insufficiency of Comparing AMEs

- **Economics Highest:** Accept, International Asian, GRE Quantitative Percentile, Elite Foreign University, Other Foreign University, #Prolific Recommenders, Missing Undergraduate GPA
- **Economics Lowest:** Covid Year, U.S. African/Hispanic/Native, GRE Verbal Percentile, GRE Analytic Percentile, Elite U.S. University, Work Experience, Missing GRE, Missing TOEFL/IELTS

A4: Graduate School Means by Program, Accept = 1

	Economics	Government	History	Linguistics	Psychology
Covid Year	0.1888	0.2188	0.2059	0.2200	0.2308
Age	25.0815	27.7292	27.2794	27.1000	25.7692
Female	0.4850	0.5000	0.5588	0.5800	0.6154
U.S. African/Hispanic/Native	0.0386	0.1458	0.1765	0.0400	0.0769
U.S. Asian	0.0429	0.0312	0.0147	0.0600	0.1154
International Asian	0.4163	0.1667	0.0882	0.3600	0.0769
International Other	0.2575	0.2292	0.3235	0.2000	0.1538
GRE Verbal Percentile	84.3980	88.8725	86.6328	77.9161	85.7729
GRE Quantitative Percentile	91.9289	68.3708	54.8113	64.5504	62.3462
GRE Analytic Percentile	69.0355	82.7222	78.0879	68.8958	74.6169
Undergraduate GPA	3.7336	3.6404	3.7204	3.7004	3.6341
TOEFL/IELTS	7.6419	7.6050	7.4730	7.6237	7.3178
Elite U.S. University	0.2833	0.3125	0.3382	0.2200	0.5385
Elite U.S. Liberal Arts College	0.1030	0.1458	0.1176	0.0200	0.1154
Elite Foreign University	0.1030	0.0521	0.0588	0.1000	0.0385
Other Foreign University	0.3863	0.2396	0.2647	0.4000	0.1154
Graduate Degree	0.5837	0.6771	0.7206	0.7400	0.4615
Work Experience	0.6309	0.8125	0.7647	0.7000	0.9231
#Professor Recommenders	2.5494	2.3750	2.8235	2.7200	2.2308
#Prolific Recommenders	1.2833	0.9167	0.6765	1.0800	0.4615
#Math Courses	7.2575				
Advanced Math	0.6781				
Missing GRE	0.0086	0.2917	0.6324	0.5800	0.1923
Missing Undergraduate GPA	0.4893	0.3125	0.3088	0.4800	0.1923
Missing TOEFL/IELTS	0.6567	0.8854	0.8529	0.7600	0.8846
Observations	233	96	68	50	26

⏩ Insufficiency of Comparing AMEs

- **Economics Highest:** International Asian, GRE Quantitative Percentile, Undergraduate GPA, TOEFL/IELTS, Elite Foreign University, #Prolific Recommenders, Missing Undergraduate GPA
- **Economics Lowest:** Covid Year, Age, Female, U.S. African/Hispanic/Native, Work Experience, Missing GRE, Missing TOEFL/IELTS

A4: Graduate School Correlation Matrix

	Covid	Age	Fem	USAHN	USAsn	IntAsn	IntOth	GREV	GREQ	GREA	GPA	T/I	EUSUni	EUSLA	EFor	OthFor	Grad	Work	Prof	Pro	MGRE	MGPA	MT/I
Covid Year	1.00																						
Age	0.01	1.00																					
Female	0.02	-0.13	1.00																				
U.S. African/Hispanic/Native	0.00	-0.04	0.04	1.00																			
U.S. Asian	-0.00	-0.06	0.05	-0.05	1.00																		
International Asian	-0.05	-0.07	0.02	-0.19	-0.14	1.00																	
International Other	0.01	0.18	-0.06	-0.17	-0.12	-0.42	1.00																
GRE Verbal Percentile	0.03	-0.06	-0.04	0.00	0.04	-0.12	-0.15	1.00															
GRE Quantitative Percentile	-0.02	-0.16	-0.10	-0.18	0.00	0.44	-0.10	0.16	1.00														
GRE Analytic Percentile	0.05	-0.06	0.03	0.08	0.08	-0.37	-0.08	0.60	-0.20	1.00													
Undergraduate GPA	-0.02	-0.29	0.07	-0.13	-0.02	0.02	0.00	0.06	0.12	0.05	1.00												
TOEFL/IELTS	-0.01	-0.07	0.01	-0.02	0.01	0.04	0.00	0.24	0.12	0.19	0.00	1.00											
Elite U.S. University	0.01	-0.12	0.02	0.12	0.14	-0.12	-0.22	0.16	-0.00	0.19	-0.01	-0.02	1.00										
Elite U.S. Liberal Arts College	-0.02	-0.02	0.04	0.04	0.04	-0.08	-0.09	0.12	-0.01	0.13	-0.02	-0.00	-0.11	1.00									
Elite Foreign University	-0.00	-0.04	-0.00	-0.06	-0.04	0.09	0.09	0.02	0.08	-0.02	0.01	0.04	-0.12	-0.06	1.00								
Other Foreign University	-0.02	0.20	-0.04	-0.23	-0.14	0.34	0.39	-0.26	0.20	-0.39	0.02	0.02	-0.43	-0.21	-0.22	1.00							
Graduate Degree	0.00	0.41	-0.08	-0.13	-0.09	0.21	0.15	-0.14	0.05	-0.20	-0.22	0.03	-0.21	-0.10	0.05	0.37	1.00						
Work Experience	0.01	0.33	0.05	0.05	0.01	-0.18	0.11	0.01	-0.15	0.11	-0.08	-0.03	-0.02	0.03	-0.03	-0.01	0.10	1.00					
#Professor Recommenders	-0.02	-0.19	-0.04	-0.04	0.01	0.11	-0.09	0.01	0.09	-0.04	0.14	0.03	-0.02	0.02	-0.00	-0.02	-0.03	-0.18	1.00				
#Prolific Recommenders	0.01	-0.06	-0.02	-0.07	-0.01	0.24	-0.10	-0.01	0.25	-0.11	0.04	0.10	-0.01	-0.01	0.06	0.08	0.17	-0.09	0.19	1.00			
Missing GRE	0.09	0.15	0.05	0.11	-0.01	-0.22	0.17	0.09	-0.33	0.17	-0.09	-0.10	-0.03	-0.04	-0.03	-0.01	0.02	0.10	-0.08	-0.20	1.00		
Missing Undergraduate GPA	-0.01	0.18	-0.04	-0.24	-0.15	0.36	0.42	-0.24	0.21	-0.38	0.02	0.04	-0.46	-0.23	0.23	0.83	0.37	-0.02	-0.03	0.10	-0.01	1.00	
Missing TOEFL/IELTS	0.05	-0.07	0.03	0.16	0.11	-0.32	-0.22	0.16	-0.24	0.32	-0.02	-0.02	0.29	0.14	-0.04	-0.58	-0.20	0.05	-0.07	-0.05	0.12	-0.58	1.00
Observations	6222																						

⏩ Insufficiency of Comparing AMEs

A4: Graduate School Logistic Regression Results

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology
Covid Year	-0.3966**	0.4332*	-0.5752*	0.5501	-0.1766
	(0.2010)	(0.2612)	(0.3479)	(0.4129)	(0.5545)
Age	-0.0995***	0.0454*	0.0380	0.0222	-0.0520
	(0.0358)	(0.0271)	(0.0327)	(0.0532)	(0.0899)
Female	0.4079**	0.6723***	0.6539**	0.1311	-0.5060
	(0.1681)	(0.2299)	(0.2671)	(0.3452)	(0.5606)
U.S. African/Hispanic/Native	0.4221	1.0081***	1.1601***	0.0120	-0.8263
	(0.5258)	(0.3545)	(0.4426)	(0.8411)	(1.1611)
U.S. Asian	-0.5304	-0.5669	-0.3260	0.8638	1.1711
	(0.4199)	(0.6504)	(1.2512)	(0.7301)	(0.7276)
International Asian	-0.8445***	-0.1853	0.9405	0.4890	-0.7827
	(0.2945)	(0.4146)	(0.7233)	(0.6729)	(0.9323)
International Other	0.5034	0.3781	1.3523***	-0.1061	0.6239
	(0.3172)	(0.4131)	(0.5086)	(0.7483)	(1.1070)
GRE Verbal Percentile	0.0155***	0.0281*	0.0753**	-0.0092	0.0607***
	(0.0060)	(0.0160)	(0.0344)	(0.0154)	(0.0218)
GRE Quantitative Percentile	0.1547***	0.0161*	0.0210*	0.0211	-0.0138
	(0.0186)	(0.0086)	(0.0116)	(0.0150)	(0.0159)
GRE Analytic Percentile	0.0179***	0.0169**	0.0030	0.0175	-0.0202
	(0.0042)	(0.0094)	(0.0134)	(0.0127)	(0.0218)
Undergraduate GPA	2.2212***	0.6968	1.3648**	1.2645	1.7772*
	(0.7210)	(0.5476)	(0.5513)	(0.9703)	(0.9487)
TOEFL/IELTS	0.3878	0.6456	1.0963**	0.9655*	-0.4656
	(0.2883)	(0.4708)	(0.5526)	(0.5456)	(0.6112)
Elite U.S. University	0.8850***	0.3043	0.6541*	-0.6057	1.4303**
	(0.2960)	(0.3135)	(0.3802)	(0.4985)	(0.6192)
Elite U.S. Liberal Arts College	1.1173***	0.6276*	1.0698**	-0.6088	2.1517**
	(0.3813)	(0.3754)	(0.5399)	(1.4186)	(1.0642)
Elite Foreign University	1.0061*	0.4758	1.7715**	1.6195*	0.3889
	(0.6060)	(0.8224)	(0.7938)	(0.8587)	(1.9889)
Other Foreign University	0.5442	0.5777	1.3733**	0.4188	-1.0737
	(0.5765)	(0.7686)	(0.6992)	(0.7157)	(1.3016)
Graduate Degree	0.0467	0.0733	0.3128	0.4049	0.4892
	(0.2238)	(0.2583)	(0.3280)	(0.4896)	(0.6399)
Work Experience	0.3932**	0.4027	0.4072	0.1491	1.8201**
	(0.1889)	(0.2993)	(0.3295)	(0.3778)	(0.8564)
#Professor Recommenders	0.0340	-0.0179	0.5511**	0.4731*	-0.0981
	(0.1317)	(0.1339)	(0.2554)	(0.2662)	(0.2660)
#Prolific Recommenders	0.2635***	0.3025**	0.1880	0.3284**	0.5901*
	(0.0854)	(0.1262)	(0.1576)	(0.1544)	(0.3145)
#Math Courses	0.1689***				
	(0.0596)				
Advanced Math	0.2319				
	(0.1700)				
Missing GRE	0.1899	-0.3259	0.6282	-0.0831	-1.1514**
	(0.7646)	(0.3100)	(0.4275)	(0.3761)	(0.5702)
Missing Undergraduate GPA	-0.4745	-0.4917	-1.5492**	-1.1097*	0.4690
	(0.4891)	(0.5440)	(0.6893)	(0.6481)	(1.1831)
Missing TOEFL/IELTS	-0.3786	0.6188	0.6745	-0.0440	-1.7224*
	(0.2432)	(0.5657)	(0.5190)	(0.5099)	(0.9271)
Concentration FE	No	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486
Pseudo-R^2	0.2627	0.1525	0.1982	0.1871	0.2475

Robust standard errors in parentheses. *p<0.10, **p<0.05, ***p<0.01.

Insufficiency of Comparing AMEs

- **Economics Positive ($p < .1$):** Female, GRE Verbal Percentile, GRE Quantitative Percentile, GRE Analytic Percentile, Undergraduate GPA, Elite U.S. University, Elite U.S. Liberal Arts College, Elite Foreign University, Work Experience, #Prolific Recommenders, #Math Courses
- **Economics Negative ($p < .1$):** Covid Year, Age, International Asian

A4: Graduate School Logistic Regression AMEs

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology	Non-Economics
Covid Year	-0.0303**	-0.0164	-0.0431*	-0.0298	-0.0075	-0.0218***
	(0.0153)	(0.0100)	(0.0261)	(0.0224)	(0.0238)	(0.0084)
Age	-0.0076***	0.0017*	0.0028	0.0012	-0.0022	0.0016*
	(0.0027)	(0.0010)	(0.0025)	(0.0029)	(0.0038)	(0.0009)
Female	0.0312**	0.0254***	0.0490**	0.0071	-0.0216	0.0207***
	(0.0129)	(0.0088)	(0.0220)	(0.0187)	(0.0239)	(0.0075)
U.S. African/Hispanic/Native	0.0323	0.0381***	0.0869***	0.0007	-0.0352	0.0384***
	(0.0402)	(0.0137)	(0.0332)	(0.0455)	(0.0487)	(0.0119)
U.S. Asian	-0.0413	-0.0215	-0.0244	0.0468	0.0499	0.0062
	(0.0322)	(0.0247)	(0.0936)	(0.0391)	(0.0313)	(0.0185)
International Asian	-0.0646***	-0.0070	0.0704	0.0265	-0.0314	0.0069
	(0.0224)	(0.0157)	(0.0543)	(0.0363)	(0.0363)	(0.0132)
International Other	0.0385	0.0143	0.1013***	-0.0057	0.0266	0.0238*
	(0.0243)	(0.0157)	(0.0385)	(0.0406)	(0.0471)	(0.0132)
GRE Verbal Percentile	0.0012***	0.0011*	0.0056**	-0.0005	0.0026***	0.0015***
	(0.0005)	(0.0006)	(0.0026)	(0.0008)	(0.0010)	(0.0005)
GRE Quantitative Percentile	0.0118***	0.0006*	0.0016*	0.0011	-0.0006	0.0005**
	(0.0014)	(0.0003)	(0.0009)	(0.0008)	(0.0007)	(0.0003)
GRE Analytic Percentile	0.0014***	0.0008*	0.0002	0.0009	-0.0009	0.0007**
	(0.0003)	(0.0004)	(0.0010)	(0.0007)	(0.0009)	(0.0003)
Undergraduate GPA	0.1699***	0.0264	0.1022**	0.0684	0.0757*	0.0488***
	(0.0551)	(0.0268)	(0.0423)	(0.0522)	(0.0418)	(0.0176)
TOEFL/IELTS	0.0297	0.0244	0.0821**	0.0523*	-0.0198	0.0326***
	(0.0219)	(0.0179)	(0.0417)	(0.0301)	(0.0259)	(0.0123)
Elite U.S. University	0.0677***	0.0149	0.0490*	-0.0328	0.0610**	0.0233**
	(0.0225)	(0.0119)	(0.0281)	(0.0264)	(0.0271)	(0.0098)
Elite U.S. Liberal Arts College	0.0855***	0.0238*	0.0801**	-0.0330	0.0867*	0.0420***
	(0.0288)	(0.0143)	(0.0402)	(0.0761)	(0.0480)	(0.0136)
Elite Foreign University	0.0770*	0.0180	0.1327**	0.0877*	0.0166	0.0479**
	(0.0463)	(0.0311)	(0.0591)	(0.0481)	(0.0850)	(0.0241)
Other Foreign University	0.0416	0.0219	0.1028*	0.0227	-0.0458	0.0273
	(0.0441)	(0.0291)	(0.0527)	(0.0391)	(0.0556)	(0.0221)
Graduate Degree	0.0036	0.0028	0.0234	0.0219	0.0208	0.0106
	(0.0171)	(0.0098)	(0.0247)	(0.0262)	(0.0273)	(0.0089)
Work Experience	0.0301**	0.0152	0.0372	0.0081	0.0776**	0.0231**
	(0.0144)	(0.0113)	(0.0248)	(0.0205)	(0.0368)	(0.0089)
#Professor Recommenders	0.0026	-0.0007	0.0413**	0.0250*	-0.0042	0.0063
	(0.0101)	(0.0051)	(0.0190)	(0.0142)	(0.0113)	(0.0047)
#Prolific Recommenders	0.0202***	0.0114**	0.0141	0.0178**	0.0251*	0.0133***
	(0.0065)	(0.0048)	(0.0118)	(0.0082)	(0.0130)	(0.0038)
#Math Courses	0.0129***					
	(0.0045)					
Advanced Math	0.0177					
	(0.0137)					
Missing GRE	0.0145	-0.0123	0.0470	-0.0045	-0.0491*	-0.0062
	(0.0585)	(0.0118)	(0.0319)	(0.0204)	(0.0252)	(0.0088)
Missing Undergraduate GPA	-0.0363	-0.0186	-0.1160**	-0.0601*	0.0200	-0.0291
	(0.0375)	(0.0206)	(0.0524)	(0.0355)	(0.0505)	(0.0179)
Missing TOEFL/IELTS	-0.0290	0.0234	0.0505	-0.0024	-0.0734*	0.0108
	(0.0188)	(0.0215)	(0.0388)	(0.0276)	(0.0417)	(0.0135)
Concentration FE	No	Yes	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486	4144
Wald Statistic | DF | p-Value	115.8837	23	0.0000			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01

- **Economics Positive ($p < .1$):** Female, GRE Verbal Percentile, GRE Quantitative Percentile, GRE Analytic Percentile, Undergraduate GPA, Elite U.S. University, Elite U.S. Liberal Arts College, Elite Foreign University, Work Experience, #Prolific Recommenders, #Math Courses
- **Economics Negative ($p < .1$):** Covid Year, Age, International Asian

Insufficiency of Comparing AMEs

A4: Graduate School Logistic Regression Results (Synthetic)

Dependent Variable – Accept	Economics	Government	History	Linguistics	Psychology
Covid Year	-0.1997	-0.0323	0.3028	-0.4791	-0.5946
	(0.1633)	(0.2215)	(0.3078)	(0.3477)	(0.4520)
Age	-0.0509**	-0.0198	0.0576*	-0.0164	0.1203***
	(0.0201)	(0.0174)	(0.0307)	(0.0357)	(0.0400)
Female	0.1941	0.0923	0.1389	0.2595	0.2119
	(0.1402)	(0.2004)	(0.2526)	(0.3255)	(0.4310)
U.S. African/Hispanic/Native	0.3053	-0.1457	-0.2379	-1.6401	0.5413
	(0.3724)	(0.4303)	(0.4670)	(1.0720)	(0.5306)
U.S. Asian	0.5663	0.6016	-0.7934	-0.4305	0.7122
	(0.3642)	(0.4452)	(1.1675)	(0.7936)	(0.8863)
International Asian	-0.1428	0.1630	0.5780	-0.4965	0.1868
	(0.2169)	(0.2803)	(0.3740)	(0.4275)	(0.5866)
International Other	-0.1231	0.1395	0.1097	-0.1602	-1.0807
	(0.2311)	(0.3046)	(0.3237)	(0.3832)	(0.7583)
GRE Verbal Percentile	0.0167***	0.0247**	0.0164	-0.0029	0.0132
	(0.0048)	(0.0117)	(0.0127)	(0.0100)	(0.0184)
GRE Quantitative Percentile	0.0233***	-0.0027	0.0026	0.0304**	0.0097
	(0.0090)	(0.0051)	(0.0088)	(0.0137)	(0.0099)
GRE Analytic Percentile	0.0042	0.0004	-0.0073	-0.0042	0.0069
	(0.0031)	(0.0059)	(0.0092)	(0.0101)	(0.0103)
Undergraduate GPA	0.2292	0.6327	1.3552**	-1.3171**	-0.1002
	(0.4745)	(0.4985)	(0.5796)	(0.5584)	(0.6152)
TOEFL/IELTS	-0.4673**	0.5141	-0.7648	0.2446	-0.7354
	(0.2157)	(0.4185)	(0.6814)	(0.7498)	(0.8025)
Elite U.S. University	-0.4850*	-0.2243	0.0572	0.4764	0.3941
	(0.2560)	(0.3258)	(0.3421)	(0.4788)	(0.4782)
Elite U.S. Liberal Arts College	0.2429	0.1759	0.2358	0.4860	0.8652
	(0.3049)	(0.4042)	(0.4964)	(0.9631)	(0.7261)
Elite Foreign University	-0.1622	-1.1929	0.7404	0.4801	3.0530***
	(0.3015)	(0.7484)	(0.5668)	(0.7841)	(0.8673)
Other Foreign University	-0.1087	0.2043	0.2721	0.1731	0.3023
	(0.1979)	(0.2732)	(0.3391)	(0.4292)	(0.5926)
Graduate Degree	-0.0198	0.1770	-0.1601	-0.8698***	-0.2769
	(0.1624)	(0.2439)	(0.2765)	(0.3320)	(0.3814)
Work Experience	0.2525*	-0.0391	-0.0736	0.2652	-1.1936**
	(0.1441)	(0.2396)	(0.2629)	(0.3417)	(0.4805)
#Professor Recommenders	-0.1059	0.0911	-0.1828	0.2181	-0.4923**
	(0.0909)	(0.1138)	(0.1473)	(0.1817)	(0.1956)
#Prolific Recommenders	0.0039	0.2189**	-0.1743	0.0673	0.5134**
	(0.0656)	(0.1108)	(0.1723)	(0.1642)	(0.2213)
#Math Courses	0.0366				
	(0.0421)				
Advanced Math	0.1961				
	(0.1416)				
Missing GRE	-0.1993	-0.2029	-0.0011	-0.0433	-0.6040
	(0.4328)	(0.2195)	(0.2689)	(0.3503)	(0.4017)
Missing Undergraduate GPA	0.1531	0.0895	-0.2455	0.6298**	-0.3574
	(0.1548)	(0.2291)	(0.3015)	(0.3125)	(0.4342)
Missing TOEFL/IELTS	-0.2950*	0.4898	0.5752	0.1346	-2.0781***
	(0.1640)	(0.3480)	(0.5466)	(0.4345)	(0.7633)
Concentration FE	No	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486
Pseudo-R^2	0.0524	0.0366	0.0753	0.1046	0.2230

Robust standard errors in parentheses. *p<0.10, **p<0.05, ***p<0.01.

- **Economics Positive ($p < .1$):** GRE Verbal Percentile, GRE Quantitative Percentile, Work Experience
- **Economics Negative ($p < .1$):** Age, TOEFL/IELTS, Elite U.S. University, Missing TOEFL/IELTS

⏩ Insufficiency of Comparing AMEs

A4: Graduate School Logistic Regression AMEs (Synthetic)

Dependent Variable ~ Accept	Economics	Government	History	Linguistics	Psychology	Non-Economics
Covid Year	-0.0204	-0.0014	0.0102	-0.0298	-0.0345	-0.0089
	(0.0167)	(0.0099)	(0.0291)	(0.0216)	(0.0262)	(0.0086)
Age	-0.0052**	-0.0009	0.0054*	-0.0010	0.0070***	0.0004
	(0.0021)	(0.0008)	(0.0029)	(0.0022)	(0.0023)	(0.0008)
Female	0.0198	0.0041	0.0131	0.0162	0.0123	0.0099
	(0.0143)	(0.0089)	(0.0239)	(0.0204)	(0.0251)	(0.0078)
U.S. African/Hispanic/Native	0.0312	-0.0065	-0.0225	-0.1021	0.0314	-0.0137
	(0.0381)	(0.0192)	(0.0441)	(0.0673)	(0.0306)	(0.0147)
U.S. Asian	0.0579	0.0268	-0.0750	-0.0368	0.0413	0.0241
	(0.0372)	(0.0199)	(0.1101)	(0.0495)	(0.0516)	(0.0179)
International Asian	-0.0146	0.0073	0.0546	-0.0309	0.0108	0.0075
	(0.0222)	(0.0125)	(0.0353)	(0.0263)	(0.0399)	(0.0112)
International Other	-0.0126	0.0062	0.0104	-0.0100	-0.0627	-0.0014
	(0.0236)	(0.0136)	(0.0306)	(0.0237)	(0.0443)	(0.0106)
GRE Verbal Percentile	0.0017***	0.0011**	0.0015	-0.0002	0.0008	0.0008**
	(0.0005)	(0.0005)	(0.0012)	(0.0006)	(0.0010)	(0.0004)
GRE Quantitative Percentile	0.0024***	-0.0001	0.0003	0.0019**	0.0006	0.0003
	(0.0009)	(0.0002)	(0.0008)	(0.0009)	(0.0006)	(0.0002)
GRE Analytic Percentile	0.0004	0.0000	-0.0007	-0.0003	0.0004	0.0000
	(0.0003)	(0.0003)	(0.0009)	(0.0006)	(0.0005)	(0.0002)
Undergraduate GPA	0.0234	0.0282	0.1261**	-0.0820**	-0.0058	0.0130
	(0.0485)	(0.0223)	(0.0553)	(0.0358)	(0.0357)	(0.0158)
TOEFL/IELTS	-0.0478**	0.0229	-0.0723	0.0152	-0.0427	0.0050
	(0.0231)	(0.0187)	(0.0645)	(0.0466)	(0.0469)	(0.0187)
Elite U.S. University	-0.0496*	-0.0100	0.0054	0.0297	0.0229	0.0036
	(0.0261)	(0.0145)	(0.0324)	(0.0297)	(0.0274)	(0.0108)
Elite U.S. Liberal Arts College	0.0248	0.0078	0.0223	0.0303	0.0502	0.0217
	(0.0312)	(0.0180)	(0.0469)	(0.0602)	(0.0418)	(0.0160)
Elite Foreign University	-0.0166	-0.0532	0.0700	0.0299	0.1721***	0.0158
	(0.0308)	(0.0337)	(0.0537)	(0.0490)	(0.0484)	(0.0179)
Other Foreign University	-0.0111	0.0091	0.0257	0.0108	0.0175	0.0116
	(0.0202)	(0.0122)	(0.0321)	(0.0267)	(0.0344)	(0.0103)
Graduate Degree	-0.0020	0.0079	-0.0151	-0.0542***	-0.0161	-0.0093
	(0.0166)	(0.0109)	(0.0261)	(0.0206)	(0.0220)	(0.0085)
Work Experience	0.0258*	-0.0017	-0.0070	-0.0165	-0.0602**	-0.0093
	(0.0148)	(0.0107)	(0.0248)	(0.0213)	(0.0275)	(0.0086)
#Professor Recommenders	-0.0108	0.0041	-0.0173	0.0136	-0.0288***	-0.0027
	(0.0093)	(0.0051)	(0.0140)	(0.0114)	(0.0110)	(0.0042)
#Prolific Recommenders	0.0004	0.0098*	-0.0165	0.0042	0.0298**	0.0083*
	(0.0067)	(0.0050)	(0.0164)	(0.0102)	(0.0128)	(0.0045)
#Math Courses	0.0037					
	(0.0043)					
Advanced Math	0.0201					
	(0.0145)					
Missing GRE	-0.0204	-0.0090	-0.0001	-0.0027	-0.0350	-0.0098
	(0.0443)	(0.0098)	(0.0254)	(0.0218)	(0.0239)	(0.0079)
Missing Undergraduate GPA	0.0157	0.0040	-0.0232	0.0392*	-0.0207	0.0019
	(0.0158)	(0.0102)	(0.0285)	(0.0201)	(0.0254)	(0.0083)
Missing TOEFL/IELTS	-0.0302*	0.0218	0.0544	0.0084	-0.1205***	0.0063
	(0.0168)	(0.0156)	(0.0517)	(0.0272)	(0.0434)	(0.0130)
Concentration FE	No	Yes	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486	4144
Wald Statistic | DF | p-Value	37.4658	23	0.0260			

Robust standard errors in parentheses. *p<0.10, **p<0.05, ***p<0.01.

- **Economics Positive ($p < .1$):** GRE Verbal Percentile, GRE Quantitative Percentile, Work Experience
- **Economics Negative ($p < .1$):** Age, TOEFL/IELTS, Elite U.S. University, Missing TOEFL/IELTS

Insufficiency of Comparing AMEs

A4: Graduate School Lasso Logistic Regression Results

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology
Covid Year	-0.3330	-0.3123	-0.3498	-0.2378	
Age	-0.0788	0.0266	0.0200		
Female	0.3352	0.5096	0.5068		-0.0654
U.S. African/Hispanic/Native	0.2638	0.8041	0.7157		
U.S. Asian	-0.2802	-0.2918		0.0413	0.1155
International Asian	-0.6744	-0.0726		0.0465	
International Other	0.4729	0.1946	0.7240	-0.1087	
GRE Verbal Percentile	0.0139	0.0252	0.0272		0.0343
GRE Quantitative Percentile	0.1349	0.0119	0.0187	0.0154	
GRE Analytic Percentile	0.0166	0.0208		0.0080	
Undergraduate GPA	1.9977	0.4621	0.7414	0.1714	
TOEFL/IELTS	0.4009	0.3295	0.5199	0.5151	
Elite U.S. University	0.6886	0.2709	0.3267		0.4606
Elite U.S. Liberal Arts College	0.8956	0.5199	0.5985		0.4470
Elite Foreign University	0.3724		0.0414	0.4343	
Other Foreign University					
Graduate Degree			0.0978		
Work Experience	0.2835	0.3447	0.3273		0.5559
#Professor Recommenders			0.3739	0.2322	
#Prolific Recommenders	0.2243	0.2362	0.1651	0.2213	0.0632
#Math Courses	0.1493				
Advanced Math	0.2272				
Missing GRE		-0.2304		-0.0527	-0.5416
Missing Undergraduate GPA					
Missing TOEFL/IELTS	-0.2364	0.2852	0.0796		
Concentration FE	No	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486
Pseudo-R^2	0.2642	0.1456	0.1589	0.1486	0.1362
CV Pseudo-R^2	0.2297	0.0912	0.0592	0.0850	0.0537

⏩ Insufficiency of Comparing AMEs

- **Economics Positive:** Female, U.S. African/Hispanic/Native, International Other, GRE Verbal Percentile, GRE Quantitative Percentile, GRE Analytic Percentile, Undergraduate GPA, TOEFL/IELTS, Elite U.S. University, Elite U.S. Liberal Arts College, Elite Foreign University, Work Experience, # Prolific Recommenders, #Math Courses, Advanced Math
- **Economics Negative:** Covid Year, Age, U.S. Asian, International Asian, Missing TOEFL/IELTS

References I

Anderson, A., J. Rosen, J. Rust & K.-P. Wong (2025). "Disequilibrium Play in Tennis". In: *Journal of Political Economy* 133.1, pp. 190–251.

Angrist, J., M. Diederichs & G. Ellison (2026). "Will The University Endowment Tax Slow Scientific Progress? Evidence from Elite Economics PhD Programs". In: *NBER Working Paper 35244*.

Athey, S., L. F. Katz, A. B. Krueger, S. Levitt & J. Poterba (2007). "What Does Performance in Graduate School Predict? Graduate Economics Education and Student Outcomes". In: *American Economic Review* 97.2, pp. 512–520.

Athey, S. & M. Luca (2019). "Economists (and Economics) in Tech Companies". In: *Journal of Economic Perspectives* 33.1, pp. 209–230.

Bai, J., M. Esche, W. B. MacLeod & Y. Shi (2022). "Subjective Evaluations and Stratification in Graduate Education". In: *NBER Working Paper 30677*.

Baron, J., W. Dobbie, R. Lombardo, A. Rambachan & J. Ryan (2026). "Human Decisions and Machine Predictions in Multistage Systems". In: *Working Paper*.

Cho, S. & J. Rust (2010). "The Flat Rental Puzzle". In: *Review of Economic Studies* 77.2, pp. 560–594.

Heckman, J. J. (1979). "Sample Selection Bias as a Specification Error". In: *Econometrica* 47.1, pp. 153–161.

Hoffman, M., L. B. Kahn & D. Li (2018). "Discretion in Hiring". In: *The Quarterly Journal of Economics* 133.2, pp. 765–800.

Kleinberg, J., H. Lakkaraju, J. Leskovec, J. Ludwig & S. Mullainathan (2018). "Human Decisions and Machine Predictions". In: *The Quarterly Journal of Economics* 133.1, pp. 237–293.

Krueger, A. B. & S. Wu (2000). "Forecasting Job Placements of Economics Graduate Students". In: *The Journal of Economic Education* 31.1, pp. 81–94.

Lessem, R. & R. A. Miller (2026). "Matching Job Applicants to Vacancies: An Empirical Model of Multistage Hiring". In: *Working Paper*.

Lewis, M. (2004). *Moneyball: The Art of Winning an Unfair Game*. W.W. Norton & Company.

References II

- Madera, J. M., M. R. Hebl & R. C. Martin (2009). "Gender and Letters of Recommendation for Academia: Agentic and Communal Differences". In: *Journal of Applied Psychology* 94.6, pp. 1591–1599.
- Matějka, F. & A. McKay (2015). "Rational Inattention to Discrete Choices: A New Foundation for the Multinomial Logit Model". In: *American Economic Review* 105.1, pp. 272–298.
- Mood, C. (2010). "Logistic Regression: Why We Cannot Do What We Think We Can Do, and What We Can Do About It". In: *European Sociological Review* 26.1, pp. 67–82.
- Mullainathan, S. & J. Spiess (2017). "Machine Learning: An Applied Econometric Approach". In: *Journal of Economic Perspectives* 31.2, pp. 87–106.
- Patki, N., R. Wedge & K. Veeramachaneni (2016). "The Synthetic data vault". In: *IEEE International Conference on Data Science and Advanced Analytics (DSAA)*, pp. 399–410.
- Rambachan, A. (2024). "Identifying Prediction Mistakes in Observational Data". In: *The Quarterly Journal of Economics* 139.3, pp. 1665–1711.
- Sims, C. A. (2003). "Implications of rational inattention". In: *Journal of Monetary Economics* 50.3, pp. 665–690.
- Stock, W. A., T. A. Finegan & J. J. Siegfried (2006). "Attrition in Economics Ph.D. Programs". In: *American Economic Review* 96.2, pp. 458–466.
- Stock, W. A. & J. J. Siegfried (2015). "The Undergraduate Origins of PhD Economists Revisited". In: *The Journal of Economic Education* 46.2, pp. 150–165.
- Weitzman, M. L. (1979). "Optimal Search for the Best Alternative". In: *Econometrica* 47.3, pp. 641–654.
- Young, L. & S. Soroka (2012). "Affective News: The Automated Coding of Sentiment in Political Texts". In: *Political Communication* 29.2, pp. 205–231.